LM01 Introduction to Financial Statement Analysis

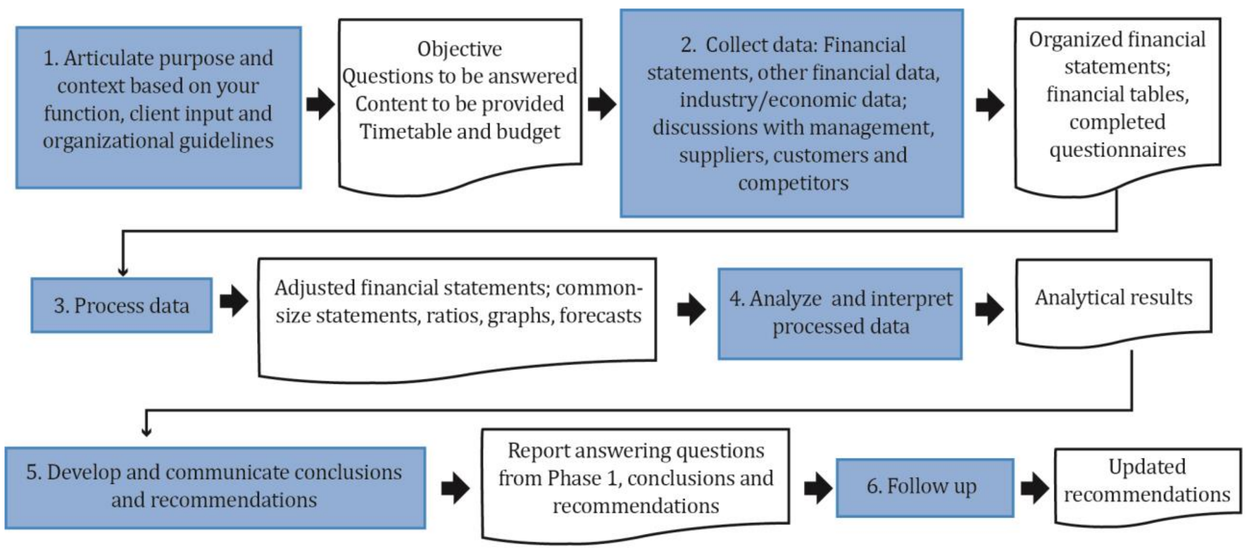
Ver 1.0

1. Introduction

Financial analysis is the process of examining a company's performance. For this purpose, financial reports are one of the most important sources of information available to a financial analyst. A financial analyst must have a strong understanding of the information provided in a company's financial reports, notes, and supplementary information.

2. Financial Statement Analysis Framework

A generic financial statement analysis framework is summarized in the figure below. The grey boxes represent phases of financial analysis while the white boxes represent outputs from each phase.



Articulate the Purpose and Context of Analysis

In this step, we understand the purpose of the analysis. For example, an equity analyst analyzes the financial reports in order to decide whether to invest in the stocks of the company or not. On the other hand, a credit analyst looks at the company in a very different light in order to judge whether it should be given a loan or not.

Next, the analyst defines the context which includes details such as the intended audience, time frame, budget, and so on. Once the purpose and the context are defined, the analyst compiles the specific questions to be answered by the analysis, decides on the content to be prepared, and finalizes the timeline and the budget.

Collect Data

Next, the analyst collects data required to answer the questions compiled in the previous step. The sources of data are financial reports and other information sources.

The output from this step includes organized financial statements, financial tables, and completed questionnaires.

Process Data

After collecting data, the analyst processes the data using appropriate analytical tools. This involves:

- Making any adjustments to the financial statements to facilitate comparison. For example, adjustments will be required to compare a company using IFRS with a company using US GAAP.
- Creating graphs, ratios, common-size statements, etc.

The output from this step includes adjusted financial statements, common-size statements, ratios, graphs, and forecasts.

Analyze/Interpret the Processed Data

The next step is to interpret the processed data and come up with a decision. For example, an equity analyst may come up with a buy, sell, or hold decision.

Develop and Communicate Conclusions/Recommendations

Next, the analyst communicates the conclusions or recommendations in the appropriate format. For example, an equity analyst will prepare a research report and send it to his firm's clients.

Follow-up

Conduct periodic reviews to check if the previous conclusions are still valid. Change the conclusions/recommendations when necessary. For example, an equity analyst may send quarterly updates on his initial buy, sell, or hold recommendation.

3. Scope of Financial Statement Analysis

In order to understand financial analysis, we first need to understand the difference between the roles of financial reporting and financial statement analysis.

Financial reporting

The role of financial reporting is to provide information about a company's performance (income statement and cash flow statement), financial position (balance sheet) and changes in financial position (statement of changes in equity).

Financial statement analysis

The role of financial statement analysis is to use the financial reports prepared by firms and combine them with other sources of information to decide if you can invest in the equity of the firm or lend money to the firm.

4. Regulated Sources of Information

Standard-setting bodies

Standard-setting bodies are private sector organizations that help develop financial reporting standards. The two important standard-setting bodies are:

- Financial Accounting Standards Board (FASB) – For the U.S. The standards developed by FASB are called U.S. GAAP (Generally Accepted Accounting Principles).
- International Accounting Standards Board (IASB) – For the rest of the world. The standards developed by IASB are called IFRS (International Financial Reporting Standards).

Standard-setting bodies simply set the standards but they do not have the authority to enforce the standards.

Regulatory Authorities

Regulatory authorities are government entities that have legal authority to enforce the financial reporting standards. For example: Securities and Exchange Commission (SEC) – for the U.S.

Regulatory authorities are also responsible for the regulation of capital markets under their jurisdiction.

The International Organization of Securities Commissions (IOSCO)

Technically not being a regulatory authority, IOSCO still regulates a significant portion of the world's financial markets. (Think of it as an umbrella organization of regulatory authorities). This organization has established objectives and principles to guide securities and capital market regulation.

Core objectives

- Protect investors.
- Ensure fairness, efficiency, and transparency in markets.
- Reduce systemic risk.

Principles

- There should be full, accurate, and timely disclosure of financial results and risks.
- Financial statements should be of a high and internationally acceptable quality.

Primary Financial Statements

Instructor's Note: The financial statements mentioned below will be covered in a lot more detail in later readings. At this stage, you simply need to understand the basics.

The curriculum presents this information in terms of SEC filings – e.g. Forms 10-K, 20-F, DEF-14A, 10-Q etc. You are not expected to memorize these forms. Also, exam questions are unlikely to test country-specific regulations. For these reasons, we have simplified the explanation in this section for better comprehension.

The primary financial statements are the balance sheet, the income statement, the cash flow statement, and the statement of changes in owners' equity.

Balance Sheet (Statement of Financial Position)

The balance sheet reports the firm's financial position at a specific point in time. It has the following elements:

- *Assets* – What the company owns.
- *Liabilities* – What the company owes.
- *Owners' equity* – What the shareholders of the company own. Depending on the form of the organization, owners' equity may be referred to as "partners' capital" or "shareholders' equity" or "shareholders' funds", or "net assets".

The relationship between the elements can be shown as:

Assets = Liabilities + Owners' equity

The capital structure of a company represents the combination of liabilities and equity used to finance its assets. Both financial position and capital structure are useful in credit analysis.

Income statement

The income statement reports the financial performance of the firm over a period of time. It has the following elements:

- *Revenues* – Income generated by selling goods and services.
- *Expenses* – Costs incurred for producing goods and services.
- *Net income* – Resulting profit or loss.

The relationship between the elements can be shown as:

Net income = Revenues - Expenses

Cash flow statement

The cash flow statement reports the sources and uses of cash for the firm over a period of time. It has the following elements:

- *Operating cash flows* – Cash flows from day-to-day activities.
- *Investing cash flows* - Cash flows associated with the acquisition and disposal of long-term assets, such as property and equipment.
- *Financing cash flows* - Cash flows from activities related to obtaining or repaying capital.

Statement of changes in owner's equity

It reports the changes in the owners' investment in the firm over time. It has the following elements:

- *Paid in capital* – Amount raised from owners.
- *Retained earnings* – Firm's profits that have been retained (i.e., not paid out as dividends).

Along with these required financial statements (mentioned above), a company typically provides additional information in its financial reports. This includes footnotes, management's commentary, and auditor's report.

Footnotes

They provide additional details about the information presented in financial statements. This includes important information about the accounting methods, estimates, and assumptions. They also contain information regarding acquisitions and disposals, commitments and contingencies, legal proceedings, employee stock options and other benefits, related party transactions and business, and geographic segments.

Management's commentary or Management Discussion and Analysis (MD&A)

It provides an assessment of the data reported in the financial statements from the management's perspective. Examples of content include trends and significant events affecting the company's operations, liquidity and capital resources, off-balance sheet obligations, and planned capital expenditures.

Auditor's report

An audit is an independent review of a firm's financial statements. It enables the auditor to express an opinion on the fairness and reliability of the financial reports. An audit report can contain one of the following opinions:

- *Unqualified Opinion* - Reasonable assurance that financial statements are fairly presented. This is also referred to as an "unmodified" or a "clean" opinion. (This is the opinion that you would like to see.)
- *Qualified Opinion* - Some misstatement or exception to accounting standards.
- *Adverse Opinion* - Financial statements materially depart from accounting standards and are not presented fairly.

  If the auditor is not able to issue an opinion for reasons such as a scope limitation, a disclaimer of opinion is issued.

For listed companies, the audit report also includes a discussion of Key Audit Matters (international) and Critical Audit Matters (US). Key Audit Matters and Critical Audit Matters include issues having a higher risk of misstatement, involving significant management judgment and estimates.

Other sources:

- Interim reports – Quarterly or semiannual reports prepared by the firm. These reports are not audited.
- Proxy statements – Statements distributed to shareholders about matters that are to be put to a vote, e.g. management and director compensation.

5. Comparison of IFRS with Alternative Financial Reporting Systems

A significant percentage of listed companies use either IFRS or US GAAP. An analyst must be cautious when comparing financial measures between companies reporting under IFRS and companies reporting under US GAAP. If needed, specific adjustments need to be made to achieve comparability.

US GAAP uses standards issued by FASB while IFRS uses standards issued by IASB. While the two organizations are working towards convergence, significant differences still remain. Exhibit 6 from the curriculum presents some of the differences.

Basis for Comparison	**US GAAP**	**IFRS**
Developed by	Financial Accounting Standards Board (FASB)	International Accounting Standards Board (IASB)
Based on	Rules	Principles
Interest paid	Cash Flows from Operating Activities	Cash Flows from Financing Activities or Cash Flows from Operating Activities
Inventory valuation	First in, First out (FIFO); Last in, First out (LIFO); and Weighted Average Method	FIFO and Weighted Average Method
Development cost	Treated as an expense	Capitalized, only if certain conditions are satisfied
Reversal of Inventory Write-down	Prohibited	Permissible, if specified conditions are met

Monitoring Developments in Financial Reporting Standards

Analysts must be aware that reporting standards are evolving rapidly. They need to monitor developments in financial reporting and assess their implications for security analysis and valuation.

A financial analyst can remain aware of developments in financial reporting standards by monitoring three sources:

- new products or transactions
- actions of standard setters and groups representing users of financial statements
- company's disclosures regarding critical accounting policies and estimates.

New Products or Types of Transactions

New products or transactions can emerge as a result of economic events such as new businesses (e.g. fintech) or a newly developed financial instrument (e.g. crypto currencies). Analysts should investigate whether these products or transactions were created for the purpose of 'window dressing' financial reports.

If needed, an analyst can obtain additional information from the company's management, which should be able to describe the economic purpose, financial statement reporting, significant estimates, judgments used in determining the reporting, and future cash flow implications for these items.

Evolving Standards and the Role of CFA Institute

The IASB and FASB both provide extensive information on their websites about new standards and proposed changes to the standards. Furthermore, both bodies seek feedback from the financial analyst community on how the standards can be improved further.

CFA Institute actively works in supporting improvements to financial reporting. CFA Institute volunteers serve on several liaison committees that meet on a regular basis to make recommendations to the IASB and FASB.

6. Other Sources of Information

In addition to the information required by regulatory authorities, analysts can also obtain information from the following sources.

- Issuer sources (other than regulatory filings such as annual and quarterly reports and proxy statements)
    - Earnings call
    - Presentations and events, such as investor days.
    - Press releases
    - Discussion with management, investor relations, or other company personnel.
    - Company website
- Public third-party sources
    - Free industry whitepapers or analyst reports from a consultancy
    - Economic or industry indicators from governments and other organizations, e.g. retail sales and price indexes.
    - General news outlets
    - Industry-specific news outlets
    - Social media
- Proprietary third-party sources
    - Analyst reports and communications
    - Reports and data from platforms such as Bloomberg
    - Reports and data from consultancies such as Rystad for the energy industry

- Proprietary primary research
    - Surveys, conversations, product comparisons, and other studies commissioned by the analyst or conducted directly.

Summary

LO: Describe the steps in the financial statement analysis framework.

The financial statement analysis framework consists of the following six steps:

1. Define the purpose and context of the analysis.
   - Define the purpose and context of the analysis based on your function, client inputs, and organizational guidelines.
   - Determine the time frame and the resources available for the task.
2. Collect data.
   - Collect data from financial statements and other information sources.
3. Process the data.
   - Make adjustments to financial statements.
   - Create graphs, ratios, common-sizes statements, etc.
4. Analyze and interpret the data.
5. Develop and communicate conclusions and recommendations.
6. Follow up.
   - Conduct periodic reviews to check if previous conclusions are still valid.

LO: Describe the roles of financial statement analysis.

The role of financial reporting is to provide information about a company's performance (income statement and cash flow statement), financial position (balance sheet) and changes in financial position (statement of changes in equity).

The role of financial statement analysis is to use the financial reports prepared by firms and combine them with other sources of information to decide if you can invest in the equity of the firm or lend money to the firm.

LO: Describe the importance of regulatory filings, financial statement notes and supplementary information, management's commentary, and audit reports.

The balance sheet reports the firm's financial position at a specific point in time. It shows the firm's assets, liabilities, and owners' equity.

The income statement reports the financial performance of the firm over a period of time. It shows the firm's revenues, expenses, and net income.

The cash flow statement reports the sources and uses of cash for the firm over a period of time. It shows the firm's operating, investing, and financing cash flows.

Statement of changes in owner's equity reports the changes in the owners' investment in the firm over time. It shows the firm's paid in capital and retained earnings.

The notes (also called footnotes) are important as they disclose information about the accounting policies, methods, and estimates used to prepare the financial statements. They contain important information regarding acquisitions and disposals, commitments and contingencies, legal proceedings, employee stock options and other benefits, related party transactions and business, and geographic segments.

Management's commentary comprises of subjective information where management is presenting its view and interpretation of the data it has reported. Examples of content include trends and significant events affecting the company's operations, liquidity and capital resources, off-balance sheet obligations, and planned capital expenditures.

An audit is an independent review of a firm's financial statements. It enables the auditor to express an opinion on the fairness and reliability of the financial reports. An audit report can contain one of the following opinions:

- <u>Unqualified Opinion</u> - Reasonable assurance that financial statements are fairly presented.
- <u>Qualified Opinion</u> - Some misstatement or exception to accounting standards.
- <u>Adverse Opinion</u> - Financial statements are not presented fairly.

LO: Describe implications for financial analysis of alternative financial reporting systems and the importance of monitoring developments in financial reporting standards.

Differences still exist between IFRS and US GAAP that affect financial reporting. Analysts must be aware of the areas where accounting standards have not converged.

Analysts must be aware that reporting standards are evolving rapidly. They need to monitor developments in financial reporting and assess their implications for security analysis and valuation.

LO: Describe information sources that analysts use in financial statement analysis besides annual and interim financial reports.

In addition to the information required by regulatory authorities, analysts can also obtain information from the following sources:

- Earnings calls
- Investor day events
- Press releases
- Company website and company visits
- Speaking with the management
- Industry white papers
- Analyst reports
- Economic information from government
- Analysts can also generate their own information through surveys, conversations, and product evaluations.

LM02 Analyzing Income Statements

Ver 1.0

1. Introduction

The income statement presents information on the financial results of a company's business activities over a period of time. It is also known as the 'statement of operations', 'statement of earnings', or 'profit and loss (P&L) statement'. The basic equation underlying the income statement is:

Income - Expenses = Net Income

Equity analysts carefully analyze a company's income statements for use in valuation models while fixed-income analysts analyze income statements to measure a company's debt servicing ability.

Components of the income statement

The components of an income statement are:

Revenues: Income generated from the sale of goods and services in the normal course of the business. Net revenue is the total revenue minus products that were returned and amounts that are unlikely to be collected.

Expenses: Costs incurred to generate revenues. Expenses may be grouped and reported in different formats, subject to some specific requirements.

Gains and losses: Amounts generated from non-operating activities.

Net income: Net income can be calculated as Net income = Revenues – Expenses + Gains – Losses.

A sample income statement is shown below:

Multi-step format		
$ million	2018	2017
Sales	35,310	31,600
Cost of sales	10,300	9,060
Gross Profit	25,010	22,540
Gain from sale of equipment	900	860
Administrative expenses	3,400	2,900
Advertising expense	1,000	900
Depreciation	960	850
Other expenses	6,500	6,100
Operating Income (EBIT)	14,050	12,650
Interest Expense	10	70
Profit before tax (EBT)	14,040	12,580
Tax Expense	3,945	3,300
Profit after tax	10,095	9,280

2. Revenue Recognition

General Principles

Under the accrual method of accounting, revenue should be recognized when earned and not necessarily when cash is received. Let us consider three simple examples to illustrate this point.

- If a company sells goods for $100 cash in Period 1, can it recognize revenue in Period 1? The answer is yes. Revenue is recorded in the period it is earned, i.e., when goods or services are delivered.
- What if the company sells goods on credit in Period 1 and expects to receive cash in Period 2? Can revenue be recognized in Period 1? The answer is that revenue is recorded in Period 1. In addition, since the goods are sold on credit, an asset called accounts receivable is created.
- What if an advance payment is received in Period 1 but goods and services are to be delivered in Period 2. When will the revenue be recognized? The revenue will be recognized in Period 2 because that is when delivery of goods will take place. In this case, the company will record a liability called unearned revenue when the advance payment is received.

Companies must disclose their revenue recognition policies in the notes to their financial statements, and analysts should read these carefully to understand how and when a company recognizes revenue.

Accounting Standards for Revenue Recognition

In May 2014, the IASB and FASB issued converged standards for revenue recognition. The standards take a principles-based approach to revenue recognition issues. The core principle behind the converged standard is that revenue should be recognized to "depict the transfer of promised goods or services to customers in an amount that reflects the consideration to which the entity expects to be entitled in an exchange for those goods or services."

According to the standard, the following five steps must be followed in order to recognize revenue:

1. Identify the contract(s) with a customer.
2. Identify the performance obligations in the contract.
3. Determine the transaction price.
4. Allocate the transaction price to the performance obligations in the contract.
5. Recognize revenue when (or as) the entity satisfies a performance obligation.

When revenue is recognized, a contract asset is added to the balance sheet. If an advance is received but performance obligations have not been met, then a contract liability is added to the seller's balance sheet.

The entity will recognize revenue when it is able to satisfy performance obligation by transferring control of the goods or service to the customer. The following factors can be considered to determine whether the customer has gained control:

- Entity has a present right to payment
- Customer has legal title
- Customer has physical possession
- Customer has the significant risks and rewards of ownership
- Customer has accepted the good or service

Analysts can encounter many companies with complex revenue recognition policies. Several examples adapted from real companies are presented in the example below.

Instructor's Note: Understand the core concepts covered in this example. On the exam you will probably be tested on a short mini-scenario resembling the scenarios below.

Example: Applying the Converged Revenue Recognition Standards
(*This is Example 1 from the curriculum.*)

Principal Versus Agent

MegaDigital is an online marketplace that sells goods and delivers them quickly to customers. For some sales, MegaDigital acts as a principal in which it controls the product before the goods are transferred to the customer. In other sales, MegaDigital acts as an agent in which it arranges for the transfer of a product controlled by a third-party seller. In transactions in which MegaDigital is the principal, revenue is recorded as the total amount of considerations received for the transfer of the product. In transactions in which MegaDigital is the agent, it records revenue only for the portion of the considerations, which amounts to its fee or commission. This can have a significant impact on common size and ratio analysis. Revenue is lower but profit margins are higher for sales for which MegaDigital is an agent.

Assume MegaDigital sells a particular product as a principal for USD100 that it purchased for USD70. Additionally, there are USD10 of other selling, general, and administrative costs. The margins would be:

Sales	USD100	100%
Cost of Sales	70	70%
Gross Profit	30	30%
SG&A	10	10%
Net Profit	20	20%

If MegaDigital acts an agent for the same item with the same retail price, MegaDigital would receive a commission of USD30 and still incurs USD10 of other costs. Margins would be:

Sales	USD30	100%

Cost of Sales	0	0%
Gross Profit	30	100%
SG&A	10	33%
Net Profit	20	67%

For companies selling both as a principal and agent, such as many e-commerce companies, an analyst would need to evaluate the relative proportion of principal versus agent sales to evaluate and forecast overall margins. This is especially important if the mix of principal and agent sales is expected to change.

Franchising/Licensing

Mahjong Pizza both operates and franchises pizza delivery restaurants around the world. Revenue recognition standards require that the company disaggregate revenue from contracts with customers into categories that depict how the nature, amount, timing, and uncertainty of revenue and cash flows are affected by economic factors. Companies must present revenues disaggregated in consolidated statements of income to satisfy this requirement. Mahjong Pizza presents the following disaggregated revenue items:

- company-owned stores revenues,
- franchise royalties and fees, and
- supply chain revenues.

Company-owned stores revenues are of retail sales of food at stores that Mahjong owns and operates.

Franchise royalties and fees are comprised of fees from third-party franchisees that are licensed to operate Mahjong restaurants. Each franchisee is generally required to pay fees equal to 5.5 percent of restaurant sales. The company recognizes the royalty fee as revenue, not the total sales of the franchisees' restaurants.

Upfront fees for opening new units are initially recognized as deferred revenue and subsequently amortized to revenue on a straight-line basis over the term of each respective franchise agreement, typically 10 years.

Supply chain revenues are primarily composed of sales of food, equipment, and supplies to franchisees. Revenues are recognized upon delivery or shipment of the related products to franchisees, based on shipping terms.

Software as a Service or License

CReaM Software and Services is a technology company providing customer relationship management software and services to business, government and not-for-profit organizations. Organizations may purchase a software license and install it on their own systems. Alternatively, they may subscribe to CReaM's cloud services platform through which they can access CReaM's software over the internet for a monthly subscription fee.

Under IFRS 15, if a company provides a license to use software where the company will take possession of the software for installation on their own system, the company will report revenue either over the term of the license or at the time of the transfer of the license. Companies should report the revenue from the license over the term of the license, if under the contract or the company's normal business activities:

- the software provider will continue to undertake activities that significantly affect the software (e.g., upgrades/enhancements)
- the rights expose the customer to positive or negative impacts from those activities, and
- the activities do not result in a transfer of goods or services.

If these criteria are not met, then the revenue is recognized when the license is transferred to the customer. CReaM's annual report footnotes state:

"Software revenues include revenues associated with term and perpetual software licenses that provide the customer with a right to use the software as it exists when made available. Revenues from term and perpetual software licenses are generally recognized at the point in time when the software is made available to the customer. Revenue from software support and updates is recognized as the support and updates are provided, which is generally ratably over the contract term."

Under the terms of CReaM's license, the software is sold "as is" and revenue is recognized at the time of the license transfer. CReaM, however, also provides a support contract for updates for which revenue is recognized over the contract term.

CReaM's cloud clients have access to constantly updated software. CReaM reports:

"Cloud services allow customers to use the Company's software without taking possession of the software. Revenue is generally recognized over the contract term. Substantially all of the Company's subscription service arrangements are non-cancelable and do not contain refund-type provisions."

In the case of CReaM, an analyst must understand the composition of revenue between licensed software in which case revenue is recognized upfront versus software as a service in which case revenue is recognized over time.

Long-Term Contracts

Armored Vehicles Inc. (AVI) manufactures weapons systems and vehicles for military customers. The company enters long-term contracts that generally extend over several years. Performance on the contracts is satisfied over time. Under IFRS 15, a performance obligation is satisfied over time if one of the following criteria is met:

- The customer simultaneously receives and consumes the benefits provided by the entity's performance as the entity performs (e.g., routine service contracts).
- The entity's performance creates or enhances an asset that the customer controls as the asset is created or enhanced (e.g., refurbishment of a factory owned and controlled by the customer or building a road for a governmental agency).
- The entity's performance does not create an asset with alternative use to the entity and the entity has an enforceable right to payment for performance completed to date (e.g., construction of a large unique asset that may not be able to be sold to another customer such as a weapons system).

AVI recognizes long-term contract revenue over the contract term as the work progresses, either as products are produced or as services are rendered because of the continuous transfer of control to the customer. For its military contracts, this continuous transfer of control to the customer is supported by clauses in the contract that allow the customer to unilaterally terminate the contract for convenience, pay for costs incurred plus a reasonable profit, and take control of any work in process.

Under IFRS 15, the extent of progress towards completion may be measured by output methods (e.g., appraisals or units completed) or input methods (e.g., costs incurred relative to estimated total costs). AVI reports that its accounting for long-term contracts involves a judgmental process of estimating total sales, costs and profit for each performance obligation. Cost of sales is recognized as incurred. The amount reported as revenues is determined by adding a proportionate amount of the estimated profit to the amount reported as cost of sales. Recognizing revenue as costs are incurred provides an objective measure of progress on the long-term contract and thereby best depicts the extent of transfer of control to the customer.

As an example, AVI has a contract to produce a weapons system for a total price of USD10 million. The expected total costs to produce the system is USD7 million and the estimated profit is USD3 million. The system will take two years to produce. In Year 1 of the contract, AVI incurs USD4.2 million of costs representing 60 percent of total estimated costs. AVI would recognize revenue of USD6 million and profit of USD1.8 million in Year 1 (both 60 percent of expected revenue and profits).

If in Year 2, the system is completed with actual total cumulative costs of USD7.5 million, the company would report revenue of USD4 million and costs of USD3.3 million for a Year 2 profit of USD0.7 million and cumulative profit of USD2.5 million.

Bill and Hold Arrangements

In addition to the long-term contracts discussed previously, AVI produces custom armored vehicles that some customers may not be able to take possession of immediately (because, for example, a lack of storage space). IFRS 15 provides that in such a "bill and hold" arrangement AVI can determine when it has satisfied its performance obligation based on when a customer obtains control of the product. Under IFRS 15, this is when all the following criteria are met:

- The reason for the bill and hold arrangement must be substantive (e.g., the customer has requested the arrangement).
- The product must be identified separately as belonging to the customer.
- The product currently must be ready for physical transfer to the customer.
- The entity cannot have the ability to use the product or to direct it to another customer.

In AVI's case, each vehicle is identified by a unique vehicle identification number and upon completion, title and risk of loss has passed to the customer. AVI recognizes revenue when the product is ready for delivery to the customer but is directed by the customer to hold delivery.

Disclosure requirements

The converged standard mandates the following disclosure requirements:

- Companies must disclose information about contracts with customers after segregating them into different categories of contracts. The categories may be based on the geographic region, the type of product, the type of customer, pricing terms, etc.
- Companies must disclose information related to revenue recognition. For example, any change in judgments, remaining performance obligations, and transaction price allotted to those obligations, and balances of contract-related assets and liabilities.

3. Expense Recognition

Expenses are 'decreases in economic benefit during the accounting period in the form of outflows or depletion of assets or incurrences of liabilities that result in decreases in equity.' For example, if a company pays rent, its cash reduces and the rent is recognized as an expense.

General Principles

Matching principle

The most important principle of expense recognition is the matching principle, under which the expenses incurred to generate revenue are recognized in the same period as revenue. For example, if some goods bought in the current year remain unsold at the end of the year, they are not included in the cost of goods sold for the current year. If they are sold in the next year, they will be included in the cost of goods sold for the next year.

Periodic costs

Expenses that cannot be tied directly to generation of revenues are called periodic costs. They are expensed in the period incurred. For example, the rent paid for office premises are simply expensed in the period for which the rent was paid.

Capitalization versus Expensing

Capitalizing: In general, when a company acquires a long-lived tangible or intangible asset, its cost is capitalized if the asset is expected to provide economic benefits beyond a year. The company records an asset in an amount equal to the acquisition cost plus any other cost to get the asset ready for its intended use.

Capitalizing results in spreading the cost of acquiring an asset over a specified period of time instead of immediately expensing it. All other costs to make the asset ready for intended use are also capitalized. Capitalizing leads to higher profitability in the period when the asset is purchased. The effect of capitalizing an expenditure on the financial statements is summarized below:

Effect of capitalization on financial statements	
Initially when an expenditure is capitalized.	Balance sheet: non-current assets increase by the capitalized amount.
	Statement of cash flows: investing cash flow decreases.
Subsequent periods over the asset's useful life.	Income statement: depreciation or amortization expense. Net income decreases.
	Balance sheet: non-current assets (carrying value of the asset) decreases. Retained earnings decreases. Equity decreases.

Expensing: The cost of an asset is expensed if it has uncertain or no impact on future earnings and provides economic benefit only in the current period. Immediate recognition of an asset's cost as an expense on the income statement results in lower profitability in the current period and higher profits in the future.

Effect of expensing on financial statements	
When an expenditure is expensed.	Income statement: Net income decreases by the after-tax amount of the expenditure. No depreciation/amortization expense.
	Balance sheet: No asset is recorded. Lower retained earnings due to lower net income.
	Statement of cash flow: Operating cash flow decreases.

The table below summarizes the effects of capitalizing versus expensing on various financial statement items.

	Capitalizing	*Expensing*
Total assets	Higher	Lower
Equity	Higher	Lower
Income variability	Lower	Higher
Net income (1st year)	Higher	Lower
Net income (later)	Lower	Higher
CFO	Higher	Lower
CFI	Lower	Higher
D/E	Lower	Higher
Interest coverage (initially)	Higher	Lower
Interest coverage (later)	Lower	Higher
ROA and ROE (initially)	Higher	Lower
ROA and ROE (later)	Lower	Higher

Capitalization of Interest Costs

When an asset requires a long period of time to get ready for its intended use, the interest costs associated with constructing or acquiring the asset are capitalized.

If the interest expenditure is incurred in connection with constructing an asset for the company's own use, capitalized interest is reported as part of the asset's cost on the balance sheet; in the future, it is reported as part of the asset's depreciation expense in the income statement.

If the interest expense is incurred in connection with the construction of an asset to be sold, such as by a real estate construction company, the capitalized interest is recorded on the company's balance sheet as inventory. When the asset is sold, the capitalized interest is expensed as part of the cost of sales.

Effect of Capitalized Interest on Financial Statements:

- Higher net income and greater interest coverage ratios during the period of capitalization.
- Higher asset values and depreciation lead to lower net income, EBIT and interest coverage ratio in the subsequent periods.

To provide a true picture of a company's interest coverage, interest coverage ratios should be calculated using the entire amount of interest expenditure, including both the capitalized and expensed portions.

Also, if a company is depreciating interest that was capitalized in a previous period, income should be adjusted to remove the effect of that depreciation.

Capitalization of Internal Development Costs

Accounting standards require companies to capitalize software development costs after a product's feasibility is established. Because determining feasibility requires judgment, capitalization practices may differ between companies.

Effect on Financial Statements:

- Income statement - Expensing rather than capitalizing development costs results in lower net income in the current period.
- Cash flow statement – Expensing rather than capitalizing development costs results in lower net operating cash flows and higher net investing cash flows.

While comparing a company that mostly expenses its software development costs with a company that mostly capitalizes its software development costs, analysts can make the following adjustments.

For the company that capitalizes:

- Adjust the income statement to include software development costs as an expense and exclude amortization of prior year's software development costs.
- Adjust the balance sheet to exclude capitalized software. This will decrease assets and equity.
- Adjust the cash flow statement to decrease operating cash flows and decrease cash used in investing by the amount of the current period development costs.

Implications for Financial Analysts: Expense Recognition

If a company's policies result in early recognition of expenses, it can be considered a conservative approach. On the other hand, if a company's polices delay the recognition of expenses, it can be considered an aggressive approach. Using this as well as other information contained in the footnotes or disclosures, an analyst can recognize whether a company's expense recognition policy is conservative or not. The analyst should also recognize that it is possible that two companies in the same industry have very different expense recognition policies.

4. Non-Recurring Items

Analysts are generally trying to estimate and assess future earnings of a company. Hence, reporting standards require firms to separate income and expense items that are likely to continue in the future, from items that are not likely to continue. (You will be more interested in items that will continue as compared to one-time items.)

Unusual or Infrequent Items

Both IFRS and US GAAP allow recognition of items that are unusual or infrequent. These items are also called exceptional items i.e., items not "inherent" to the company's current activities. Examples include restructuring charges and gains/losses from sale of equipment, receipts from a legal case, costs of integrating an acquisition, and impairment of intangible assets, etc. These items are shown as part of a company's continuing operations but are presented separately.

While forecasting future earnings, an analyst should assess whether the items reported as unusual or infrequent are likely to reoccur. Analysts should not simply ignore all unusual items.

Discontinued Operations

A discontinued operation is an operation which a company has disposed of or plans to dispose of. Net income from discontinued operations is shown (as a separate line item on the income statement) net of tax after net income from continuing operations.

Assets and liabilities related to the discontinued operations are aggregated and recognized on the balance sheet as held for sale.

Since the discontinued operation will no longer provide earnings to the company, an analyst may exclude discontinued operations when forecasting future earnings.

Changes in Accounting Policy

At times, new accounting standards may require companies to change accounting policies. An example can be changing the inventory valuation method from last in, first out (LIFO) to first in, first out (FIFO). Companies are allowed to adopt standards prospectively (in the future) or retrospectively.

- Retrospective application means that the financial statements for previous years are presented as if the newly adopted accounting principle had been used throughout the period. A change in accounting policy is applied retrospectively. For example, if a company shifts from LIFO valuation method to FIFO valuation method, this change will require a retrospective application.
- Prospective application means that only the financial statements for the period of change and for future periods are changed. Financial statements for previous years are not changed. At times, new standards might require companies to change accounting estimates such as the useful life of a depreciable asset. Changes in accounting estimates are applied prospectively.
- Correction of an error for a prior period is another possible adjustment which requires a restatement of the four major financial statements. If a company is making corrections very often, this gives a negative signal and investors will avoid investing in such a company.

Modified retrospective approach: According to new revenue recognition standards, companies can also use "modified retrospective" method of adoption. Under this approach, companies can adjust opening balances of retained earnings (and other applicable accounts) for the cumulative impact of the new standard. They are not required to revise previously reported financial statements.

5. Earnings Per Share

Simple versus Complex Capital Structure

Earnings per share (EPS) is a very important profitability measure. It depicts the earnings per ordinary share. Some basic terminologies related to EPS are:

- Potentially dilutive securities: Securities that can be converted into ordinary shares are called potentially dilutive securities. This includes convertible bonds, convertible preferred stock, and employee stock options.
- Simple capital structure: If a company has no potentially dilutive securities it is said to have a simple capital structure.
- Complex capital structure: If a company has potentially dilutive securities it is said to have a complex capital structure.
- Dilutive securities: A potentially dilutive security that decreases EPS when exercised is called a dilutive security.
- Antidilutive security: A potentially dilutive security that increases EPS when exercised is called an antidilutive security.

Basic EPS

Basic EPS is the amount of income available to common shareholders divided by the weighted average number of common shares outstanding over a period. Basic EPS is calculated as:

$$\text{Basic EPS} = \frac{\text{Net income} - \text{Preferred dividends}}{\text{Weighted average number of shares outstanding}}$$

In this calculation we do not consider the effect of any potentially dilutive securities.

Weighted average number of shares outstanding is the number of shares outstanding during the year, weighted by the portion of the year they were outstanding. Stock splits and stock dividends are applied retroactively to the beginning of the year, so the old shares are converted to new shares for consistency.

Example

During 2018, Company ABC had a net income of $100,000. It paid $22,000 as dividends to its preference shareholders and $12,000 as dividends to its common shareholders. The number of common shares outstanding during 2018 was as follows:

Shares as of January 1, 2018:	10,000
Additional shares issue on July 1, 2018:	2,000

Calculate the basic EPS of the company for 2018.

Solution:

We had 10,000 shares outstanding for the first 6 months and 12,000 shares outstanding for the last 6 months.

Therefore, weighted average number of shares outstanding = 10,000 x 6/12 + 12,000 x 6/12 = 11,000 shares.

$$\text{Basic EPS} = \frac{\$100{,}000 - \$22{,}000}{11{,}000} = \$7.09$$

Note: We ignore dividend paid to common shareholders.

Diluted EPS: The If-Converted Method

In this calculation, we consider the effect of potentially dilutive securities. If a firm has a complex capital structure it has to report both basic and diluted EPS. Diluted EPS is calculated as:

$$\text{Diluted EPS} = \frac{\text{Net Income} + \text{After tax interest} - \text{Preferred dividend} + \text{convertible preferred dividends}}{\text{Weighted Average Shares} + \text{New shares if convertible debt is converted}}$$

For 'preference shares', we need to subtract preference share dividends from the numerator and add new shares issued from conversion to the denominator.

Example

During 2018, Company ABC had a net income of \$100,000. It paid \$22,000 dividends to its preference shareholders and \$12,000 dividends to its common shareholders. It had 2,200 preference share and 11,000 common shares outstanding during 2018. Each preference share is convertible into 2 shares of common stock. Calculate the diluted EPS for the company.

Solution:

Number of common shares issued upon conversion = 2,200 x 2 = 4,400

$$\text{Diluted EPS} = \frac{\text{NI} + \text{conv debt int }(1-\text{t}) - \text{pref div} + \text{conv pref div}}{\text{Wt avg shares} + \text{New shares issued}}$$

$$\text{Diluted EPS} = \frac{\$100{,}000 + 0 - \$22{,}000 + \$22{,}000}{11{,}000 + 4{,}400} = \$6.5$$

For 'convertible bonds', we need to add the after-tax interest cost savings to the numerator and new shares issued from conversion to the denominator.

Example

During 2018, Company ABC had a net income of $100,000. The capital structure of the company for 2018 was as follows:

11,000 common shares

1,000 convertible bonds with par value of $100 and 10% coupon; convertible to 5,000 shares

The tax rate of the company is 30%.

Calculate diluted EPS.

Solution:

Number of common shares issued upon conversion = 5,000

Interest payable on the bonds = 100 x $1,000 x 10% = $10,000

$$\text{Diluted EPS} = \frac{\text{NI} + \text{conv debt int } (1 - \text{t}) - \text{pref div} + \text{conv pref div}}{\text{Wt avg shares} + \text{New shares issued}}$$

$$\text{Diluted EPS} = \frac{\$100{,}000 + \$10{,}000 \text{ x } 0.7 - \$0 + \$0}{11{,}000 + 5{,}000} = \$6.69$$

For ‘stock options’**,** we use the ‘Treasury Stock Method’, which assumes that the hypothetical funds received by the company from the exercise of options are used to purchase shares of the company’s common stock at the average market price over the reporting period. Thus, the numerator is unchanged and the number of shares to be added to the denominator = the number of shares created by exercising the options – number of shares hypothetically repurchased with the proceeds of the exercise.

Example

During 2018, Company ABC had a net income of $100,000. It paid $22,000 dividends to its preference shareholders and $12,000 dividends to its common shareholders. The capital structure of the company for 2018 was as follows:

11,000 common shares

1,000 stock options outstanding, that have an exercise price of $20.

During 2018, the average market price for the company’s share was $25.

Calculate the diluted EPS.

Solution:

Number of common shares issued upon conversion = 1,000

Cash proceeds from the exercise of options = 1,000 x 20 = $20,000

Number of shares that can be purchased at the average market price with these funds = \$20,000/25 = 800

Net increase in common shares outstanding = 1,000 – 800 = 200

$$\text{Diluted EPS} = \frac{\text{NI} + \text{conv debt int } (1 - \text{t}) - \text{pref div} + \text{conv pref div}}{\text{Wt avg shares} + \text{New shares issued}}$$

$$\text{Diluted EPS} = \frac{\$100{,}000 + \$0 - \$22{,}000 + \$0}{11{,}000 + 200} = \$6.96$$

Other Issues with Diluted EPS and Changes in EPS

Some potentially convertible securities could be antidilutive. Including them in the calculations would result in an EPS that is higher than the company's basic EPS. Such securities should not be included in the calculation of diluted EPS.

Instructor's Note:

Assess each instrument individually and determine if it is dilutive or not. Only instruments which are dilutive must be included in the diluted EPS calculation.

Changes in EPS
In general, an EPS can increase either due to an increase in net income, a decrease in the number of shares outstanding, or a combination of both.

6. Income Statement Ratios and Common-Size Analysis

Common-Size Analysis of the Income Statement

Common-size income statement presents each line item on the income statement as a percentage of revenue. This format standardizes the income statements and helps remove the effects of company size. They are useful to comparisons across time periods and across companies.

Income Statement Ratios

The income statement is used to calculate **income statement ratios** to evaluate a firm's profitability. The commonly used ratios are:

Gross profit margin = Gross profit / Revenue

Operating profit margin = Operating profit / Revenue

Net profit margin = Net profit / Revenue

High margin ratios are desirable. A firm can increase its margins by either increasing selling price or by lowering costs, or both.

An example of a common size income statement is shown below.

	2018	**%**	**2017**	**%**
Revenue	$100,000	100%	$110,000	100%
Cost of goods sold	$60,000	60%	$65,000	59%
Gross profit	**$40,000**	**40%**	**$45,000**	**41%**
SG&A	$10,000	10%	$11,000	10%
Depreciation expense	$10,000	10%	$11,000	10%
Operating profit	**$20,000**	**20%**	**$23,000**	**21%**
Interest expense	$5,000	5%	$5,500	5%
Earnings before taxes	$15,000	15%	$17,500	16%
Taxes (10%)	$1,500	1.5%	$1,750	1.6%
Net income	**$13,500**	**13.5%**	**$15,750**	**14.3%**

Looking at the above common-size statement, we can conclude that, the profitability margins of this company have declined in 2018 as compared to 2017.

Summary

LO: Describe general principles of revenue recognition, specific revenue recognition applications, and implications of revenue recognition choices for financial analysis.

Under the accrual method of accounting, revenue should be recognized when earned and not necessarily when cash is received.

The following five steps must be followed in order to recognize revenue:

1. Identify the contract(s) with a customer.
2. Identify the performance obligations in the contract.
3. Determine the transaction price.
4. Allocate the transaction price to the performance obligations in the contract.
5. Recognize revenue when (or as) the entity satisfies a performance obligation

Analysts can encounter many companies with complex revenue recognition policies such as:

- Principal versus agent
- Franchising/Licensing
- Software as a service or license
- Long-Term Contracts
- Bill and hold arrangements

Firms can use any revenue recognition technique provided there is a rationale behind their choice. Firms using an aggressive revenue recognition method will most likely inflate the earnings of the current period and report lower revenues in later periods. An analyst should consider the effects different revenue recognition methods can have on the financial statements of a company.

LO: Describe general principles of expense recognition, specific expense recognition applications, implications of expense recognition choices for financial analysis and contrast costs that are capitalized versus those that are expensed in the period in which they are incurred.

The most important principle of expense recognition is the matching principle, under which the expenses incurred to generate revenue are recognized in the same period as revenue.

Expenses that cannot be tied directly to generation of revenues are called periodic costs. They are expensed in the period incurred.

If an asset is expected to provide benefits only for the current period, its cost is expensed on the income statement for that period.

If an asset is expected to provide benefits over multiple periods, its cost is capitalized on the balance sheet and spread over the life of the asset.

If a company's policies result in early recognition of expenses, it can be considered a conservative approach. On the other hand, if a company's polices delay the recognition of expenses, it can be considered an aggressive approach.

LO: Describe the financial reporting treatment and analysis of non-recurring items (including discontinued operations, unusual or infrequent items) and changes in accounting policies.

Net income from discontinued operations is shown net of tax after net income from continuing operations.

Both IFRS and U.S. GAAP allow recognition of unusual or infrequent items.

Changes in accounting policies can be adopted retrospectively (the financial statements for all fiscal years are presented as if the newly adopted accounting principle had been used throughout the period) or prospectively (only the financial statements for the period of change and for future periods are changed).

LO: Describe how earnings per share is calculated and calculate and interpret a company's basic and diluted earnings per share for companies with simple and complex capital structures including those with antidilutive securities.

When a company has simple capital structure, basic EPS is calculated using the formula:

$$\text{Basic EPS} = \frac{\text{Net Income} - \text{Preferred dividends}}{\text{Weighted Average Number of Shares Outstanding}}$$

When a company has complex capital structure, diluted EPS is calculated using the formula:

$$\text{Diluted EPS} = \frac{\text{Net Income} + \text{After tax interest} - \text{Preferred dividend} + \text{convertible preferred dividends}}{\text{Weighted Average Shares} + \text{New shares if convertible debt is converted}}$$

LO: Evaluate a company's financial performance using common-size income statements and financial ratios based on the income statement.

Common-size analysis of the income statement can be performed by stating each line item on the income statement as a percentage of revenue. Common-size statements facilitate comparison across time periods as well as across companies because the standardization of each line item removes the effect of size.

Net profit margin is calculated as: Net Income / Sales. This indicates how much income a company was able to generate for each dollar of revenue.

Gross profit margin is calculated as: Gross Profit / Sales. Where gross profit is calculated as revenue minus cost of goods sold.

Operating profit margin is calculated as: Operating Profit/ Sales.

Analysts can use these profit margins to compare over time and with industry peers.

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LM03 Analyzing Balance Sheets

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1. Introduction

The balance sheet presents the financial position of a company on a particular date, in terms of three elements: assets, liabilities, and equity.

- Assets (A) are what the company owns. They are the resources controlled by the company as a result of past events and they are expected to provide future economic benefits.
- Liabilities (L) are what the company owes. They represent the obligations of a company arising from past events, the settlement of which is expected to result in a future outflow of economic benefits from the entity.
- Equity (E) represents the owners' residual interest in the company's assets after deducting its liabilities. It is also known as shareholders' equity. The accounting equation for determining equity is: E = A – L

Limitations of the balance sheet in financial analysis

- Some assets and liabilities are measured based on historical cost while some are measured based on fair value, which represents its current value as of the balance sheet date. These differences can have significant impact on reported figure.
- The value of an item reported on the balance sheet is the value at the end of the reporting period. If we are analyzing the company at a later date, these values may have changed.
- Some assets and liabilities are difficult to quantify and are not reported on the balance sheet. For example, brand, customer loyalty, human capital, etc.

A sample balance sheet is presented below:

Assets				**Liabilities**		
Current assets				**Current liabilities**		
Cash		$100,000		Accounts payable	$80,000	
Short- term investments		50,000		Salaries payable	10,000	
Accounts receivable		75,000		Interest payable	15,000	
Inventories		200,000		Taxes payable	5,000	
Prepaid insurance		25,000	$450,000	Current portion of note	40,000	$150,000
Long- term investments				**Long- term liabilities**		
Stock investments		$40,000		Notes payable	$110,000	
Cash value of insurance		10,000	50,000	Bank Loan	35,000	
				Mortgage obligation	75,000	
Property, plant & equip.				Deferred income taxes	80,000	300,000
Land		$25,000		Total Liabilities		$450,000
Buildings and equipment	$150,000					
Less: Accum. depreciation	(50,000)	100,000	125,000			
Intangible assets						
Goodwill			275,000	**Stockholders' equity**		
				Capital stock	$300,000	
Other assets				Retained earnings	160,000	
Receivable from employee			10,000	Total stockholders' equity		460,000
Total			$910,000	Total liabilities and equity		$910,000

Current assets are those assets that are expected to be used up or converted to cash within one year or in one operating business cycle, whichever is greater. When the entity's normal operating cycle cannot be clearly identifiable, its duration is assumed to be one year.

All assets that are not classified as current are considered to be non-current or long-term assets.

Current liabilities are those liabilities which are expected to be settled within one year or in one operating business cycle, whichever is greater.

All liabilities that are not classified as current are considered to be non-current or long-term liabilities

This learning module covers the financial reporting and disclosure requirements related to:

- Intangible assets
- Goodwill
- Financial instruments
- Non-current liabilities

Finally, we will also learn how to analyze a balance sheet using financial ratios and common-size analysis.

2. Intangible Assets

Intangible assets refer to identifiable non-monetary assets that lack physical substance. Examples include patents, licenses, and trademarks.

IFRS allows companies to report intangible assets using either a cost model or a revaluation model. US GAAP allows only the cost model.

For each intangible asset, a company determines whether its useful life is finite or indefinite.

- An intangible asset with a finite useful life is amortized on a systematic basis over the best estimate of its useful life. The amortization and useful life estimate is reviewed at least annually.
- The principles of impairment for an intangible asset with a finite useful life are the same as for property, plant and equipment (PP&E).
- An intangible asset with an indefinite useful life is not amortized. Instead, it is tested for impairment at least annually.

Intangible Assets Developed Internally

Costs to internally develop intangible assets are generally expensed when incurred, although there are exceptions.

For internally developed intangible assets, there are two phases: the research phase and the development phase.

<u>Research phase</u> refers to the period during which commercial feasibility of an intangible asset is yet to be established. It is defined as “original and planned investigation undertaken with the prospect of gaining new scientific or technical knowledge and understanding.”

<u>Development phase</u> refers to the period during which the technical feasibility of completing an intangible asset has been established with the intent of either using or selling the asset.

The treatment for the two phases varies slightly under IFRS and US GAAP as outlined below:

Under IFRS:

- Research costs are expensed.
- Development costs can be capitalized if technical feasibility and the intent to sell the asset are established.

Under US GAAP:

- Both research and development costs are expensed.

Intangible Assets Purchased Externally

In contrast to internally created intangibles, acquired or purchased intangible assets are capitalized and reported as separately identifiable intangibles as long as they are based on contractual rights (such as a licensing agreement), other legal rights (such as patents), or can be separated and sold (such as a customer list).

3. Goodwill

Goodwill is an unidentifiable intangible asset. It is created when one company is purchased by another company. If the purchase price is greater than fair value at acquisition, then the excess amount is recognized as an asset on the acquirer’s balance sheet and referred to as goodwill.

Let us consider a simple example. Company A buys Company T for $100 million. The book value of Company T’s assets and liabilities are $125 million and $75 million respectively. The fair value of Company T’s assets and liabilities are $160 million and $75 million respectively. What is the goodwill? In this case, the purchase price is $100 million and the net fair value is $160 - $75 million = $85 million. Hence, goodwill is ($100 million - $85 million) $15 million. Note that the book values of assets and liabilities are not used in the goodwill calculation.

Under both IFRS and US GAAP, goodwill is capitalized (i.e., shown as an asset on the balance sheet). Goodwill is not amortized but is tested for impairment annually. If goodwill is impaired, it is written down and the impairment loss is shown on the income statement.

The recognition and impairment of goodwill can have a significant impact on the comparability of financial statements between companies. Therefore, analysts often make adjustments to remove the impact of good will such as:

- Excluding goodwill from balance sheet used to compute financial ratios
- Excluding goodwill impairment losses from income data used to examine income trends.

4. Financial Instruments

IFRS defines a financial instrument as a contract that gives rise to a financial asset of one company and a financial liability or equity instrument of another entity. Financial assets include stocks and bonds, derivatives, loans and receivables.

Financial assets can be measured either at fair value or amortized cost. The measurement basis depends on how financial asset is categorized. The major categories for financial assets are:

- Measured at Cost or Amortized Cost: Under IFRS, financial assets are measured at amortized cost if the asset's cash flows occur on specified dates and consist solely of principal and interest, and if the business model is to hold the asset to maturity. For example, investment in a long-term bond. Unrealized gains and losses are not recorded anywhere.
- Measured at Fair value through profit or loss (FVTPL) under IFRS or Held-for-Trading under US GAAP: This category of asset is acquired primarily for the purpose of selling in the near term and is likely to be held for only a short period of time. Unrealized gains and losses are shown in the income statement.
- Measured at Fair value through other comprehensive income (FVTOCI) under IFRS or available-for-sale under US GAAP: This category of asset is expected neither to be held till maturity nor traded in the near term. Unrealized gains and losses are shown in other comprehensive income.
  Unlike IFRS, the US GAAP category available-for-sale applies only to debt securities and is not permitted for investments in equity securities.

The table below summarizes where gains and losses associated with the financial asset are recognized in the financial statements of the company.

Asset Category	**Treatment**
Measured at Fair Value through Profit and Loss	• Measured at fair value. • Unrealized gains shown on Income Statement.
Measured at Fair Value through Other Comprehensive Income	• Measured at fair value. • Unrealized gains/losses shown in other comprehensive income (OCI).
Measured at Cost or Amortized Cost	• Measured at cost or amortized cost. • Unrealized gains not recorded anywhere.

Realized gains for all categories are shown on the income statement of the company. An important concept related to these assets is **mark-to-market**. It is the process whereby the value of a financial instrument is adjusted to reflect current value based on market prices. Let us illustrate the different accounting treatments for each of these categories through a simple example.

Example

Company owners contribute $100,000, which is invested in a 20-year bond with a 5% coupon paid semi-annually. After six months, the company receives the first coupon payment of $2,500. At this stage, the market price has increased to $102,000. Show the balance sheet and income statement treatment under each of the three categorizations.

Solution:

The accounting treatment under the three categories is summarized below:

	Measured at Fair Value through Profit and Loss	**Measured at Fair Value through Other Comprehensive Income**	**Measured at Cost or Amortized Cost**
Balance Sheet			
Cash	$2,500	$2,500	$2,500
Cost of securities	$100,000	$100,000	$100,000
Unrealized gains/losses	$2,000	$2,000	
PIC	$100,000	$100,000	$100,000
RE	Up by $4,500	Up by $2,500	Up by $2,500
OCI		Up by $2,000	
Income Statement			
Interest income	$2,500	$2,500	$2,500
Unrealized gain	$2,000		

For Measured at Fair Value through Profit and Loss, unrealized gains and cash from coupon payments are shown on the asset side of the balance sheet. On the equity side, paid-in capital remains the same at $100,000. Retained earnings increase by $4,500 (sum of coupon payment of $2,500 and unrealized gain of 2,000). On the income statement unrealized gain of $2,000 and interest income of $2,500 is recognized.

For Measured at Fair Value through Other Comprehensive Income, the accounting treatment is the same as HFT except for unrealized gains. For AFS, the unrealized gain is shown as part of other comprehensive income (OCI). It is not shown on the income statement.

For Measured at Cost or Amortized Cost, the asset is valued at amortized cost. Therefore, the unrealized gain of $2,000 is not shown on the balance sheet or income statement. Only the coupon payment of $2,500 is shown on the balance sheet as cash and on the income statement as interest income.

5. Non-Current Liabilities

All liabilities that are not classified as current are considered to be non-current or long-term liabilities.

Long-term financial liabilities - Include loans, notes and bonds payable. These are usually reported at amortized cost on the balance sheet.

For example, if a company issues $10 million in bonds at 97.50% of par value (i.e., at a discount to par), the bonds will be reported as a liability of $9.75 million on the date of issue. The $250,000 discount will be amortized over the bond's life, resulting in a book value of $10 million at maturity.

Deferred tax liabilities - Arise from temporary timing difference between a company's taxable income and reported income. They are defined as the amounts of income taxes payable in future periods due to temporary taxable differences.

6. Ratios and Common-Size Analysis

Common-Size Analysis of the Balance Sheet

Balance sheet analysis can help us evaluate a company's liquidity and solvency. A balance sheet can be used to analyze a company's capital structure and ability to pay liabilities.

In a vertical common-size balance sheet, all balance sheet items are expressed as a percentage of total assets. Common-size statements are useful in comparing a company's balance sheet composition over time (time-series analysis) and across companies in the same industry. An example of a common-size balance sheet is shown below for a fictitious company - Everest Inc.

ASSETS	**2021**	**2020**
Cash and cash equivalents	10.81%	13.12%
Short-term marketable securities	1.24%	0.62%
Other financial assets	1.24%	1.21%
Accounts receivable	7.50%	4.80%
Inventory	25.32%	25.97%
Other current assets	3.37%	2.14%

Property, plant and equipment	38.76%	40.06%
Investment property	6.18%	6.22%
Intangible assets	0.22%	0.35%
Deferred tax assets	0.11%	0.08%
Goodwill	0.91%	1.09%
Long-term loans	4.32%	4.32%
Other non- current assets	0.03%	0.02%
Total	100.00%	100.00%
EQUITY and LIABILITIES		
Short-term borrowing	0.46%	0%
Accounts payable	6.49%	6.22%
Accrued expenses	4.22%	3.38%
Deferred revenue	1.30%	1.21%
Other current liabilities	24.29%	24.42%
Long-term borrowings	0.23%	0.31%
Deferred tax liabilities	4.01%	4.20%
Other long-term liabilities	0.12%	0.13%
Stockholder's equity	58.88%	60.13%
Equity and Liabilities	100.00%	100.00%

Balance Sheet Ratios

Balance sheet ratios are those involving balance sheet items only. Liquidity ratios tell us about a company's ability to meet current liabilities, while solvency ratios tell us about a company's ability to meet long-term and other obligations. They also help us evaluate a company's financial risk and leverage. The following table summarizes some liquidity ratios. The last column shows the relevant ratios for Everest Inc. for 2021.

Liquidity Ratios	**Calculation**	**Ratios for Everest Inc. for 2021**
Current	Current assets ÷ Current liabilities	1.35
Quick (acid test)	(Cash + Marketable securities + Receivables) ÷ Current liabilities	0.53
Cash	(Cash + Marketable securities) ÷ Current liabilities	0.33

Solvency ratios help to evaluate:

- a company's ability to meet long-term and other liabilities.

- a company's financial risk and leverage. The following table summarizes some solvency ratios:

Solvency Ratios	**Calculation**	**Ratios for Everest Inc. for 2021**
Long-term debt-to-equity	Total long-term debt ÷ Total equity	0.004
Debt-to-equity	Total debt ÷ Total equity	0.012
Total debt-to-assets	Total debt ÷ Total assets	0.007
Financial leverage	Total assets ÷ Total equity	1.69

It is important for analysts to remember that ratio analysis requires judgment. For example, current ratio is only a rough measure of liquidity. In addition, ratios are sensitive to end of period financing and operating decisions that can potentially impact current asset and current liability amounts. Analysts should also evaluate ratios in the context of a company's industry. This requires an examination of the entire operations of a company, its competitors, and the external economic and industry setting.

Summary

LO: Explain the financial reporting and disclosures related to intangible assets.

Intangible assets refer to identifiable non-monetary assets that lack physical substance.

For each intangible asset, a company determines whether its useful life is finite or indefinite.

- An intangible asset with a finite useful life is amortized on a systematic basis over the best estimate of its useful life. The amortization and useful life estimate is reviewed at least annually.
- The principles of impairment for an intangible asset with a finite useful life are the same as for property, plant and equipment (PP&E).
- An intangible asset with an indefinite useful life is not amortized. Instead, it is tested for impairment at least annually.

For internally developed intangible assets,

Under IFRS:

- Research costs are expensed.
- Development costs can be capitalized if technical feasibility and the intent to sell the asset are established.

Under US GAAP:

- Both research and development costs are expensed.

LO: Explain the financial reporting and disclosures related to goodwill.

Goodwill arises when one company is purchased by another company. If the purchase price is greater than fair value at acquisition, then the excess amount is recognized as an asset on the acquirer's balance sheet and referred to as goodwill. Under both IFRS and U.S. GAAP, accounting goodwill is capitalized.

Goodwill is not amortized; instead, it is tested for impairment at least annually.

LO: Explain the financial reporting and disclosures related to financial instruments.

Financial assets include investment securities, derivatives, loans, and receivables. The following table summarizes measurement of different categories of financial assets:

Asset Category	Treatment
Measured at Fair Value through Profit and Loss	• Measured at fair value. • Unrealized gains shown on Income Statement.
Measured at Fair Value through Other Comprehensive Income	• Measured at fair value. • Unrealized gains/losses shown in other comprehensive income (OCI).

Measured at Cost or Amortized Cost	• Measured at cost or amortized cost. • Unrealized gains not recorded anywhere.

LO: Explain the financial reporting and disclosures related to non-current liabilities.

Long-term financial liabilities such as loans, notes and bond payables are reported at amortized cost on the balance sheet.

Deferred tax liabilities arise from temporary timing difference between a company's taxable income and reported income. They are defined as the amounts of income taxes payable in future periods due to temporary taxable differences.

LO: Calculate and interpret common-size balance sheets and related financial ratios.

In a common-size balance sheet, all balance sheet items are expressed as a percentage of total assets. Common-size statements are useful in comparing a company's balance sheet composition over time and across companies in the same industry.

Balance sheet ratios are those involving balance sheet items only. Liquidity ratios tell us about a company's ability to meet current liabilities while solvency ratios tell us about a company's ability to meet long-term and other obligations. The following table summarizes these ratios:

Liquidity Ratios	**Calculation**
Current	Current assets ÷ Current liabilities
Quick (acid test)	(Cash + Marketable securities + Receivables) ÷ Current liabilities
Cash	(Cash + Marketable securities) ÷ Current liabilities
Solvency Ratios	**Calculation**
Long-term debt-to-equity	Total long-term debt ÷ Total equity
Debt-to-equity	Total debt ÷ Total equity
Total debt-to-assets	Total debt ÷ Total assets
Financial leverage	Total assets ÷ Total equity

LM04 Analyzing Statements of Cash Flows I

Ver 1.0

1. Introduction

The cash flow statement provides important information about a company's cash receipts and payments during an accounting period. It is a vital information source that assists users to evaluate a company's liquidity, solvency, and financial flexibility.

Under both IFRS and US GAAP, cash flows are categorized as operating, investing, or financing activities on the cash flow statement.

Operating activities: These are activities related to the normal operations of a company. Examples include:

- Cash inflows such as cash collected from sales, commissions, royalties, etc.
- Cash outflows such as cash payments for inventory, salaries, and operating expenses.
- Cash payments and receipts related to trading securities (securities that are not bought as investments).

Investing activities: These are activities associated with acquisition and disposal of long-term assets. Examples include:

- Cash from sale of property, plant, and equipment.
- Cash spent to purchase property, plant, and equipment.
- Cash payments and receipts related to investment securities (not trading securities).

Financing activities: These are activities related to obtaining or repaying capital. Examples include:

- Issuance or repurchase of a company's own preferred or common stock.
- Issuance or repayment of debt.
- Dividend payments to shareholders.

2. Linkages Between the Financial Statements

Link between the Cash Flow Statement and the Balance Sheet

Cash is an asset and is reported on the balance sheet. The cash flow statement explains the change in cash during an accounting period. This can be illustrated through a simple scenario. Assume that beginning cash is 1,100. This is reported on the balance sheet. The cash flow statement will show the cash receipts and cash payments. For the scenario presented below, the cash receipts equal 3,200 and the cash payments equal 3,300 which means that the net change in cash is -100. This explains how the cash balance went from 1,100 at the start of the period to 1,000 at the end of the period.

Beginning Balance Sheet 1 Jan 2015	*Statement of Cash Flows for Year Ended 31 December 2015*		*Ending Balance Sheet at 31 Dec 2015*
Beginning Cash	Plus: Cash Receipts	Less: Cash Payments	Ending Cash

1,100	3,200	3,300	1,000

Link between Cash Flow Statement, Balance Sheet, and Income Statement

Let's consider an example of how items on the balance sheet are related to the income statement and the cash flow statement. Suppose the beginning accounts receivable is 200, the revenue during the year is 5,000 and the cash collected from customers is 4,800. What is the ending accounts receivables? The table below makes it easy to compute the missing amount. We see that the ending accounts receivables will be 400.

Balance Sheet at 1 Jan 2015	*Income Statement*	*Statement of Cash Flows*	*Balance Sheet at 31 Dec 2015*
Beginning A/R	Plus: Revenue	Less: Cash Collected from Customers	Ending A/R
200	5,000	4,800	400

This example clearly shows that receivables (balance sheet item), revenue (income statement item) and cash collected from customers (cash flow item) are related as follows:

Ending receivables = Beginning receivables + Revenue – Cash collected from customers

3. The Direct Method for Cash Flows From Operating Activities

Under IFRS and US GAAP, there are two acceptable formats for reporting cash flow from operating activities: indirect and direct.

- The **indirect method** shows how cash flow from operations can be obtained from reported net income through a series of adjustments.
- The **direct method** shows the specific cash inflows and outflows that result in reported cash flow from operating activities.

Indirect Format Sample

With the indirect method, we start with net income and make several adjustments for non-cash, non-operating items to arrive at the cash flow from operations. Shown below is a sample of the indirect format for a fictitious company called K2 Corp.

Net income	2,775
Depreciation	1,000
Gain on sale of equipment	(200)
Increase in accounts receivable	(150)
Increase in inventory	(600)
Increase in pre-paid expenses	(30)

Increase in accounts payable	300
Increase in wages payable	10
Increase in tax payable	5
Increase in other accrued liabilities	100
Decrease in interest payable	(10)
Cash flow from operations	*3,200*

<u>Direct Format Sample</u>

In the direct format, we look at the specific cash inflows and outflows that resulted in cash flow from operating activities. This method is encouraged by both IFRS and US GAAP.

Cash from customers	24,850
Cash paid to suppliers	(10,300)
Cash paid to employees	(7,990)
Cash paid for other operating expenses	(1,930)
Cash paid for interest	(510)
Cash paid for taxes	(920)
Cash flow from operations	*3,200*

Notice that while the presentation formats are different, the cash flow from operations number is the same under both methods.

Operating Activities: Direct Method

Only cash flow from operating activities is presented differently under the two methods. Presentation of cash flow from investing activities and cash flow from financing activities is the same under both methods.

<u>Direct method</u>:

In the direct method, we take each item from the income statement and convert it to its cash equivalent by removing the impact of accrual accounting.

The rules to adjust are:

- Increase in assets is use of cash (-ve adjustment) and decrease in asset is source of cash (+ve adjustment).
- Increase in liability is source of cash (+ve adjustment) and decrease in liability is use of cash (-ve adjustment).

Cash collected from customers: Adjust sales for changes in accounts receivable and unearned revenue.

Cash for inputs: Adjust COGS for changes in inventory and accounts payable.

Cash operating expenses: Adjust SG&A for changes in related accrued liabilities or prepaid expenses.

Cash interest paid: Adjust interest expense for changes in interest payable.

Cash taxes paid: Adjust tax expense for changes in tax payable and changes in deferred tax assets and liabilities.

Example

Consider a company that reported sales of $10 million. Accounts receivable for the year went up from $2 million to $4 million. Unearned revenue went up from $1 million to $2 million. Calculate cash collected from customers.

Solution:

Δ Accounts receivable = $2 million. This is an asset and increase in asset is use of cash so –ve adjustment.

Δ Unearned revenue = $1million. This is a liability and increase in liability is source of cash so +ve adjustment.

Cash collected from customers = + $10 million - $2million + $1million = $9 million.

Example

Consider a company with COGS of $20 million for a particular period. During this period inventory increased by $4 million and accounts payable went up by $2 million. Calculate the cash paid for inputs.

Solution:

Δ Inventory = $4 million. This is an asset and increase in asset is use of cash so –ve adjustment.

Δ Accounts payable = $2 million. This is a liability and increase in liability is source of cash so +ve adjustment.

Cash paid for inputs = - $20 million - $4 million + $2 million = - $22 million

4. The Indirect Method for Cash Flows from Operating Activities

Indirect method shows how cash flow from operations can be obtained from reported net income as a result of a series of adjustments.

The steps are:

- Begin with net income.
- Add back all non-cash charges to income and subtract all non-cash components of revenue (For example, add depreciation and amortization).

- Subtract any gains that resulted from financing or investing cash flows (For example, gain on the sale of an equipment).
- Add or subtract changes to related balance sheet operating accounts.
- Decrease in operating assets (source of cash) should be added and increase in operating assets (use of cash) should be subtracted.
- Similarly, increase in current liabilities (source of cash) should be added and decrease in current liabilities (use of cash) should be subtracted.

Example

Consider a company with net income of $100 million in 2001. Depreciation expense is $10 million. Gain on sale of equipment is $4 million. Increase in A/R is $8 million. Increase in A/P is $4 million. Increase in inventory is $10 million. Calculate CFO using the indirect method.

Solution:

Net income	+100
Add non-cash charges (depreciation)	+ 10
Less gain on sale of equipment	- 4
Less increase in A/R	-8
Add increase in A/P	+4
Less increase in inventory	-10
Total	**92**
CFO = $ 92 million	

5. Conversion from the Indirect to Direct Method

Instructor's note: The probability of getting tested on this topic on the exam is low.

The operating cash flow from indirect method can be converted to direct by using the three-step process:

- Aggregate all the revenues and expenses.
- Remove all non-cash items from aggregated revenues and expenses and break up remaining items into relevant cash flow items.
- Convert accrual amounts to cash flow amounts by adjusting for changes in corresponding working accounts.

6. Cash Flows from Investing Activities

CFI is calculated by examining the change in the gross asset account that results from investing activities. Typically, this change results from purchases or sale of equipment (long-term assets). To determine the cash inflow from the sale of equipment, we need to use the expression shown below.

$$\text{Cash from sale of equipment} = \text{Historical cost of equipment sold} - \text{Accumulated depreciation on equipment sold} + \text{Gain on sale of equipment}$$

where:

$$\text{Historical cost of equipment sold} = \text{beginning balance} + \text{equipment purchased} - \text{ending balance equipment}$$

Accumulated depreciation on equipment sold = beginning value of depreciation + depreciation expense – ending value of depreciation

Example

The balance sheet extract for Jackal Labs Ltd shows the machinery and accumulated depreciation balances for the years 2011 and 2012.

	2011	**2012**
Machinery (Gross)	$80 million	$91 million
Accumulated depreciation	$25 million	$31 million

Further information provided is as follows:

Gain on sale of machinery	$1.5 million
Depreciation expense for 2012	$7 million
Capital expenditure on machinery	$14 million

What is the cash received from sale of equipment?

Solution:

Cash from sale of machinery = historical cost of equipment sold – accumulated depreciation on equipment sold + gain on sale of equipment

We know the gain is $1.5 million. Calculate the other components in the equation:

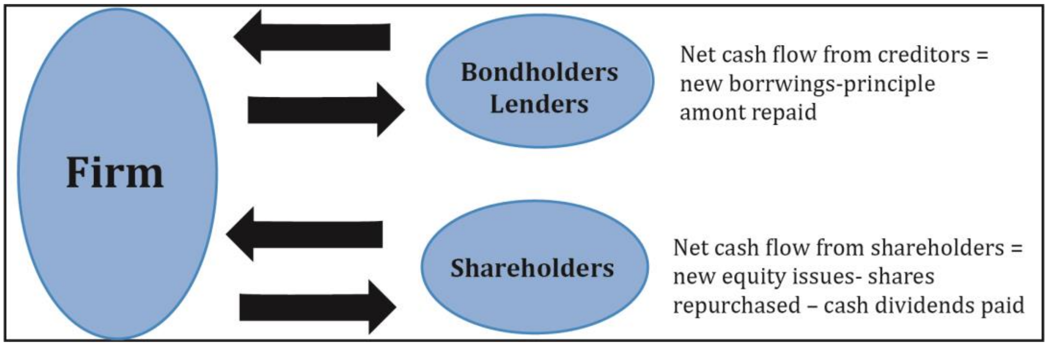
Historical cost of equipment sold = beginning balance + equipment purchased – ending balance of equipment = 80 + 14 – 91 = $3 million

Accumulated depreciation on equipment sold = beginning value of depreciation + depreciation expense – ending value of depreciation = 25 + 7– 31 = $1 million

Cash from sale of machinery = 3 – 1 + 1.5 = 3.5

7. Cash Flows from Financing Activities

Cash flow from financing activities refers to cash flows between the firm and the suppliers of capital. Suppliers of capital include creditors, bondholders, and shareholders. Similar to investing activities, the presentation of cash flows from financing activities is also identical under both methods. The figure below summarizes the calculation of net cash flows from creditors, bondholders, and shareholders.



It can be calculated using the following formulae:

(1) CFF = Net cash flow from creditors + Net cash flow from shareholders

(2) Net cash flow from creditors = New borrowings – Principal repaid

(3) Net cash flow from shareholders = New equity issued – Shares repurchased – Cash dividends

Example

The following information is available about company ABC for 2001.

New borrowings	\$10 million
Principal repaid	\$5 million
New equity issued	\$5 million
Shares repurchased	-
Dividends paid	\$2 million

Calculate CFF.

Solution:

Net cash flow from creditors = New borrowings – Principal repaid = 10 – 5 = \$5 million

Net cash flow from shareholders = New equity issued – Shares repurchased – Cash dividends = 5 – 0 -2 = \$3 million

CFF = Net cash flow from creditors + Net cash flow from shareholders = 5 + 3 = \$8 million

8. Differences in Cash Flow Statements Prepared Under US GAAP Versus IFRS

The reporting of interest paid/received and dividends paid/received is different between IFRS and US GAAP. The differences between the two standards are summarized in the table below.

Cash flow	**IFRS**	**US GAAP**
Interest received	Operating or investing	Operating
Interest paid	Operating or financing	Operating
Dividends received	Operating or investing	Operating
Dividends paid	Operating or financing	Financing

In addition to the points made above, IFRS and US GAAP also have some differences with respect to bank overdrafts, taxes paid, and the format of the cash flow statement. These are outlined in the table below.

Cash Flow	**IFRS**	**US GAAP**
Bank overdrafts	Considered part of cash equivalents.	Not considered part of cash equivalents and classified as financing.
Taxes paid	Generally categorized as operating, but a portion can be allocated to investing or financing if it can be specifically identified with these categories.	Operating.
Format of statement	Both direct and indirect formats are allowed but the direct format is encouraged.	Both direct and indirect formats are allowed but the direct format is encouraged. A reconciliation of net income to cash flow from operating activities must be provided regardless of method used.

Summary

LO: Describe how the cash flow statement is linked to the income statement and the balance sheet.
Cash is an asset. The cash flow statement ultimately shows the change in cash during an accounting period. The beginning and ending balances of cash are shown on the balance sheet and the bottom of the cash flow statement reconciles beginning cash with ending cash.

Because a company's operating activities are reported on an accrual basis in the income statement, any differences between the accrual basis and cash basis for accounting result in an increase or decrease in some asset or liability on the balance sheet.

LO: Describe the steps in the preparation of direct and indirect cash flow statements, including how cash flows can be computed using income statement and balance sheet data.
CFO can be computed using the direct method or the indirect method.

Direct method:

In the direct method, we take each item from the income statement and convert it to its cash equivalent by removing the impact of accrual accounting.

The rules to adjust are:

- Increase in assets is use of cash (-ve adjustment) and decrease in asset is source of cash (+ve adjustment).
- Increase in liability is source of cash (+ve adjustment) and decrease in liability is use of cash (-ve adjustment).

Indirect method:

Indirect method shows how cash flow from operations can be obtained from reported net income as a result of a series of adjustments.

The steps are:

- Begin with net income.
- Add back all non-cash charges to income and subtract all non-cash components of revenue (For example, add depreciation and amortization).
- Subtract any gains that resulted from financing or investing cash flows (For example, gain on the sale of an equipment).
- Add or subtract changes to related balance sheet operating accounts.
- Decrease in operating assets (source of cash) should be added and increase in operating assets (use of cash) should be subtracted.
- Similarly, increase in current liabilities (source of cash) should be added and decrease in current liabilities (use of cash) should be subtracted.

CFI is calculated by determining changes in the gross asset account that result from the purchase or sale of equipment.

CFF is the sum of net cash flows from creditors and net cash flows from shareholders.

LO: Demonstrate the conversion of cash flows from the indirect to direct method.
The operating cash flow from indirect method can be converted to direct by using the three-step process:

- Aggregate all the revenues and expenses.
- Remove all non-cash items from aggregated revenues and expenses and break up remaining items into relevant cash flow items.
- Convert accrual amounts to cash flow amounts by adjusting for working capital changes.

LO: Contrast cash flow statements prepared under International Financial Reporting Standards (IFRS) and US generally accepted accounting principles (US GAAP).

Cash flow	**IFRS**	**U.S. GAAP**
Interest received	Operating or investing	Operating
Interest paid	Operating or financing	Operating
Dividends received	Operating or investing	Operating
Dividends paid	Operating or financing	Financing

LM05 Analyzing Statements of Cash Flows II

Ver 1.0

1. Introduction

This learning module covers the tools and techniques for analyzing the statement of cash flows such as:

- Evaluating the sources and uses of cash
- Common size analysis
- Calculation of free cash flow measures
- Cash flow ratios

2. Evaluating Sources and Uses of Cash

Evaluation of the cash flow statement should involve the following:

- Evaluate where the major sources and uses of cash flow are between operating, investing, and financing activities. Major sources of cash for a company can vary with its stage of growth. For example, for a mature company it is expected that operating activities are the primary source of cash flows. However, analysts must analyze whether operating cash flows are positive and cover capital expenditures for all companies.
- Evaluate the primary determinants of operating cash flow. Analysts should compare operating cash flow with net income. If a company has large net income but poor operating cash flow, it may be a sign of poor earnings quality. In addition, analysts need to look at consistency of operating cash flows.
- Evaluate the primary determinants of investing cash flow. This is useful for letting the analyst know how much is being invested for the future in property, plant, and equipment and how much is put aside in liquid investments.
- Evaluate the primary determinants of financing cash flow. This helps understand if the company is raising capital or repaying capital and which capital sources are being used.

3. Ratios and Common-Size Analysis

In common-size analysis of a company's cash flow statement, there are two alternative approaches. In the first approach, we express each line item of cash inflow (outflow) as a percentage of total inflows (outflows). An example of a common-size cash flow statement using this approach is shown below for a fictitious company - K2 Corp.

Inflows	**Actual**	**% of Total Inflow**
Net cash provided by operating activities	3,200	80 %
Sale of Equipment	800	20%
Total	4,000	100%
Outflows	**Actual**	**% of Total Outflow**

Purchase of equipment	1,500	36.58%
Retirement of long-term debt	500	12.19%
Retirement of common stock	325	7.9%
Dividend payments	1,775	43.29%
Total	4,100	100%
Net increase (decrease) in cash	(100)	

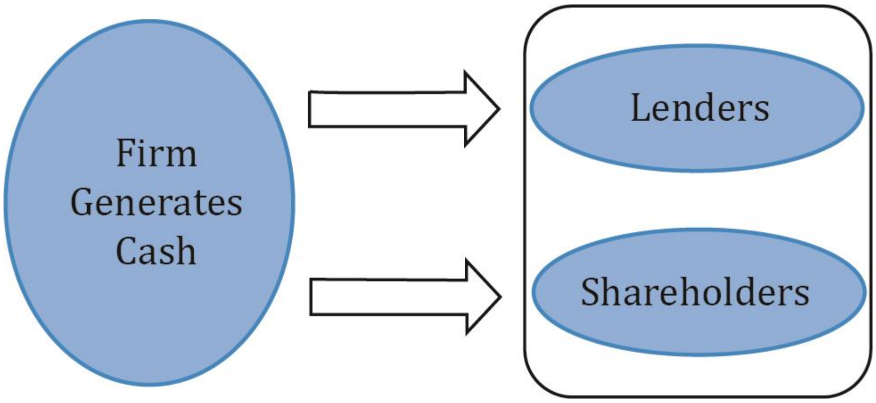
In the second approach, we express each line item as a percentage of revenue. An example of such a statement is shown below for K2 Corp. In this example, we have assumed total revenue is 10,000.

Cash flow	**Actual**	**% of Total Revenue**
Cash flow from operating activities		
Net Income	4,000	40%
Depreciation expense	500	5%
Increase in accounts receivable	(500)	(5%)
Increase in inventory	(1,000)	(10%)
Decrease in prepaid expenses	1,000	10%
Increases in accounts payable	500	5%
Increases in accrued liabilities	500	5%
Net cash provided by operating activities	**5,000**	**50%**
Cash flow from investing activities		
Cash received from sale of equipment	2,000	20%
Cash paid for purchase of equipment	(5,000)	(50%)
Net cash used for investing activities	**(3,000)**	**(30%)**
Cash flow from financing activities		
Sale of bonds	1,000	10%
Cash dividends	(2,000)	(20%)
Net cash used for financing activities	**(1,000)**	**(10%)**
Net increase in cash	**1,000**	**10%**

The common-size cash flow statement makes it easier to see trends in cash flow rather than just looking at the total amount. The second approach is useful for the analyst in forecasting future cash flows.

4. Free Cash Flow Measures

Free cash flow to firm (FCFF) is the cash flow available to all the suppliers of capital to a company after all operating expenses have been paid and necessary investments in working capital and fixed capital have been made. The suppliers of capital include both lenders (debt) and equity shareholders (equity). This is illustrated in the figure below:



The formula for computing FCFF from net income is:

$$FCFF = NI + NCC + Int(1 - \text{Tax rate}) - FCInv - WCInv$$

where:
NI = Net Income
NCC = non-cash charges
Int = Interest expense
FCInv = Fixed capital investment/expenditures
WCInv = working capital expenditures

FCFF can also be computed from CFO as:

$$FCFF = CFO + Int(1 - \text{Tax rate}) - FCInv$$

While FCFF indicates how much cash is available to all suppliers of capital, **free cash flow to equity** (FCFE) is the cash flow available to the company's stockholders after all operating expenses and borrowing costs (principal and interest) have been paid and necessary investments in working capital and fixed capital have been made. The formula for computing FCFE is as follows:

$$FCFE = CFO - FCInv + \text{Net borrowing}$$

5. Cash Flow Statement Analysis: Cash Flow Ratios

There are several ratios useful for the analysis of the cash flow statement. These ratios generally fall into cash flow performance (profitability) ratios and cash flow coverage (solvency) ratios. The calculation and interpretation of these ratios are summarized in the tables below.

Performance Ratios	**Calculation**	**What It Measures**
Cash flow to revenue	CFO ÷ Net revenue	Operating cash generated per dollar of revenue.

Cash return on assets	CFO ÷ Average total assets	Operating cash generated per dollar of asset investment.
Cash return on equity	CFO ÷ Average shareholders' equity	Operating cash generated per dollar of owner investment.
Cash to income	CFO ÷ Operating income	Cash generating ability of operations.
Cash flow per share	(CFO – Preferred dividends) ÷ Number of common shares outstanding	Operating cash flow on a per-share basis.

Coverage Ratios	**Calculation**	**What It Measures**
Debt coverage	CFO ÷ Total debt	Financial risk and financial leverage.
Interest coverage	(CFO + Interest paid + Taxes paid) ÷ Interest paid	Ability to meet interest obligations.
Reinvestment	CFO ÷ Cash paid for long-term assets	Ability to acquire assets with operating cash flows.
Debt payment	CFO ÷ Cash paid for long-term debt repayment	Ability to pay debts with operating cash flows.
Dividend payment	CFO ÷ Dividends paid	Ability to pay dividends with operating cash flows.
Investing and financing	CFO ÷ Cash outflows for investing and financing activities	Ability to acquire assets, pay debts, and make distributions to owners.

Summary

LO: Analyze and interpret both reported and common-size cash flow statements.

Operating cash flow

- A healthy firm should generate positive cash flows from operating activities. (This is not applicable for startups).
- Positive operating cash flow generated by liquidating non-cash working capital items (like liquidating inventory and receivables or increasing payables) is not sustainable.
- Earnings that are significantly greater than operating cash flows indicate that aggressive accounting policies are being followed.

Investing cash flow

- Cash outflows can result from investments in property, plant and equipment, or other assets.
- Increasing outflows is an indication of growth.
- Decreasing outflows may indicate a reduction of capital expenditure and reduction in growth.

Financing cash flow

- Tells us if the company is generating cash by issuing debt or equity.
- Also tells us if the company is using cash to repay debt, reacquire stock, or pay dividends.

Common-Size format

- There are two approaches:
  - In the first approach, we express each line item of cash inflow (outflow) as a percentage of total inflows (outflows).
  - In the second approach, we express each line item as a percentage of revenue.
- Common-size cash flow statement makes it easier to identify trends in cash flows.
- It also helps us in forecasting future cash flows.

LO: Calculate and interpret free cash flow to the firm, free cash flow to equity, and performance and coverage cash flow ratios.

FCFF = NI + NCC + Int (1 - Tax rate) – FCInv – WCInv

Or

FCFF = CFO + Int (1 - Tax rate) – FCInv

FCFE = CFO – FCInv + Net borrowing

Commonly used cash flow ratios are:

		Calculation	What It Measures
Performance Ratios	Cash flow to revenue	CFO ÷ Net revenue	Operating cash generated per dollar of revenue.
	Cash return on assets	CFO ÷ Average total assets	Operating cash generated per dollar of asset investment.
	Cash return on equity	CFO ÷ Average shareholders' equity	Operating cash generated per dollar of owner investment.
	Cash to income	CFO ÷ Operating income	Cash generating ability of operations.
	Cash flow per share	(CFO – Preferred dividends) ÷ Number of common shares outstanding	Operating cash flow on a per-share basis.
Coverage Ratios	Debt coverage	CFO ÷ Total debt	Financial risk and financial leverage.
	Interest coverage	(CFO + Interest paid + Taxes paid) ÷ Interest paid	Ability to meet interest obligations.
	Reinvestment	CFO ÷ Cash paid for long-term assets	Ability to acquire assets with operating cash flows.
	Debt payment	CFO ÷ Cash paid for long-term debt repayment	Ability to pay debts with operating cash flows.
	Dividend payment	CFO ÷ Dividends paid	Ability to pay dividends with operating cash flows.
	Investing and financing	CFO ÷ Cash outflows for investing and financing activities	Ability to acquire assets, pay debts, and make distributions to owners.

LM06 Analysis of Inventories

Ver 1.0

1. Introduction

Inventories are assets held by a company to produce finished goods for sale. They are shown as a current asset on the balance sheet; and can represent a significant part of the total assets for many companies.

Manufacturing and merchandising companies (Ex: Nike, Caterpillar) generate sales and profit through the sale of inventory. An important measure in calculating profits is cost of goods sold, i.e., how much cost the company incurred from procuring raw materials to converting it to a finished product, and finally selling it.

There is no universal inventory valuation method. The four inventory valuation methods used by companies are:

First In First Out (FIFO)

- The cost of the first item purchased is assumed to be the cost of the first item sold.
- Ending inventory is based on the cost of the most recent purchases.

Last In First Out (LIFO)

- The cost of the last item purchased is assumed to be the cost of the first item sold.
- Ending inventory is based on the cost of the earliest purchases.

Weighted average cost

- Each item in the inventory is valued using an average cost of all items in the inventory.
- COGS and inventory values are between their FIFO and LIFO values.

Specific identification

- Each unit sold is matched with the unit's actual cost.
- This method is usually used for items that are unique in nature, for example, jewelry.

All four methods are permitted under U.S. GAAP. However, IFRS does not permit LIFO method.

2. Inventory Valuation

Holding inventory for a prolonged period results in the risk of spoilage, obsolescence, or decline in prices, and the cost of inventory may not be recoverable in such circumstances. We define some terms first before looking at the differences in how inventory is measured under IFRS and GAAP.

Net realizable value: Estimated selling price under ordinary business conditions minus estimated costs necessary to get the inventory in condition for sale. NRV is from a seller's perspective.

Net realizable value = estimated sales price – estimated selling costs

Market value: Current replacement cost subject to lower or upper limits. Market value has upper limit of net realizable value and lower limit of NRV less a normal profit margin. Market value is from a buyer's perspective.

Market value limits = (NRV - normal profit margin, NRV)

The following table describes how inventory is measured under IFRS and GAAP:

Inventory measurement under IFRS and US GAAP	
IFRS	*US GAAP*
Lower of cost or net realizable value.	Lower of cost or market value.
If NRV is less than the balance sheet cost, the inventory is "written down" to NRV. The loss in value is reflected in the income statement in cost of goods sold. Inventory write-down has a negative effect on profitability, liquidity, and solvency ratios and positive effect on activity ratios.	If cost exceeds market, inventory is written down to market value on the balance sheet and the loss is recognized.
If value recovers subsequently, inventory can be written up and gain is recognized in the income statement. The amount of gain is limited to loss previously recognized.	If value recovers subsequently, no write up is allowed. There is no reversal of write-downs. This may motivate companies not to record inventory write-downs unless the decline is permanent as it affects profitability ratios.
Commodities and agricultural goods prices can be reported above historical cost.	Commodities and agricultural goods prices can be reported above historical cost.

An inventory write-down reduces both profit and carrying amount of inventory on the balance sheet, which, in turn, affects the ratios. The following table shows the effect of inventory write-downs on various financial ratios:

Ratio	*Effect*	*Reason*
Liquidity ratios		
Current ratio	Lower	Current assets decrease due to lower inventory.
Activity ratios		
Inventory turnover	Higher	COGS increases assuming inventory write-downs are reported as part of cost of sales. Average inventory decreases. Lower inventory carrying amounts make it appear as if the company is managing its inventory

		effectively, but write-downs reflect poor inventory management.
Days of inventory on hand	Lower	Inventory turnover is higher.
Profitability ratios		
Net profit margin	Lower	Cost of sales is higher. Sales stay the same.
Gross profit margin	Lower	Cost of sales is higher. Sales stay the same.

Companies that use weighted average, specific identification, and FIFO are more likely to have inventory write-downs than companies using the LIFO method.

3. The Effects of Inflation and Deflation on Inventories, Costs of Sales, and Gross Margin

First In, First Out (FIFO)

Under First In, First Out:

- Oldest goods purchased or manufactured are assumed to be sold first.
- Newest goods purchased or manufactured remain in ending inventory.
- When prices are increasing or stable, cost assigned to items in inventory is higher than the cost of items sold.

The following example illustrates how cost of goods sold and inventory are accounted for in each period:

Assume you bought four pencils. The first two pencils were worth $1 each and the next two pencils were worth $2 each. Before you start selling, your inventory consists of four pencils.

In period 1, you sell two pencils. The cost of pencils sold in period 1 is $2 (two pencils of $1 each). The pencils that were bought first are considered sold. Inventory at the end of period 1 is $4 and looks like this (2 pencils of $2 each):

As you could see, cost of pencils sold in period 1 was $2 (cheaper pencils bought initially) whereas the cost of pencils in inventory was $4. In period 2, you again sell two pencils. The cost of pencils sold in period 2 is $4. Inventory at the end of period 2 is 0.

Advantage of using FIFO is that it is less subject to manipulation. The ending inventory is valued based on most recent purchases. COGS is based on earliest purchase costs. Therefore, in an inflationary environment, FIFO results in higher net income.

Weighted Average Cost

Under weighted average cost method, each item in inventory is valued using an average cost of all items in the inventory.

$$\text{Weighted average cost} = \frac{\text{Total cost of units available for sale}}{\text{Total units available for sale}}$$

Let's use the pencils example again to illustrate how inventory is calculated using the WAC method.

Total cost of pencils available for sales = $6

Total number of pencils available for sale = 4

Weighted average cost per pencil = $6/4 = $1.5

WAC Method	
Item	*WAC*
Cost of 2 pencils sold in period 1	$3
Inventory for 2 pencils at the end of period 1	$3
Cost of 2 pencils sold in period 2	$3
Inventory at the end of period 2	$0

Last In, First Out (LIFO)

Under Last In, First Out method:

- The newest items purchased or manufactured are assumed to be sold first.
- Oldest goods purchased or manufactured remain in ending inventory.
- The cost of goods sold reflects the cost of goods purchased or manufactured recently; the value of inventory reflects the cost of older goods purchased.

Let's continue with the pencils example to see how inventory is accounted for in LIFO:

Unlike FIFO, at the end of period 1, LIFO inventory consists of the first two pencils:

LIFO Method	
Item	*LIFO*
Cost of 2 pencils sold in period 1	$4
Inventory for 2 pencils at the end of period 1	$2
Cost of 2 pencils sold in period 2	$2
Inventory at the end of period 2	$0

LIFO is not allowed under IFRS; it is allowed only under US GAAP. Companies use LIFO during inflation to reduce taxes as cost of goods sold (COGS) is high.

Under LIFO, ending inventory is valued using earliest purchase costs. Therefore, in an inflationary environment, LIFO ending inventory will be less than current costs. Also, LIFO COGS will be higher than FIFO COGS leading to a lower net income.

Calculation of Cost of Sales, Gross Profit, and Ending Inventory

Based on the inventory valuation method used by a company, the allocation of inventory costs between cost of goods sold on the income statement and inventory on the balance sheet varies in periods of changing prices.

Continuing with the pencils example, assume each of the pencils was sold for $5. The table below summarizes the cost of goods sold, inventory ending value, and gross profit under each of the methods:

Inventory Accounting under Various Methods			
Item	FIFO (in $)	LIFO (in $)	WAC (in $)
COGS for period 1	2	4	3
Gross profit for period 1	8	6	7
Inventory at end of period 1	4	2	3
COGS for period 2	4	2	3
Gross profit for period 2	6	8	7
Inventory at end of period 2	0	0	0

Some points to be noted:

- The total gross profit and COGS for all the periods combined is the same under each of the methods.
- As the prices of pencils were increasing, the ending inventory was highest and COGS was lowest under FIFO.
- Similarly, the ending inventory was lowest and COGS was highest under LIFO.

The Effects of Inflation and Deflation

The allocation of total cost of goods available for sale to COGS and ending inventory varies under different inventory valuation methods. The following table compares LIFO vs. FIFO for different parameters when prices are rising and inventory levels are stable:

LIFO vs. FIFO with rising prices and stable inventory levels		
	LIFO	*FIFO*
COGS	Higher	Lower
Taxes	Lower	Higher
Earnings before taxes (EBT)	Lower	Higher

Earnings after taxes (Net Income)	Lower	Higher
Ending inventory	Lower	Higher
Working capital (CA-CL)	Lower	Higher
Cash flow (after tax)	Higher	Lower

Instructor's Note:

Weighted average costs provide results between FIFO and LIFO. Some tips for remembering the table above are listed below:

1. Remember the pencils example of $1, $1, $2, and $2. Deducing the LIFO values from this example for COGS, net income, and ending inventory becomes simpler.
2. FIFO is the opposite of LIFO.
3. Cash flow (after tax) is higher under LIFO as taxes paid are lower.
4. Companies following US GAAP prefer LIFO because the taxes paid are lower.
5. LIFO gives a better income statement and FIFO a better balance sheet as they reflect economic reality or recent costs. Under LIFO, cost of goods sold in income statement shows the most recent costs reflecting better quality. Similarly, under FIFO ending inventory on the balance sheet shows the most recent costs, reflecting better quality.

LIFO Reserve

The LIFO reserve is the difference between the reported LIFO inventory carrying amount and the inventory amount that would have been reported if the FIFO method had been used instead. The equation for LIFO reserve is given by:

LIFO reserve = FIFO inventory value – LIFO inventory value

LIFO Liquidation

In periods of rising inventory, the carrying amount of inventory under FIFO will exceed the carrying amount of inventory under LIFO. LIFO reserve is equal to the difference between LIFO inventory and FIFO inventory. LIFO reserve may increase for two reasons:

- The number of inventory units manufactured or purchased exceeds the number of units sold.
- Increasing difference between the older costs used to value inventory under LIFO and the more recent costs used to value inventory under FIFO.

If a firm is liquidating its inventory or if the prices are declining, the LIFO reserve will decline.

When the number of units sold in a period exceeds the number of units purchased/manufactured, it is called 'LIFO liquidation'. In LIFO liquidation, the costs from older LIFO layers will flow to COGS and it can be used by the management to manipulate earnings and margins. The gross profits increase because the older inventory carrying amounts are used for COGS while sales are at current prices. An increase in gross profit accompanied by a decrease in LIFO reserve must be used as a warning sign.

4. Presentation and Disclosure

The efficiency and effectiveness of inventory management can be evaluated using the following ratios:

- Inventory Turnover
- Days of Inventory on hand
- Gross Profit Margin

An analyst must understand that the choice of inventory valuation method can impact several financial ratios and make comparisons between two firms difficult. He needs to be particularly careful when comparing an IFRS and US GAAP firm.

Presentation and Disclosure

IFRS requires the following financial statement disclosures concerning inventory:

- The accounting policies used to measure inventory, including the cost formula.
- The total carrying amount of inventories and the carrying amount in classification (for example, merchandise, raw materials, production supplies, work in progress, and finished goods appropriate to entity).
- The carrying amount of inventories carried at fair value less costs to sell.
- The amount of inventories recognized as an expense in the period (cost of sales).
- The amount of any reversal of any write-down recognized as a reduction in cost of sales in the period.
- What led to the reversal of a write-down in the inventories?
- Carrying amount of inventories pledged as a security for liabilities.

Disclosures under U.S. GAAP are similar to IFRS except that it does not permit reversal of write down of inventories. In addition, any income from liquidation of LIFO inventory must be disclosed.

Inventory Ratios

The choice of inventory valuation method impacts various components of the financial statements such as cost of goods sold, net income, current assets, and total assets. As a result, it affects the financial ratios containing these items. Analysts must consider the differences in valuation methods when evaluating a company's performance over time or in comparison to other companies.

The table below summarizes the impact of valuation method on inventory-related ratios in an inflationary environment:

Ratio	*Numerator*	*Denominator*	*Impact on ratio*
Inventory turnover	Cost of goods sold is higher under LIFO.	Average inventory is lower under LIFO.	Higher under LIFO.
Days of inventory	No. of days are the same.	Higher under LIFO.	Lower under LIFO.
Total asset turnover	Revenue is the same.	Lower average total assets under LIFO.	Higher under LIFO.
Current ratio	Ending inventory is lower under LIFO so current assets are lower.	Current liabilities are the same.	Lower under LIFO.
Cash ratio	Cash is higher under LIFO because taxes are lower.	Current liabilities are the same.	Higher under LIFO.
Gross profit margin	Gross profit is lower under LIFO as COGS is higher.	Revenue is the same.	Lower under LIFO.
Return on assets	Net income is lower under LIFO as COGS is higher.	Lower average total assets under LIFO.	Lower under LIFO.
Debt to equity	Debt is the same.	Lower equity under LIFO. Equity = assets – liabilities. Total assets under LIFO are lower as ending inventory is lower.	Higher under LIFO.

The ratios that are important in evaluating a company's management of inventory are inventory turnover, number of days of inventory, and gross profit margin. A high inventory turnover implies that a company is utilizing inventory efficiently.

Inventory Analysis

Retailers normally report inventory in a single account. Whereas, manufacturing companies usually classify inventory into three separate accounts: raw materials, work in progress, and finished goods. These classifications can provide insights into a company's future sales and profits.

For example, a significant increase in the raw materials or work in progress inventory may be considered as a sign of increased demand, and higher future sales and profits. On the other hand, a significant increase in the finished goods inventory may be considered as a sign of slowing demand, and lower future sales and profits.

Analysts should also compare the growth rate of finished goods inventory to the growth rate of sales. For example, if growth of inventories is greater than the growth of sales, this could indicate slowing demand.

Summary

LO: Describe the measurement of inventory at the lower of cost and net realizable value and its implications for financial statements and ratios.

Net realizable value: It is calculated as estimated selling price under ordinary business conditions minus estimated costs necessary to get the inventory in condition for sale.

Net realizable value = estimated sales price – estimated selling costs

Market value: It is the current replacement cost subject to lower or upper limits. Market value has upper limit of net realizable value and lower limit of NRV less a normal profit margin.

Market value limits = (NRV - normal profit margin, NRV)

Under IFRS, inventories are valued at the lower of cost or net realizable value. Inventory write-ups are allowed but only to the extent a previous write-down to net realizable value was recorded.

Under US GAAP, inventories are valued at lower of cost or market. If cost exceeds market, inventory is written down. No subsequent write-ups are allowed.

When inventory is written down from cost to net realizable value:

- It decreases inventory, assets, and equity.
- It increases asset turnover, debt to equity ratio, and the debt to assets ratio.
- It results in a loss on the income statement, which decreases net income, net profit margin, return on assets, and return on equity.

LO: Calculate and explain how inflation and deflation of inventory costs affect the financial statements and ratios of companies that use different inventory valuation methods.

The following table compares LIFO and FIFO when prices are rising and inventory levels are stable. (We get the opposite effect during periods of falling prices)

LIFO vs. FIFO with rising prices and stable inventory levels		
	LIFO	*FIFO*
COGS	Higher	Lower
Taxes	Lower	Higher
Earnings before taxes (EBT)	Lower	Higher
Earnings after taxes (Net Income)	Lower	Higher
Ending inventory	Lower	Higher
Working capital (CA-CL)	Lower	Higher
Cash flow (after tax)	Higher	Lower

For weighted average costs, all values will be between those for the LIFO and FIFO methods.

LO: Describe the presentation and disclosures relating to inventories and explain issues that analysts should consider when examining a company's inventory disclosures and other sources of information.

IFRS requires the following financial statement disclosures concerning inventory:

- The accounting policies used to measure inventory, including the cost formula.
- The total carrying amount of inventories and the carrying amount in classification.
- The carrying amount of inventories carried at fair value less costs to sell.
- The amount of inventories recognized as an expense in the period (cost of sales).
- The amount of any reversal of any write-down recognized as a reduction in cost of sales in the period.
- What led to the reversal of a write-down in the inventories?
- Carrying amount of inventories pledged as a security for liabilities.

Disclosures under U.S. GAAP are similar to IFRS except that U.S. GAAP does not permit reversal of write-down of inventories. In addition, any income from liquidation of LIFO inventory must be disclosed.

Analysis of Inventory

- If finished goods inventory is increasing while raw material and work in progress inventory is decreasing, then this may indicate decreasing demand for the product.
- If raw material and work in progress inventory is increasing, then this may indicate increasing demand for the product.
- If finished goods inventory is increasing at a greater rate than increases in sales, then this may indicate decreasing demand for the product.

LM07 Analysis of Long-Term Assets

Ver 1.0

1. Introduction

Long-term assets are defined as those assets that are expected to provide future economic benefits extending more than one year.

These assets include:

- **Tangible assets** also known as fixed assets or property, plant, and equipment. Examples include land, buildings, furniture, machinery, etc.
- **Intangible assets** lack physical substance. Examples include patents, trademarks, etc.

While a "economic" balance sheet would include a wide range of assets such as a company's reputation and its trained, experienced workforce, "accounting" balance sheets prepared in accordance with IFRS and US GAAP allow for the recognition of a narrow range of assets.

There are two important questions in accounting for a long-term asset:

- What cost should be shown on the balance sheet?
- How should this cost be allocated over the life of an asset?

2. Acquisition of Intangible Assets

Intangible assets lack physical substance. Classic examples include software, customer lists, patents, copyrights, and trademarks.

Under IFRS, identifiable intangible assets must meet three definitional criteria. They must be:

- Identifiable – either capable of being separated from the entity or arising from contractual or legal rights
- Under the control of the company
- Expected to generate future economic benefits

In addition, two recognition criteria must be met:

- It is probable that the expected future economic benefits of the asset will flow to the company.
- The cost of the asset can be reliably measured

Accounting for an intangible asset depends on how it is acquired.

Acquired in a Business Combination

This refers to a situation where one company buys another company and, in the process, acquires intangible assets.

- Both IFRS and US GAAP require the use of acquisition method in accounting for business combinations. (This method will be studied in detail at Level II.)
- Under the acquisition method, identifiable intangible assets such as patents, copyrights, and trademarks are recorded at their fair value.

- Goodwill is an intangible asset that cannot be identified separately. It is recorded when one business acquires another business. If the purchase price exceeds the fair value of the net identifiable assets (both the tangible assets and the identifiable intangible assets, minus liabilities) acquired. For example, Tan Hospitals Inc. acquires Man Equipments Inc. for $100 million. The fair value of Man Equipments' net assets equal $95 million. In this case, the excess of $5 million will be recorded as goodwill.

Purchased in Situations Other than Business Combinations

This refers to a situation where an identifiable intangible asset is purchased, e.g. buying a patent from an inventor. The identifiable intangible asset is recorded at fair value which is assumed to be equal to the purchase price.

Developed Internally

Costs to internally develop intangible assets are generally expensed when incurred, although there are exceptions. The differences in whether the costs are capitalized or expensed affect financial statement ratios as outlined below:

- Balance sheet: A company that develops intangible assets internally will expense costs and record lower assets compared to a company that acquires such assets through purchase.
- Statement of cash flows: The costs of internally developing intangible assets are classified as operating cash flows, while the cost of acquiring intangible assets is classified as investing cash flows.

For internally developed intangible assets, there are two phases: the research phase and the development phase.

Research phase refers to the period during which commercial feasibility of an intangible asset is yet to be established. It is defined as "original and planned investigation undertaken with the prospect of gaining new scientific or technical knowledge and understanding."

Development phase refers to the period during which the technical feasibility of completing an intangible asset has been established with the intent of either using or selling the asset.

The treatment for the two phases varies slightly under IFRS and US GAAP as outlined below:

Under IFRS:

- Research costs are expensed.
- Development costs can be capitalized if technical feasibility and the intent to sell the asset are established.

Under US GAAP:

- Both research and development costs are expensed, but there are exceptions for software development.

- Software for sale: Costs incurred to develop a software product for sale are expensed until the product's feasibility is established, and capitalized after the product's feasibility has been established. Determining feasibility involves judgment.
- Software for internal use: All development costs should be capitalized.

Example

Acme Inc. starts an internal software development project on January 1, 2012. It incurs expenditures of $10,000 per month during the fiscal year ended December 31, 2012. By March 31, it is clear that the product will be developed successfully and will be used as intended. How are the software development costs recorded before and after March 31 according to IFRS and US GAAP?

Solution:

IFRS: Under IFRS all costs are expensed until feasibility is established if the software is developed for internal use. So, $30,000 (period from January 1 to March 31, 2012) is expensed and $90,000 is capitalized (from April 1 to December 31, 2012).

U.S GAAP: The entire cost of $120,000 should be capitalized.

3. Impairment and Derecognition of Assets

Impairment of Assets

Impairment charges reflect an unexpected decline in the fair value of an asset to an amount lower than its carrying amount (Whereas depreciation and amortization charges allocate the cost of a long-term asset over its useful life.)

Under IFRS

- An asset is impaired when its carrying value exceeds the recoverable amount.
- The recoverable amount is the greater of (fair value less selling costs) and the (present value of expected cash flows from the asset, i.e., the value in use).
- If impaired, the asset is written down to the recoverable amount.
- Subsequent loss recoveries are allowed, but they cannot exceed the historical cost.

Under US GAAP,

- An asset is impaired if its carrying value is greater than the asset's undiscounted future cash flows.
- If impaired, the asset is written down to the fair value.
- Subsequent loss recoveries are not allowed.

<u>Impact of Financial Statements</u>

When an asset is impaired the impact in that period is:

- The value of the asset is written down.

- Activity ratios such as sales/assets are higher.
- Income is lower due to impairment expense.
- Therefore, profitability ratios are lower.
- Cash flows are not impacted (ignoring taxes).

The impact in subsequent periods is:

- Higher income because of reduced depreciation expense.
- Therefore, profitability ratios are higher.
- Activity ratios such as sales/assets are higher.

Example

Given the following data, what is the reported value under IFRS and US GAAP:

- Carrying amount = $8,000
- Undiscounted expected future cash flows = $9,000
- Present value of expected future cash flows = $6,000
- Fair value if sold = $7,000
- Costs to sell = $200

Solution:

IFRS: Recoverable amount = greater of ($7,000 - $200, $ 6,000) = $6,800

Impairment loss = $8,000- $6,800 = $1,200

Write down the value of asset from $8,000 to $6,800 in the balance sheet and record a loss of $1,200 in the income statement.

US GAAP: Is the asset impaired? No, since the carrying amount of $8,000 is less than the undiscounted future cash flows of $9,000.

<u>Other Impairment Scenarios</u>

Impairment of Intangible Assets with Indefinite Lives: Intangible assets with indefinite lives are not amortized. They are carried on the balance sheet at historical cost, but they are tested annually for impairment.

Impairment of Long–term Assets Held for Sale: A long-term asset is reclassified as held for sale, if the management's intent is to sell it and its sale is highly probable. For example, if a company owns a machine with the intent of using it but now intends to sell it, then it should be reclassified as held for sale. Held-for-sale assets are not depreciated or amortized. At the time of reclassification, the asset should be tested for impairment and any impairment loss should be recognized.

The impairment loss can be reversed under IFRS and US. GAAP if the value of the asset recovers in the future. However, this reversal is limited to the original impairment loss.

Therefore, the carrying value of the asset after reversal cannot exceed the carrying value before the impairment was recognized.

Derecognition of Assets

A company derecognizes an asset (i.e., removes it from the financial statements) when the asset is disposed of or is expected to provide no future benefits from either use or disposal.

The four ways in which an asset can be derecognized (removed from a company's financial statements) are as follows:

- *Selling the asset*: The difference between the sales proceeds and the carrying value of the asset is reported as a gain or loss on the income statement.
- *Abandoning the asset*: The carrying value of the asset is removed from the balance sheet and a loss is recognized in that amount in the income statement.
- *Exchanging the asset*: The carrying value of the old asset is compared to the fair value of the new asset and a gain or loss is reported.
- *Distributed to owners in a spin-off*: In a spin-off, typically, an entire cash generating unit of a company with all its assets is spun off and does not result in any gain or loss.

Impact on Financial Statements

A derecognition can result in either a gain or loss on the income statement.

- A loss will lead to lower net income and assets.
- A gain will lead to higher net income and assets.

4. Presentation and Disclosure

Under IFRS, for each class of property, plant, and equipment, a company must disclose the measurement bases, the depreciation method, the useful lives (or, equivalently, the depreciation rate) used, the gross carrying amount, the accumulated depreciation at the beginning and end of the period, and a reconciliation of the carrying amount at the beginning and end of the period.

Under U.S. GAAP, the requirements are less exhaustive. A company must disclose the depreciation expense for the period, the balances of major classes of depreciable assets, accumulated depreciation by major classes or in total, and a general description of the depreciation method(s) used in computing depreciation expense with respect to the major classes of depreciable assets.

The disclosures related to impairment losses also differ under IFRS and US GAAP. Under IFRS, a company must disclose for each class of assets the following:

- The amounts of impairment losses and reversals of impairment losses recognized in the period and where those are recognized on the financial statements.
- The main classes of assets affected by impairment losses and reversals of impairment losses and the main events and circumstances leading to recognition of these

impairment losses and reversals of impairment losses.

Under US GAAP, reversal of impairment losses for assets held for use is not permitted. The company must disclose the following:

- The description of the impaired asset, what led to the impairment;
- The method of determining fair value;
- The amount of the impairment loss, and where the loss is recognized on the financial statements.

5. Using Disclosures in Analysis

Ratios used to analyze fixed asset include: fixed asset turnover, asset age, and remaining useful life.

$$\text{Fixed asset turnover} = \frac{\text{Total revenue}}{\text{Average net fixed assets}}$$

The higher this ratio, the higher the amount of sales a company is able to generate with a given amount of investment in fixed assets. A higher asset turnover ratio is often interpreted as an indicator of greater efficiency.

$$\text{Average age of a company's assets} = \frac{\text{Accumulated depreciation}}{\text{Depreciation expense}}$$

$$\text{Average remaining life of a company's assets} = \frac{\text{Net PPE}}{\text{Depreciation expense}}$$

The older the assets and the shorter the remaining life, the more a company may need to reinvest to maintain productive capacity.

Assuming straight-line depreciation and no salvage value, we can state the following relationships:

Estimated total useful life = Time elapsed since purchase (Age) + Estimated remaining life

Estimated total useful life = Historical cost ÷ annual depreciation expense

Historical cost = Accumulated depreciation + Net PPE

The information presented in a company's disclosures and these relationships can be used to conduct an analysis on the company's long-lived assets.

Summary

LO: Compare the financial reporting of the following types of intangible assets: purchased, internally developed, and acquired in a business combination.

Purchased intangible assets

- The cost of a finite-lived intangible asset is amortized over its useful life.
- Indefinite-lived intangible assets are not amortized. They are tested for impairment at least annually.

Internally developed intangible assets

- Under IFRS, research costs are expensed but development costs may be capitalized.
- Under US GAAP, both research and development costs are expensed. (Except in the case of software created for sale to others.)

Intangible assets acquired in a business combination

- Acquired intangible assets such as patents, copyrights, and trademarks are recorded at their fair value; similar to long-lived tangible assets.

LO: Explain and evaluate how impairment and derecognition of property, plant, and equipment and intangible assets affect the financial statements and ratios.

<u>Impairment</u>

Under IFRS

- An asset is impaired when its carrying value exceeds the recoverable amount.
- The recoverable amount is the greater of (fair value less selling costs) and the present (value of expected cash flows from the asset i.e. the value in use).
- If impaired, the asset is written down to the recoverable amount.
- Subsequent loss recoveries are allowed, but they cannot exceed the historical cost.

Under US GAAP,

- An asset is impaired if its carrying value is greater than the asset's undiscounted future cash flows.
- If impaired, the asset is written down to the fair value.
- Subsequent loss recoveries are not allowed.

Impact on financial statements

When an asset is impaired the impact in that period is:

- The value of the asset is written down.
- Activity ratios such as sales/assets are higher.
- Income is lower due to impairment expense.
- Therefore, profitability ratios are lower.

- Cash flows are not impacted (ignoring taxes).

The impact in subsequent periods is:

- Higher income because of reduced depreciation expense.
- Therefore, profitability ratios are higher.
- Activity ratios such as sales/assets are higher.

<u>Derecognition</u>

The three ways in which an asset can be derecognized (removed from a company's financial statements) are:

- Selling the asset: The difference between the sales proceeds and the carrying value of the asset is reported as a gain or loss on the income statement.
- Abandoning the asset: The carrying value of the asset is removed from the balance sheet and a loss is recognized in that amount in the income statement.
- Exchanging the asset: The carrying value of the old asset is compared to the fair value of the new asset and a gain or loss is reported.
- Distributed to owners in a spin-off: In a spin-off, typically, an entire cash generating unit of a company with all its assets is spun off and does not result in any gain or loss.

Impact on financial statements

- This can result in either a gain or loss on the income statement.
- A loss will lead to lower net income and assets.
- A gain will lead to higher net income and assets.

LO: Analyze and interpret financial statement disclosures regarding property, plant, and equipment and intangible assets.

An analyst can use financial statement disclosures to calculate the following:

Average age = accumulated depreciation/ annual depreciation expense.

Total useful life = historical cost/ annual depreciation expense.

Remaining useful life = ending PP&E/ annual depreciation expense.

LM08 Topics in Long-Term Liabilities and Equity

Version 1.0

1. Introduction

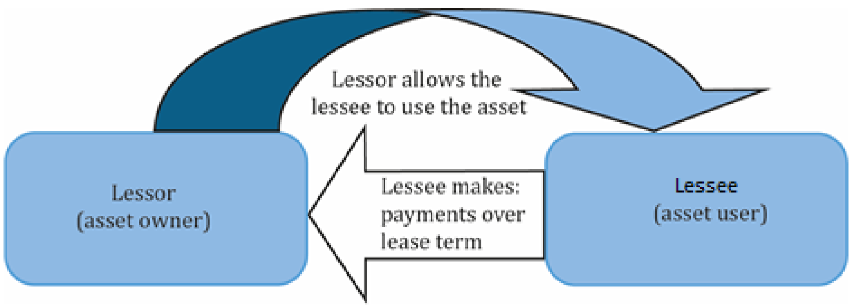
The learning module covers:

- Financial reporting of leases from the perspective of lessors and lessees
- Financial reporting of defined contribution, defined benefit, and stock-based compensation plans
- Presentation and disclosures relating to long-term liabilities and share-based compensation

2. Leases

A lease is a contract in which a lessor grants the lessee the exclusive right to use a specific underlying asset for a period of time in exchange for payments.

Below is a pictorial representation of what constitutes a lease. An asset's owner is called a lessor. The entity or person wishing to use the asset is called the lessee. The lessor allows the lessee to use the asset for a pre-determined period. In return, the lessee makes periodic payments to the lessor over the period for the right to use the asset. This period can be as long as 20 years, or as short as a month.



Advantages of Leasing

Following are some of the advantages to leasing an asset compared to purchasing it.

- Less cash is needed upfront. Leases typically require little to no down payment.
- Since leases are a form of secured borrowing, they generally have low interest rates.
- Lower risks associated with ownership such as obsolescence.

Lease Classification as Finance or Operating

Leases are classified as finance or operating. A finance lease is similar to purchasing an asset while an operating lease is similar to renting an asset.

A lease is classified as a finance lease if <u>any</u> of the following five criteria are met.

1. The lease transfers ownership of the underlying asset to the lessee.
2. The lessee has an option to purchase the underlying asset and is reasonably certain it will do so.

3. The lease term is for a major part of the asset's useful life.
4. The present value of the sum of the lease payments equals or exceeds substantially all of the fair value of the asset.
5. The underlying asset has no alternative use to the lessor.

If none of the criteria are met, then the lease is classified as an operating lease.

Financial Reporting of Leases

The financial reporting of leases depends on:

- Whether the party is the lessee or lessor
- Whether the party reports with IFRS or US GAAP
- Whether the lease is a finance or operating

US GAAP and IFRS share the same accounting treatment for lessors but differ for lessees.

IFRS has a single accounting model for both operating leases and finance lease lessees, while US GAAP has different accounting models for each.

Lessee Accounting—IFRS

Under IFRS, there is a single accounting model for both finance and operating leases for lessees.

- At inception, recognize a lease liability and corresponding right-of-use (ROU) asset on the balance sheet, both equal to the present value of lease payments.
- The lease liability is subsequently reduced by each lease payment using the effective interest method. Each lease payment is composed of:
  - Interest expense = Lease liability x discount rate
  - Principal repayment = Lease payment – Interest expense
- The right-of-use asset is amortized, often on a straight-line basis, over the lease term.

  (Although the lease liability and ROU begin with the same carrying value, their balance sheet values tend to diverge over time because of the differences in the calculation of principal repayments that reduces the lease liability and the amortization expense that reduces the ROU asset.)

The following list shows how the lease transaction affects the financial statements.

- <u>Balance sheet</u>: The lease liability is reported net of principal repayments and the ROU asset is reported net of accumulated amortization
- <u>Income statement</u>: Interest expense and amortization expense are shown separately.

- Cash flow statement: The principal repayment component is reported as cash outflow under financing activities. The interest expense can be reported under either operating or financing activities.

Example: Lessee Accounting - IFRS

(*This is based on Example 2 from the curriculum.*)

A company is offered the following terms to lease a machine: five-year lease with an implied interest rate of 10% and an annual lease payment of EUR100,000 per year payable at the end of each year. PV = EUR379,079. The asset will be amortized over the five-year lease term on a straight-line basis. The company reports under IFRS.

What would be the impact of this lease on the company's:

1. balance sheet at the beginning of the year?
2. income statement during the following year?
3. statement of cash flows during the following year?

Solution to 1:

The company will report a lease liability and a ROU asset of EUR379,079.

Solution to 2:

In Year 2, the company will report an interest expense of EUR31,699 and an amortization expense of EUR 75,816.

The calculations are shown in the tables below:

	Lease Payment	Interest Expense (10% × Lease Liability)	Principal Repayment (Payment – Interest)	Lease Liability
	FO.1	FO.2	FO.3	FO.4
Year 0				379,079
Year 1	100,000	37,908	62,092	316,987
Year 2	100,000	31,699	68,301	248,685
Year 3	100,000	24,869	75,131	173,554
Year 4	100,000	17,355	82,645	90,909
Year 5	100,000	9,091	90,909	0
Total	500,000	120,921	379,079	

	Amortization Expense	ROU Asset
	Straight-Line F.1	F.2
Year 0		379,079
Year 1	75,816	303,263

Year 2	75,816	227,447
Year 3	75,816	151,631
Year 4	75,816	75,816
Year 5	75,816	0
Total	379,079	

Solution to 3:

In Year 2, principal repayment of EUR68,301 will be reported as a cash outflow under financing activities. The interest expense of EUR31,699 may be reported under operating or financing activities depending on the company's reporting policies.

Lessee Accounting—US GAAP

Under US GAAP, there are two accounting models for lessees: one for finance leases and another for operating leases.

The finance lease accounting model is the same as the lease accounting model for IFRS.

The operating lease accounting model is different:

- At inception, recognize a lease liability and corresponding right-of-use asset on the balance sheet, both equal to the present value of lease payments.
- As with the previous method, the lease liability is subsequently reduced by each lease payment using the effective interest method
- But the amortization of the right-of-use asset is different, it is calculated as the lease payment less the interest expense.

  (Since the principal repayment and amortization are calculated in the same way, the lease liability and the ROU asset will always equal each other.)

The following list shows how the lease transaction affects the financial statements.

- Balance sheet: The lease liability is reported net of principal repayments and the ROU asset is reported net of accumulated amortization
- Income statement: Interest expense and amortization expense are shown together as a single operating expense on the income statement. They are not reported separately.
- Cash flow statement: The entire lease payment is reported as cash outflow under operating activities. The interest and principal components are not reported separately.

For a US GAAP company classifying a lease as an operating lease instead of a finance lease affects the financial ratios as shown below:

Ratio	Formula	Impact of Using an Operating Lease Instead of a Finance Lease
EBITDA margin	$\frac{\text{EBITDA}}{\text{Total revenues}}$	Lower: Lease expense is classified as an operating expense rather than interest and amortization.
Asset turnover	$\frac{\text{Total revenues}}{\text{Total assets}}$	Lower: Total assets are higher under an operating lease because the ROU asset is amortized at a slower pace in initial years.
Cash flow per share	$\frac{\text{Cash flow from operations}}{\text{Shares outstanding}}$	Lower: Cash flow from operations is lower because the entire lease payment is included in operating activities versus solely interest expense for a finance lease.

Lessor Accounting

The accounting for lessors is identical under IFRS and US GAAP. However, the accounting differs based on whether the lease is a finance lease or an operating lease.

<u>Finance lease lessors (IFRS and US GAAP)</u>

- At inception, recognize a lease receivable asset equal to the present value of future lease payments and de-recognize the leased asset, simultaneously recognizing any difference as a gain or loss.
- The lease receivable is subsequently reduced by each lease payment using the effective interest method.

The following list shows how the lease transaction affects the financial statements.

- *Balance sheet:* Lease receivable net of principal proceeds is reported on the balance sheet.
- *Income statement:* Interest income is reported on the income statement, typically as revenue.
- *Cash flow statement*: The entire cash receipt is reported under operating activities.

Operating lease lessors (IFRS and US GAAP)

- The lease contract is treated as a rental agreement.

The following list shows how the lease transaction affects the financial statements.

- *Balance sheet:* The balance sheet is not affected. The lessor continues to recognize the underlying asset and depreciate it.
- *Income statement:* Lease revenue is recognized on a straight-line basis on the income statement. The depreciation expense continues to be recognized.
- *Cash flow statement:* The entire cash receipt is reported under operating activities.

Example: Lessor Accounting

(*This is based on Example 4 from the curriculum.*)

Let's examine the previous example from the perspective of the lessor. Assume that the carrying value of the asset immediately prior to the lease is EUR350,000, accumulated depreciation is zero, and the lessor elects to depreciate it on a straight-line basis over five years.

How would the lessor's financial statements be affected by the classification of the lease as a finance or operating lease?

Solution:

Balance sheet: The difference on the balance sheet is material. The present value of lease payments is well above the carrying value of the asset. The finance lease classification therefore results in a significant increase in assets.

Balance Sheet	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
Finance lease:					
Lease receivable, net	316,987	248,685	173,554	90,909	0
Operating lease:					
Property, plant, and equipment, net	280,000	210,000	140,000	70,000	0

Income statement: The difference on the income statement is also material.

Income Statement	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
Finance lease:					
Interest revenue	37,908	31,699	24,869	17,355	9,091
Operating lease:					
Lease revenue	100,000	100,000	100,000	100,000	100,000

Cash flow statement: The cash flow statement is the same under both options.

Statement of Cash Flows	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
Finance lease:					
Cash flows from operating activities	100,000	100,000	100,000	100,000	100,000
Operating lease:					
Cash flows from operating activities	100,000	100,000	100,000	100,000	100,000

3. Financial Reporting for Postemployment and Share-Based Compensation Plans

Employee Compensation

Employee compensation packages are designed to achieve a variety of goals, including meeting employees' liquidity needs, retaining employees, and motivating employees.

Common components of employee compensation are:

- Salary: Provides for the liquidity needs of an employee.
- Bonuses: Generally provided in the form of cash. They can motivate and reward employees for short-or-long term performance.
- Non-monetary benefits such as health and life insurance premiums: Provided to facilitate employees performing their jobs.
- Defined contribution/benefit pension plan: Provides cash flow to employees in retirement.
- Share-based compensation: Aligns employee's interest with those of shareholders.

Pension Plans

Instructor's Note: Pensions are discussed in great detail at Level II.

One common post-employment benefit offered by companies to their employees is pension. Pensions and other post-employment benefits give rise to non-current liabilities reported by many companies. When companies promise its employees certain benefits after a certain period of time, they are obligated to fulfill that promise.

The accounting treatment of pensions depends on the type of pension plan. There are primarily two types of pension plans:

1. Defined contribution plan: Under this plan, a company contributes an agreed-upon amount to the plan. This contribution is recognized as a pension expense on the income statement and an operating cash outflow. Since there is no future payout or obligation, no liability is reported on the balance sheet. A liability is recognized on the balance sheet if some prior agreed-upon amount is not paid by the end of the fiscal year.
2. Defined benefit plan: Under this plan, a company promises to pay a certain amount in the future to the employees. The amount of future obligation is based on a lot of assumptions such as retirement age of its employees, last drawn salary before retirement, mortality rate, etc. For example, a company may promise an employee an annual pension payment equal to 60% of his last salary at retirement, until death. The pension obligation is the present value of future payments the company expects to make. A company fulfills this obligation by setting up a pension fund (also known as plan assets) and making

payments to this fund. The ongoing pension obligations are paid from this fund. The amount in the fund remains invested until it has to be paid to the retirees.

Accounting for Defined-Benefit Plans

Since the future obligation of defined benefit pension fund cannot be determined with certainty, accounting is more complicated than the defined contribution plan. Listed below are a few rules for you to remember for a defined benefit plan:

- If the fair value of plan assets > present value of estimated pension obligation, the plan is overfunded (has a surplus). It is called net pension asset.
- If the present value of estimated pension obligation > fair value of plan assets, the plan is underfunded. It is called net pension liability.

Under both IFRS and US GAAP, the net pension asset or liability is reported on the balance sheet. An underfunded defined benefit pension plan is reported as a non-current liability on the balance sheet.

For each period, the change in net pension asset or liability is recognized either in profit or loss or in other comprehensive income.

Under IFRS, the change in net pension asset or liability has three components:

- Employee service costs and past service costs: Recognized as pension expense in the income statement.
- *Service cost is the present value of the benefit earned by an employee for one additional year of service. It is the sum of past service costs and present value of the increase in pension benefit earned by working for one more year.*
- Net interest expense or income accrued on the beginning net pension asset or liability represents the change in value of the net defined benefit pension asset or liability: Recognized as pension expense in the income statement.
- *Net interest expense = net pension asset or liability* x *discount rate used to estimate the present value of the pension obligation.*
- Remeasurements: Recognized in other comprehensive income on the balance sheet.
- *Remeasurements = actuarial gains and losses and the actual return on plan assets minus the net interest expense or income.*

The actual return on plan assets includes interest, dividends and other income derived from the plan assets, including realized and unrealized gains or losses.

Under US GAAP, the change in net pension asset or liability has five components:

- Employees' service costs for the period.
- Interest expense accrued on the beginning pension obligation.

- Expected return on plan assets. It isa reduction in the amount of expense recognized.
- Past service costs.
- Actuarial gains or losses.

The first three are recognized in profit and loss during the period incurred. Past service costs and actuarial gains and losses are recognized in other comprehensive income in the period they occur and later amortized into pension expense. Under US GAAP, companies are also allowed to immediately recognize actuarial gains and losses in profit and loss.

Example

On 31 Dec. 2012, a company has a pension obligation of 100 and pension assets are 90. What will the company report on the balance sheet under IFRS and US GAAP?

Solution:
Funded status = pension assets – pension obligation = 90 – 100.

The company will report a net pension liability of 10.

Share-Based Compensation

Share-based compensation is intended to align incentives of management and owners. From an accounting perspective, share-based compensation is treated as an expense even though no cash changes hands when stock options/grants are issued. Both IFRS and US GAAP require us to come up with a fair value to measure the share-based compensation. Also, the nature and extent of share-based compensation and its effect on financial statements need to be disclosed.

There are several disadvantages of share-based compensation:

- Employees may have limited influence over the company's market value. Hence, it does not necessarily provide the desired incentives.
- Increased ownership may lead managers to be more risk averse or less risk averse than they should be, depending on whether the current stock price is above or below the exercise price of the stock options.
- When shares are granted to employees, the existing shareholder's ownership is diluted.

The common forms of share-based compensation are discussed below.

Stock Grants

There are three types of stock grants:

<u>Outright grants</u>

- Compensation expense is reported on the basis of fair value of the stock on the grant date.

- Compensation expense is allocated over the service period.

<u>Restricted grants</u>

- Shares have to be returned if certain conditions not met. Conditions might include staying with the company for a specified period, or meeting certain performance goals.
- Compensation expense is reported on the basis of fair value of the stock on the grant date.
- Compensation expense is allocated over the service period.

<u>Performance shares</u>

- Determined by performance measures other than change in share price. For example, accounting earnings or return on assets.
- It addresses employees' concern that share price is beyond their control and should not be used for assessment.
- Compensation expense is reported on the basis of fair value of the stock on the grant date.
- Compensation expense is allocated over the service period.

Stock Options

Like stock grants, compensation expense related to stock options is reported at fair value. The fair value has to be estimated using an appropriate valuation model. Commonly used models are the Black–Scholes option pricing model and the binomial model. The value of a stock option is sensitive to model inputs and assumptions. The table below shows the impact of an increase in an input value or assumption:

Input or Assumption	**Impact if there is an <u>increase</u> in input value or assumption**
Exercise price	Higher exercise price leads to a **lower** option value
Stock price volatility	Higher stock price volatility leads to **higher** option value
Estimated life of each award	Higher time to expiration leads to **higher** option value
Estimated dividend yield	Higher dividend yield leads to a **lower** option value
Risk free rate	Higher risk-free rate leads to a **higher** option value

In accounting for stock options, there are several important dates, including the grant date, the vesting date, the exercise date, and the expiration date. The grant date is the day that options are granted to employees. The vesting date is the date that employees can first exercise the stock options. The exercise date is the date when employees actually exercise the options and convert them to stock. If the options go unexercised, they may expire at some pre-determined future date, commonly 5 or 10 years from the grant date. The compensation expense is allocated over the service period.

Example: Stock options

A company awards 1,000,000 stock options to its executives on 1 July 2015. The estimated cost of each option is $0.50. The options require a service period of 4 years after the grant date before vesting. What is the stock option expense for 2015?

Solution:

The expense for the year = 1,000,000 x $0.50/4 x ½ = $62,500

Note: We divide by 4 because the options have a service period of 4 years. Since the options are granted in the middle of 2015, we multiply by ½ to allocate the expense to the second half of 2015.

Other Types of Share-Based Compensation

Both stock grants and stock options have the potential to dilute EPS. To avoid this, options such as Stock Appreciation Rights (SARs) are available. They compensate an employee on the basis of changes in the value of shares without requiring the employee to hold the shares. Like other forms of share-based compensation, SARs align employee's interests with shareholders.

They have the following additional advantages:

- The potential for risk aversion is limited because employees have limited downside risk and unlimited upside potential similar to employee stock options, and
- Shareholder ownership is not diluted.

A disadvantage is that SARs require a current-period cash outflow.

Similar to other share-based compensation, SARs are valued at fair value and compensation expense is allocated over the service period of the employee.

4. Presentation and Disclosure

Instructor's Note: Relevant excerpts from the curriculum are reproduced below. Read through this section once or twice. It is not very testable.

Presentation and Disclosure of Leases

The objective of lease disclosure is to provide the user of financial statements with information to assess the amount, timing and uncertainty of cash flows associated with leases.

The non-current portion of the balance sheet will typically contain a "right of use" asset and the non-current liabilities section will typically show the lease liability.

Lessee Disclosure

Lessee disclosures should include the following:

- the carrying amount of right of use assets and the end of the reporting period by class of underlying asset;
- total cash outflow for leases;
- interest expense on lease liabilities;
- depreciation charges for right-of-use assets by class of underlying asset; and
- additions to right of use assets

In addition, lessees should disclose a maturity analysis of lease liabilities and additional quantitative and qualitative information about leasing activity.

Lessor Disclosure

At a minimum, lessors should disclose:

- for finance leases, the amount of selling profit or loss; and finance income on the net investment in the lease; and income relating to variable lease payments not included in the measurement of the lease;
- for operating leases, lease income with separate disclosure for income relating to variable lease payments based on an index or rate.

Presentation and Disclosure of Postemployment Plans

Companies are required to make disclosures such as:

- the nature of benefits provided, the regulatory framework in which the plan operates, governance of the plan, and risks to which the plan exposes the entity;
- a reconciliation from the opening balance to the closing balance of the net pension asset or liability, with separate reconciliations for plan assets and the present value of the defined benefit obligation, showing service costs, interest income or expense, remeasurements, past service costs, contributions to the plan, and other components of the change;
- a sensitivity analysis showing how changes in significant assumptions (such as the discount rate used to measure the defined benefit pension obligation) would affect the amounts reported on the financial statements;
- the composition of plan assets by category, such as equity securities, fixed-income securities, and real estate; and
- indications of the effect of the defined benefit pension plans on the entity's future cash flows

Presentation and Disclosure of Share-Based Compensation

Companies are required to make disclosures such as:

- A description of each type of share-based payment arrangement, including its general terms and conditions, such as vesting requirements, the maximum term of options granted and the method of settlement (i.e., cash or equity)

- Details about the number and weighted average exercises price of options, including:
  - the number outstanding at the beginning of the period,
  - granted during the period,
  - forfeited during the period,
  - exercised during the period,
  - expired during the period,
  - outstanding at the end of the period, and
  - exercisable at the end of the period.
- For other equity instruments granted during the period (i.e., other than share options), the number and weighted average fair value of those equity instruments at the measurement date, and information on how that fair value was measured.

Summary

LO: Explain the financial reporting of leases from the perspectives of lessors and lessees.

<u>Lessee's perspective</u>:

Under IFRS, there is a single accounting model for both finance and operating leases for lessees.

- Recognize a lease liability and corresponding right-of-use asset on the balance sheet, both equal to the present value of lease payments.
- The liability is subsequently reduced using the effective interest method
- The right-of-use asset is amortized, often on a straight-line basis over the lease term.
- Interest expense and amortization expense are shown separately on the income statement.
- The principal repayment component is reported as cash outflow under financing activities. The interest expense can be reported under either operating or financing activities.

Under US GAAP, there are two accounting models for lessees: one for finance leases and another for operating leases.

The finance lease accounting model is the same as the lease accounting model for IFRS.

The operating lease accounting model is different:

- Recognize a lease liability and corresponding right-of-use asset on the balance sheet, both equal to the present value of lease payments.
- The liability is subsequently reduced using the effective interest method
- But the amortization of the right-of-use asset is the lease payment less the interest expense.
- Interest expense and amortization expense are shown together as a single operating expense on the income statement.
- The entire lease payment is reported as cash outflow under operating activities.

<u>Lessor's perspective</u>:

Finance lease lessors (IFRS and US GAAP)

- Recognize a lease receivable asset equal to the present value of future lease payments and de-recognize the leased asset, simultaneously recognizing any difference as a gain or loss.
- The lease receivable is subsequently reduced by each lease payment using the effective interest method.
- Interest income is reported on the income statement, typically as revenue.

- The entire cash receipt is reported under operating activities on the statement of cash flows.

Operating lease lessors (IFRS and US GAAP)

- The balance sheet is not affected: the lessor continues to recognize the underlying asset and depreciate it.
- Lease revenue is recognized on a straight-line basis on the income statement.
- The entire cash receipt is reported under operating activities on the statement of cash flows.

LO: Explain the financial reporting of defined contribution, defined benefit, and stock-based compensation plans.

Defined Contribution Plans

- The amount of contribution into the plan is specified. However, the amount of pension that is ultimately paid by the plan is not defined and it depends on the performance of the plan's assets.
- The cash payment made into the plan is recognized as pension expense on the income statement.

Defined Benefit Plans

- The amount of pension that is ultimately paid by the plan is defined, usually according to a benefit formula.
- Under both IFRS and US GAAP, companies must report the difference between the defined benefit pension obligation and the pension assets as an asset or liability on the balance sheet. An underfunded defined benefit pension plan is reported as a non-current liability on the balance sheet.
- Under IFRS, the change in the defined benefit plan net asset or liability is recognized as a cost of the period. Two components of the change (service cost and net interest expense or income) are recognized in the income statement and one component (re-measurements) is recognized in other comprehensive income.
- Under US GAAP, the change in the defined benefit plan net asset or liability is also recognized as a cost of the period. Three components of the change (current service costs, interest expense on the beginning pension obligation, and expected return on plan assets) are recognized in the income statement and two components (past service costs and actuarial gains and losses) are recognized in other comprehensive income. Under US GAAP, companies are also allowed to immediately recognize actuarial gains and losses in profit and loss.

Stock Based Compensation Plans

- Share-based compensation is treated as an expense even though no cash changes hands when stock options/grants are issued.

- Both IFRS and US GAAP require us to come up with a fair value to measure the share-based compensation. Also, the nature and extent of share-based compensation and its effect on financial statements need to be disclosed.

LO: Describe the financial statement presentation of and disclosures relating to long-term liabilities and share-based compensation.

Refer to Section 4 of the notes.

LM09 Analysis of Income Taxes

Version 1.0

1. Introduction

One of the key concepts we will discuss in this learning module is deferred tax assets and liabilities. Deferred tax assets and liabilities are created because of differences between how and when transactions are recognized for financial reporting purposes relative to tax reporting.

2. Differences Between Accounting Profit and Taxable Income

Some common terms related to financial reporting are defined below:

- Accounting profit: It is also known as pretax income or earnings before tax (EBT) and appears on the income statement. In simple terms, this is before taxes are calculated. Accounting profit is based on accounting standards.
- Income tax expense: Tax expense, or tax benefit, appears on a company's income statement, which is created using financial reporting standards. It is calculated based on the accounting profit (profit before tax) using a given tax rate.
- Carrying value**:** The net value of an asset or liability reported on the balance sheet according to accounting principles.

Some common terms related to tax reporting are defined below:

- Taxable income: It is the portion of income that is subject to income taxes under the tax laws where the company is operating.
- Income tax payable: Income tax payable is calculated on a company's taxable income using the applicable tax rate. This is the amount that is generally paid to the tax authorities and it appears on the balance sheet. Since it results in a cash outflow, firms minimize taxes payable by showing higher expenses and lower taxable income.
- Tax Base of an Asset: Tax base of an asset is the amount that will be deductible for tax purposes in future periods as economic benefits are realized. It is used to calculate tax payable and is analogous to the carrying amount (net book value) concept. Tax base is the amount allocated to asset for tax purposes whereas carrying amount is based on accounting principles.

Why are accounting profit and taxable income different?

Both report income before deducting tax expense, yet they are different because accounting profit is based on accrual method of accounting (revenues reported when earned and expenses when incurred). On the other hand, taxable income is usually based on cash-basis accounting (revenue recognized when cash is collected and expense reported when cash is paid).

Accounting profit and taxable income differ when:

- Revenues and expenses are recognized in one period for accounting purposes and a different period for tax purposes.
- The carrying amount and tax base of assets/liabilities differ.
- Gain/loss of assets/liabilities in the income statement is different than tax return.
- Some revenues/expenses recognized in the income statement are not considered for tax purposes.

The following table shows the distinction between accounting profit/ taxable income and income tax expense/taxes payable.

Income Statement for Everest Inc.			**Tax Return for Everest Inc.**		
$ million	**2011**	**2012**	$ million	**2011**	**2012**
Revenue	100	100	Revenue	100	100
Cash Expenses	50	50	Cash Expenses	50	50
Depreciation (SL)	25	25	Depreciation (Acc)	40	10
Accounting profit	25	25	**Taxable income**	10	40
Income tax expense (40%)	10	10	**Taxes payable (40%)**	4	16
Profit after tax	15	15	**Profit after tax**	6	24

Instructor's Note:

The following table summarizes the analogous financial and tax reporting terms:

Financial reporting	**Tax Reporting**
Accounting profit	Taxable income
Tax expense	Income tax payable
Carrying amount	Tax base

3. Deferred Tax Assets and Liabilities

Deferred tax liabilities

Deferred tax liability (DTL) occurs when income tax expense (financial accounting) is greater than income tax payable. It is a liability because we pay less tax now, thereby creating a liability or an obligation to pay more in the future. Since the tax will be paid later, it is deferred.

Such a situation can happen when:

- Revenue is recognized on income statement before being included on tax return (accrued/unbilled revenue).
- Expenses are tax deductible before being recognized on income statement.

For example, in the sample income statement and tax return shown for Everest Inc, at the end of 2011, the income tax expense (10) is greater than the income tax payable (4), hence a DTL of 6 (10 - 4) will be recorded on the balance sheet. At the end of 2012, the DTL is reversed and it increases taxes payable by 6.

Deferred Tax Assets

Deferred tax assets (DTA) arise when income tax payable is temporarily greater than income tax expense. In other words, taxable income is higher than accounting profit. Since tax is paid in advance, it is considered an asset; it can be viewed as a prepaid expense.

Such a situation can happen when:

- Revenue is taxed before being recognized on income statement (unearned revenue).
- Expense is recognized on the income statement before being tax deductible.

Consider the following income statement and tax return for Atlas Inc.

Income Statement for Atlas Inc.			**Tax Return for Atlas Inc.**		
$ million	**2011**	**2012**	$ million	**2011**	**2012**
Revenue	100	100	Revenue	120	80
Cash Expenses	50	50	Cash Expenses	50	50
Accounting profit	50	50	**Taxable income**	70	30
Income tax expense (40%)	20	20	**Taxes payable (40%)**	28	12
Profit after tax	30	30	**Profit after tax**	42	18

At the end of 2011, since the income tax payable (28) is greater than the income tax expense (20), a DTA of 8 (28 - 20) will be recorded on the balance sheet. At the end of 2012, the DTA is reversed and it brings down taxes payable by 8.

Any deferred tax asset or liability is the result of a temporary difference that is expected to reverse in the future. Deferred tax liability reverses when taxes are paid in the future resulting in cash outflows. Similarly, deferred tax asset reverses when tax benefits are realized in the future resulting in lower cash outflows.

Realizability of Deferred Tax Assets

A DTA may be created only if the company expects to be able to realize the economic benefit of the deferred tax asset in the future.

If it is uncertain whether future economic benefits will be realized from a temporary difference (for example, if the company is being liquidated), the temporary difference will not result in the recognition of a DTA.

If a DTA was previously recognized, but now there is sufficient doubt about whether the economic benefits will be realized, then, under IFRS, the DTA would be reversed. Under US GAAP, a DTA is reduced by creating a valuation allowance (a contra account).

Tax Base of an Asset

Asset tax base is the value of an asset according to tax rules and is used to calculate tax payable. Asset tax base is analogous to carrying amount (net book value).

<u>Example</u>

An asset is purchased for 50 and is depreciated over two years. On the financial statements the depreciation is 25 and 25. According to tax rules the depreciation is 40 and 10. Show the carrying amount and tax base at T=0, T=1, and T=2.

Time period	Carrying Amount (Financial Reporting)	Tax Base (Tax reporting)
T = 0	50	50
T = 1	25	10
T = 2	0	0

Link between Tax Base and DTL

Deferred tax liability = (Carrying amount – Tax base) x Tax rate

Assuming a 40% tax rate for the above example,

At T = 1: DTL = (25 - 10) x 0.4 = 6

At T = 2: DTL = (0 - 0) x 0.4 = 0

Instructor's Note: If the carrying amount and tax base are the same then DTL is 0. If the carrying amount is greater than the tax base, then there will be a deferred tax liability. If carrying amount is less than the tax base, then there will be a deferred tax asset.

Both DTL and DTA should be measured at the tax rate which is expected to apply when the liability is settled (reversed).

Link between Income Tax Expense, Tax Payable, and DTL

Income tax expense = Income tax payable + Change in net DTL

where net DTL = DTL – DTA and change in net DTL is the ending value of net DTL – beginning value of net DTL.

Example

In 2015, the income tax payable for a certain company is 100. During the year, DTL increased from 20 to 25 and DTA increased from 0 to 10. What is the provision for income tax in 2015?

Solution:

ITE = ITP + ΔDTL – ΔDTA = 100 + 5 – 10 = 95

Determining the Tax Base of an Asset

Asset tax base is the amount that will be deductible for tax purposes in future periods as the economic benefits become realized.

Examples:

Item	*Carrying Amount*	*Tax Base*	*Temporary Difference*
An asset is purchased for 50; for year 1 depreciation = 25 on income statement and 40 for tax purposes.	25	10	15
Capitalized development cost = 100 at the start of the year. During the year 30 was amortized. For tax purposes only 25% amortization is allowed.	70	75	-5
Research cost for the year = 100; entire cost was expensed. Tax rules require cost to be spread over 5 years.	0	80	-80
Gross accounts receivable = 100 Provision for doubtful debt = 10%. Tax authorities allow 20%.	90	80	10

Determining the Tax Base of a Liability

The tax base of a liability is the carrying amount of the liability less any amounts that will be deductible for tax purposes in the future.

Example:

Item	*Carrying Amount*	*Tax Base*	*Temporary Difference*
Customer payments received in advance = 50 Amount is taxable.	50	0	50

Since the customer pays 50 in advance. A liability called unearned revenue is created on the accounting side making the carrying amount of the liability to 50. On the tax side, 50 is shown as revenue and taxes are paid for the same. So, tax base is 0 and carrying amount is 50.

Temporary and Permanent Differences between Taxable and Accounting Profit

Permanent differences are differences between tax and financial reporting of revenue (expenses) that will **not** be reversed at some future date. These differences do not give rise to DTLs and DTAs. Examples include:

- Income or expense items not allowed by tax legislation. One example that leads to a permanent difference is when a company incurs a penalty or fine on breaking a civil or criminal law.
- Tax credits for some expenditures that directly reduce taxes.

As no deferred tax item is created for permanent differences, all permanent differences result in a difference between the company's effective tax rate and statutory tax rate.

$$\text{Reported effective tax rate} = \frac{\text{Income tax expense}}{\text{Pretax income}}$$

Example

In 2012, Acme's provision for income tax was 20 against an EBT of 100. In the same year, the tax payable was 25 and the taxable income was 110. What was Acme's effective tax rate for 2012?

Solution:

Effective tax rate = 20/100 = 20%

Temporary differences between taxable and accounting profit arise from a difference between the tax base and the carrying amount of assets and liabilities. DTLs and DTAs are only created if there is temporary difference which is expected to reverse in the future.

Some examples of situations that lead to temporary differences in carrying amount and tax base are listed below:

Temporary differences between carrying amount and tax base			
Balance Sheet Item	**Carrying amount vs. Tax Base**	**DTL or DTA**	**Example**
Asset	Carrying amount > Tax Base	DTL	Straight-line depreciation for accounting profit. Accelerated depreciation for taxable profit.
Asset	Carrying amount < Tax Base	DTA	Research cost expensed for accounting profit. Amortized for tax.
Liability	Carrying amount > Tax Base	DTA	Cash from customers before revenue recognition. Cash from customers is taxed.
Liability	Carrying amount < Tax Base	DTL	

Instructor's Note: Remember the first relation for how a DTL is created for an asset. Everything else follows.

4. Corporate Income Tax Rates

There are three types of tax rates that are relevant to analysts:

- Statutory tax rate: The tax rate established by law.
- Effective tax rate: Tax amount reported on the income-tax statement divided by the pre-tax income.
- Cash tax rate: Tax actually paid divided by pre-tax income.

Differences between cash taxes and reported taxes occur due to timing differences between accounting and tax calculations. These differences are reflected as a deferred tax asset or a deferred tax liability on the balance sheet.

Statutory tax rate and effective tax rates can be different for a number of reasons, such as tax credits, adjustments to previous years, withholding tax on dividends, etc.

The effective tax rate of a company that operates in multiple countries is the weighted average of the effective tax rates in each country. If the company is expected to report higher profits in a country with a high tax rate and lower profits in a country with a low tax rate, then its effective tax rate will increase. The effective tax rate is used to forecast earnings on the income statement and the cash tax rate is used to forecast cash flows.

To improve the quality of forecasts, the analyst should adjust for one-time events. Information provided in the notes to the financial statements can be used to identify such events.

If a company's effective tax rates are consistently lower than statutory rates or the effective tax rates reported by competitors, then analysts must scrutinize the reason when forecasting tax expense.

5. Presentation and Disclosure

Measurement of DTL

The treatment of deferred tax liability is discussed below:

- DTL should be classified as debt if the liability is expected to reverse in the future when taxes are paid.
- If it is determined that a DTL will not be reversed, then DTL should be reduced and the amount by which it is reduced should be treated as equity. There is no cash outflow expected in the future. Assume for Period 1 in the example above there is a DTL because of the different depreciation methods used for accounting and tax reporting purposes. Also assume the company is expected to grow at a rate of 30% in the foreseeable future, making the depreciation amounts higher with no reversal in sight. In such cases, the liability will be treated as equity.
- If there is uncertainty about the timing and amount of tax payments, analysts should treat DTLs as neither liabilities nor equity.

Measurement of DTA and Valuation Allowance

If it is determined that the DTA will not be realized because of insufficient future taxable income to recover the tax asset, then the DTA must be reduced.

Under US GAAP, a DTA is reduced by creating a valuation allowance (a contra account). DTA and net income decrease in the period in which a valuation allowance is established. DTA can be revalued upward by decreasing the valuation allowance which would increase earnings.

Instructor's Note

For the exam, you may think of valuation allowance in terms of depreciation. When depreciation expense goes up, net income comes down. Similarly, if valuation allowance goes up, net income comes down. Depreciation is shown as an expense on the income statement. Similarly, an increase in valuation allowance is shown as a loss on the income statement.

Changes in Income Tax Rates

The measurement of deferred tax assets/liabilities is based on current tax law. But, if there is any subsequent change in tax laws or new income tax rates, then existing deferred tax assets and liabilities must be adjusted to reflect those changes. When income tax rate changes, deferred tax assets and liabilities are calculated based on the new tax rate.

Let us take an example to see what happens to DTL. Assume the carrying amount of an asset is 25 and its tax base is 10. When the tax rate is decreased from 40% to 30%, the effect on DTL is:

$\text{DTL (old rate)} = (\text{carrying amount} - \text{tax base}) \text{ x tax rate} = (25 - 10) \text{ x } 0.4 = 6$

$\text{DTL (new rate)} = (25 - 10) * 0.3 = 4.5$

DTL changes from 6 to 4.5.

<u>Relationship between tax rate, DTL, and DTA</u>

- Decrease in tax rate reduces both deferred tax liabilities and deferred tax assets.
- Increase in tax rate increases both deferred tax liabilities and deferred tax assets.

Example

Firm A has a net deferred tax liability. The government announces a decrease in the statutory tax rate. Will this change benefit the income statement and balance sheet?

Solution:

If the government announces a decrease in the statutory tax rate, it will cause the net DTL to decrease. A lower tax rate causes the tax expense to decrease and, consequently, the net income and equity to increase.

Example

Rocky Inc. a US-based company, reports the following information:

	2014	2015
Deferred tax asset	100	100
Valuation allowance	25	20
Deferred tax asset, net of valuation allowance	75	80
Deferred tax liability	70	70
Net deferred tax asset	5	10
Tax rate	40%	40%

1. What does the decrease in valuation allowance imply about future profitability?
2. How does the reduction in valuation allowance impact income tax expense and net income?
3. What is the impact on deferred taxes if the tax rate is reduced to 35%?

Solution to 1:

The decrease in valuation allowance implies that the company is more likely to benefit from the deferred tax asset. This is probably because the company expects higher profitability in the future.

Solution to 2:

The reduction in valuation allowance causes the tax expense to be lower and the net income to be higher.

Solution to 3:

If the tax rate is reduced from 40% to 35% that reduces both the deferred tax asset and deferred tax liability. Since the company has a net deferred tax asset, a reduction in the tax rate will cause the net deferred tax asset to be lower. Consequently, the equity value will also decrease.

Instructor's Note: If the valuation allowance is equal to the deferred tax asset, this implies that a company expects no taxable income prior to the expiration of the deferred tax asset.

When a company decreases the valuation allowance, it implies a higher probability that the deferred tax asset will benefit the company.

Summary

LO: Contrast accounting profit, taxable income, taxes payable, and income tax expense and temporary versus permanent differences between accounting profit and taxable income.

Accounting profit: This is the pretax income from the income statement. It is based on accounting standards.

Taxable income: This is income subject to tax. It is based on the tax return.

Taxes payable: It is a liability on the balance sheet calculated using taxable income.

Income tax expense: It is an expense recognized in the income statement, which includes taxes payable and changes in deferred tax assets and liabilities.

Income Tax Expense = Income Tax Payable + ΔDTL - ΔDTA

Permanent differences => Effective tax rate ≠ Statutory Tax rate

Reported effective tax rate = Income tax expense/ Pretax Income

Temporary differences between carrying amount and tax base			
Item	Carrying amount vs. Tax Base	DTL/DTA	Example
Asset	Carrying amount > Tax Base	DTL	Straight-line depreciation for accounting profit. Accelerated depreciation for taxable profit.
Asset	Carrying amount < Tax Base	DTA	Research cost expensed for accounting profit. Amortized for tax.
Liability	Carrying amount > Tax Base	DTA	Cash from customers before revenue recognition. Cash from customers is taxed.
Liability	Carrying amount < Tax Base	DTL	

LO: Explain how deferred tax liabilities and assets are created and the factors that determine how a company's deferred tax liabilities and assets should be treated for the purposes of financial analysis.

Deferred tax liabilities are created when:

- Income tax expense is greater than taxes payable.
- This can occur if revenues are recognized on the income statement before being included on the tax return (e.g., credit sales).
- Or expenses are tax deductible before they are recognized on the income statement.

Deferred tax assets are created when:

- Taxes payable are greater than income tax.
- This can occur if revenues are taxable before they are recognized in the income statement (unearned revenue).
- Or when expenses are recognized in the income statement before they are tax deductible.

LO: Calculate, interpret, and contrast an issuer's effective tax rate, statutory tax rate, and cash tax rate.

There are three types of tax rates that are relevant to analysts:

- <u>Statutory tax rate</u>: The tax rate established by law.
- <u>Effective tax rate</u>: Tax amount reported on the income-tax statement divided by the pre-tax income.
- <u>Cash tax rate</u>: Tax actually paid divided by pre-tax income.

Differences between cash taxes and reported taxes occur due to timing differences between accounting and tax calculations. These differences are reflected as a deferred tax asset or a deferred tax liability on the balance sheet.

LO: Analyze disclosures relating to deferred tax items and the effective tax rate reconciliation and explain how information included in these disclosures affects a company's financial statements and financial ratios.

If deferred tax liabilities are expected to reverse in the future they can be classified as liabilities. If they are not expected to reverse in future, they can be classified as equity.

If there is any change in tax laws or new income tax rates, then existing deferred tax assets and liabilities must be adjusted to reflect those changes.

Decrease in tax rate reduces both deferred tax liabilities and deferred tax assets.

Increase in tax rate increases both deferred tax liabilities and deferred tax assets.

LM10 Financial Reporting Quality

Version 1.0

1&2. Introduction & Conceptual Overview

There are two main interrelated concepts that will be discussed in detail in this reading: financial reporting quality and earnings quality.

Financial reporting quality: High-quality financial reporting provides information that is useful to analysts in assessing a company's performance and prospects. They contain information that is relevant, complete, neutral, and free from error.

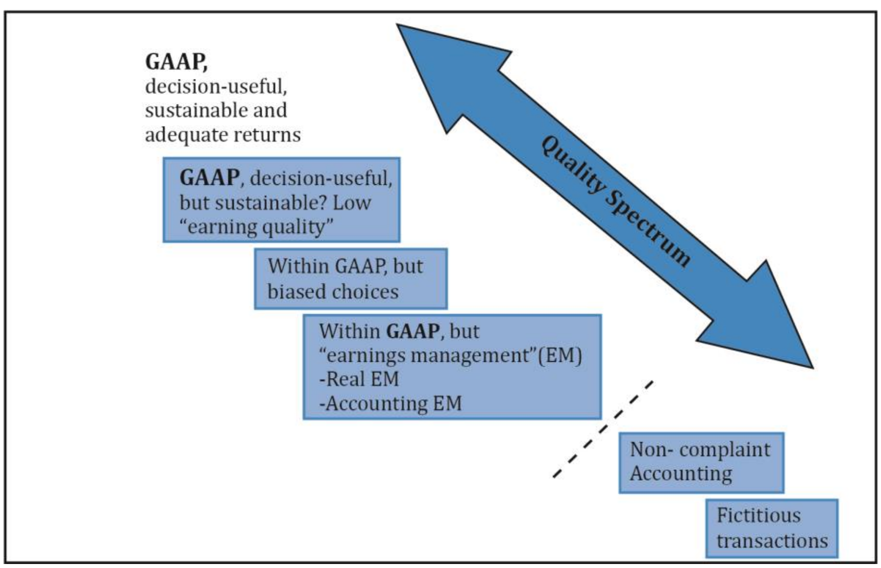
High-quality reporting helps in making the right decision as it depicts the true economic reality of a company for the reporting period. Low-quality financial reporting contains inaccurate, misleading, or incomplete information.

Earnings quality: High-quality earnings result from activities that a company will likely be able to *sustain* in the future and provide a sufficient return on the company's investment. If the return on investment is greater than the cost of funds, then it indicates high earnings quality.

Sustainability is the key here. For example, assume a company uses accrual-based earnings in a quarter. It has high accounts receivable and as a result reports high earnings, which is not sustainable in the following quarters. This implies earnings quality is low.

3 – 5. Quality Spectrum of Financial Reports

Combining the two aspects – financial reporting quality and earnings quality, we get a spectrum spanning from highest to lowest. Let us now look at the characteristics of reporting/earnings quality as we move down along the spectrum as shown in the exhibit below.



1. Reporting is GAAP compliant and decision useful. The earnings are also sustainable and adequate.

2. Reporting is GAAP compliant and decision useful. However, earnings quality is low, i.e., the earnings are not sustainable or adequate.
3. Reporting is GAAP compliant, but the reporting choices and estimates used while preparing the reports are biased.
4. Reporting is GAAP compliant, but the amount of earnings is actively managed. The intent is to increase/decrease/smooth reported earnings.
5. Reporting is not GAAP compliant, although the reports are based on the company's actual economic activities.
6. Reporting is not GAAP compliant and the reports contain numbers that are fictitious.

Non-GAAP reporting of financial metrics which is not in compliance with generally accepted accounting principles such as US GAAP and IFRS includes both financial metrics and operating metrics. Non-GAAP earnings are sometimes referred to as underlying earnings, adjusted earnings, recurring earnings, or core earnings.

6. Differentiate between Conservative and Aggressive Accounting

The choice of accounting methods used can distort the economic reality. Unbiased financial reporting is the ideal, but investors may prefer conservative accounting choices as a positive surprise is acceptable. Whereas the management may prefer aggressive accounting choices.

<u>Aggressive accounting</u>: It refers to biased accounting choices that aim to improve the reported earnings or financial position in the period under review.

<u>Conservative accounting</u>: It refers to biased accounting choices that aim to decrease the reported earnings or financial position in the reporting period.

Some managers use aggressive accounting when earnings are below targets and conservative accounting when earnings are above targets, to artificially smooth earnings.

When a company makes conservative choices, it implies that:

- revenue is recognized only when earned and when collections are reasonably certain.
- expenses/losses are recognized when probable.
- earnings will be understated in the current period.

7. Context for Assessing Financial Reporting Quality

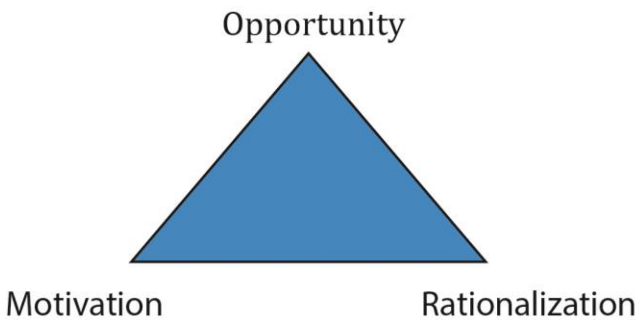
Motivations

Managers may be motivated to issue financial reports that are not high quality in order to:

- mask poor performance.
- boost the stock price.
- increase personal compensation.
- avoid violation of debt covenants.

Conditions Conducive to Issuing Low-Quality Financial Reports

The three conditions conducive for issuing low-quality financial reports are presented below:



Opportunity: It can be the result of weak internal controls, ineffective board of directors, and accounting standards that allow a range of choices.

Motivation: It can result from pressure to meet some criteria for some personal reasons.

Rationalization: It can result from justifying a wrong choice as seen in Enron's case. Enron's CFO sought board approvals, legal and accounting opinions for misstated financial statements.

8. Mechanisms that Discipline Financial Reporting Quality

Market Regulatory Authorities

Regulators in every country can play a key role in enforcing financial reporting quality. Examples of regulatory authorities include:

- the SEC (Securities Exchange Commission).
- SEBI (Securities and Exchange Board of India).
- Securities and Futures Commission in Hong Kong.

These regulatory authorities are members of an international organization called the International Organization of Securities Commissions (IOSCO), comprising 120 regulatory authorities and 80 securities market participants like the stock exchanges.

The actual regulation, however, is enforced through each individual regulatory authority in a country. The features of any regulatory regime such as the SEC that affect financial reporting quality include the following:

- *Registration requirements*: Publicly traded companies must register securities before offering securities for sale to the public. A registration document (often known as a prospectus in an Initial Public Offering) contains current financial statements, future prospects of the company, and securities being offered.
- *Disclosure requirements*: Publicly traded companies are required to make public periodic reports such as financial statements.
- *Auditing requirements*: The financial statements must be audited by an independent auditor that states the statements conform to the accounting standards.

- *Management commentaries*: Financial reports must include statements by the management. Some regulators require a management report containing "(1) a fair review of the issuer's business, and (2) a description of the principal risks and uncertainties facing the issuer."
- *Responsibility statements*: Individuals responsible for company's filings must issue a statement explicitly acknowledging responsibility and correctness of the information in the reports. Falsely certifying may be considered criminal offence and attract a jail sentence.
- *Regulatory review of filings*: Regulators conduct reviews periodically to ensure that the rules have been followed.
- *Enforcement mechanisms*: Regulators have the authority to enforce these rules, without which the rules are of no significance. These powers include fines, barring market participants, or bringing criminal charges.

Auditors

Financial statements of public companies must be audited by an independent auditor. Auditors issue opinions on the financial statements of the company and on the effectiveness of the companies' internal controls. An unqualified opinion on the financial statements indicates that the financial statements present fairly the company's performance in accordance with relevant standards. Key audit matters discuss matters of most significance in the audit of the financial statements of the current period.

However, there are some drawbacks of audited opinion:

- It is based on information provided by the company.
- Only a sample is audited, which may not reveal misstatements.
- The intent of the auditor is not to detect fraud, but to ensure that the information is presented fairly.
- The company being audited pays the audit fees. The auditor has an incentive to be lenient to the company being audited in case of a conflict of interest; particularly if the auditor's firm provides additional services to the company.

Private Contracting

We have seen earlier that managers are motivated to manipulate earnings in order to avoid violating debt covenants or triggers that may prompt investors to recover all or part of their investment.

Consider an example where a company takes a loan from a bank; there is every incentive for the company to dress up its financial reports to keep its cost of capital low. So it is in the best interest of investors, such as the bank here, to monitor the quality of financial reports and detect any misreporting.

9. Detection of Financial Reporting Quality Issues: Introduction & Presentation Choices

Analysts must be able to understand the choices that companies make in financial reporting while evaluating the overall quality of reports – both financial reporting quality and earnings quality. There is no right or wrong choice. The intent of the management is what makes the difference.

Choices exist both in how information is presented (financial reporting quality) and in how financial results are calculated (earnings quality).

- Choices in presentation are often transparent.
- Choices in the calculation of financial results are more difficult to detect.

Ways to increase performance and financial position in the reporting period include the following:

- Recognize revenue prematurely. Ex: a software services company recognizes revenue before the services are delivered to a client. Revenue and earnings will be overstated in the current period.
- Use non-recurring transactions to increase profits. Ex: selling accounts receivable, which increases earnings in an unsustainable manner.
- Defer expense to later periods. Ex: warranty expense for a sale that happened in this period should be recognized now and not put off for later. Deferring understates expense.
- Measure and report assets at higher values; and/or
- Measure and report liabilities at lower values. Equity will be overstated if assets are higher and liabilities are lower.

Ways to increase performance and financial position in a later period include the following:

- Defer current income to a later period (save income for a rainy day); and/or
- Recognize future expenses in a current period; setting the table for improving future performance. Ex: cookie jar reserve accounting. Higher expenses are reported in the current period. This allows earnings in the later period to be overstated because lower expenses are reported later.

Presentation Choices

- Companies may use "strange new metrics". Metrics are set by a company or an industry, and not by a standard-setting body. For example, website companies started using metrics such as "eyeballs" or "stickiness" to measure user engagement and operating performance; traditional valuation methods such as P/E could not justify their stock prices.
- Companies may present "pro forma earnings". These are earnings that are not prepared in accordance with any standard such as US GAAP or IFRS. Analysts must be careful

about the assumptions made in the financial reports as they may be manipulated to make the earnings look better. Ex: companies would exclude huge restructuring charges (to the tune of $3-$7 billion) in performance presentation to make earnings look good to investors.

- EBITDA is earnings before interest, taxes, depreciation, and amortization. It is often used as a proxy for operating cash flow. EBITDA is used to compare companies as the expense incurred for depreciation, amortization, and restructuring may vary with the choice of accounting method. Companies may construct their own version of EBITDA by excluding the following from net income:
    - Rental payments for operating leases.
    - Equity-based compensation, usually justified on the grounds that it is a non-cash expense.
    - Acquisition-related charges.
    - Impairment charges for goodwill or other intangible assets.
    - Impairment charges for long-lived assets.
    - Litigation costs.
    - Loss/gain on debt extinguishments.
- If companies are compared using EBITDA measure, then it must be ensured that the companies calculate EBITDA in a similar manner and the same assumptions are made.
- IFRS requires a definition and explanation of any non-IFRS measures included in financial reports.
- Similarly, if a company uses a non-GAAP financial measure, then it must also include the closest GAAP measure with prominence. It must also explain why the non-GAAP measure is a better choice to represent the company's financial condition than the GAAP measure.

10 - 12. Accounting Choices and Estimates and Their Effects

In this section, we look at the accounting methods (choices and estimates) made by the management for a desired outcome such as earnings growth or meeting the numbers.

How Choices Affect the Cash Flow Statement

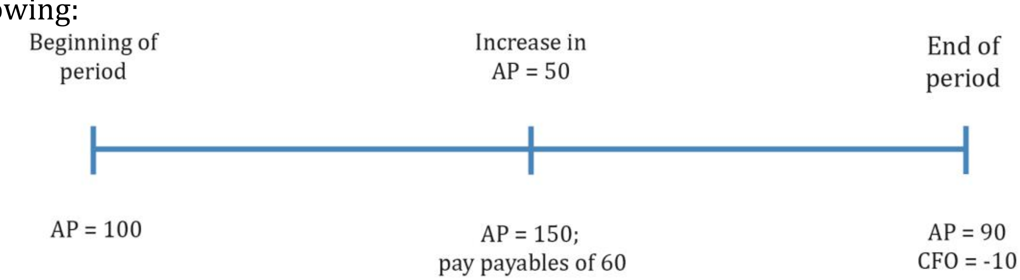
A cash flow statement has three sections:

- Cash flow from operations (CFO): This is of most interest to investors. The CFO is insulated from manipulation more than the income statement. For instance, if a large part of the earnings is from accruals, then it should raise a red flag.
- Cash flow from investing (CFI)
- Cash flow from financing (CFF)

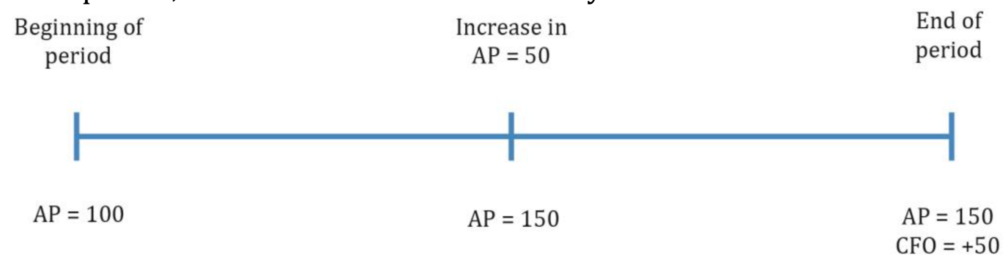
How the cash flow statement is manipulated:

- Misclassification of cash flows: Analyze the composition of CFO closely. For example, if a certain cash outflow should be classified as part of CFO but is instead shown as

CFI, or if a cash inflow must be part of CFI but shown as CFO, then it indicates manipulation.
- Payables management: Decrease in accounts payables is a use of cash. Consider the following:



At the end of the period, the payables decrease to 90 which is a decrease in the liability. It is the use of cash and decrease in CFO. Contrast this with the following if a company wanted to manipulate the CFO. It could delay the payable by stretching the credit period, which will increase the CFO by +50.



- Interest capitalization: This is due to the differences between interest payments and interest costs. Assume a company takes a loan to construct a factory. It pays an interest of 100,000 in a given period of which 70,000 is the interest expense (on the income statement) and 30,000 is capitalized interest on the loan taken. The amount that will show up as interest in CFO will be 70,000 and 30,000 in CFI.
- Flexibility in the classification of interest/dividends paid and received: Interest paid and interest/dividends received may be classified as operating cash flow. Or interest paid can be classified as a financing cash flow and interest/dividends received can be classified as investing cash flows. Dividend paid may be classified as a financing cash flow or cash flow from operating activities.

Analysts should:
- Examine the composition of the operations segment.
- Compare company's cash generation with other companies in the industry; study relationship between net income and CFO.

How Accounting Choices and Estimates Affect Earnings and Balance Sheets

This section identifies areas where choices affect financial reporting and the questions analysts must ask to assess the quality of reporting.

<u>Revenue Recognition</u>

When evaluating a company's revenue recognition practices, an analyst should ask the following questions:

- How is the revenue recognized, upon shipment or upon delivery of goods?
- Is the company engaged in "channel stuffing" – the practice of overloading a distribution channel with more product than it is normally capable of selling? For example, a washing machine manufacturer pressurizes a retailer to sell more machines through special discounts. The threat is that the retailer may return unsold units.
- Does the company engage in bill-and-hold transactions? A company bills the customer, recognizes revenue but does not ship the product.
- Does the company use rebates as part of its marketing approach? If so, how significantly do the estimates of rebate fulfillment affect net revenues? And have any unusual breaks with history occurred?
- Does the company separate its revenue arrangements into multiple deliverables of goods or services?

Long-Lived Assets: Depreciation Policies

- Companies have a choice to use one of the three depreciation methods: Straight-line, accelerated double-declining-balance method, or units-of-production method. Compared to straight line depreciation, using an accelerated depreciation increases the depreciation expense and reduces net income in the early years of an asset's life.
- Depreciation expense: Depends on the method used and the salvage value of the assets being depreciated. A high salvage value reduces the depreciation expense and increases net income.
- Analysts must consider if the estimated life spans of the associated assets make sense, or are they unusually low compared with others in the same industry? And if there have been changes in depreciable lives that have a positive effect on current earnings. Using a longer estimated useful life reduces the depreciation expense and increases the net income in the early years of an asset's life.

Inventory Costing Method

Companies cannot arbitrarily switch between inventory costing methods once the policy decision is made. For example, a cost flow assumption between FIFO vs. weighted average cost can lead to different values for income statement and balance sheet items, and eventually affect profitability. Let us take an example of a company that sells one good. There are four pieces of that good whose costs are 1, 1, 2, and 2 respectively. If two pieces are sold then, according to:

- FIFO: COGS = 2; Ending inventory = 4. FIFO understates cost and overstates ending inventor during periods of rising prices.
- Weighted average cost: COGS = 3; Ending inventory = 3. The balance sheet is a mix of old and new inventory costs. Understates inventory if costs are rising.

Analysts must examine the following:

- Does the company use a costing method that produces fair reporting results in view of its environment? How do its inventory methods compare with others in its industry? Are there differences that will make comparisons uneven if there are unusual changes in inflation?
- Does the company use reserves for obsolescence in its inventory valuation? If so, are they subject to unusual fluctuations that might indicate adjusting them to arrive at a specified earnings result?
- If a company reports under US GAAP and uses last-in, first-out (LIFO) inventory accounting, does LIFO liquidation (assumed sale of old, lower-cost layers of inventory) occur through inventory reduction programs? This inventory reduction may generate earnings without supporting cash flow, and management may intentionally reduce the layers to produce specific earnings benefits.

<u>Capitalization Policies of Intangible Assets</u>

Another example of how choices affect both the balance sheet and income statement is in the use of capitalization. If the payment benefits only the current period, then it must be classified as an expense. If it will be used in future periods, then it must be capitalized.

- Does the company capitalize expenditures related to intangibles, such as software?
- Does its balance sheet show any R&D capitalized as a result of acquisitions?
- Or, if the company is an IFRS filer, has it capitalized any internally generated development costs?

<u>Goodwill</u>

When a company acquires another company, and the acquiring company pays more than its fair value, then goodwill is created. The fair value of the assets created is based on the management's estimate. The depreciable value of assets is kept low to lower the depreciation expense, and the amount that cannot be allocated to specific assets is classified as goodwill.

The initial value of the goodwill is objective. Over time, the value of goodwill is subjective. Companies must test goodwill for impairment annually on a qualitative basis. The value of the assets reported depends on the management's intent; if the fair value of assets cannot be recovered, the company must write-down goodwill. To avoid writing down goodwill, the company may project a better future performance.

<u>Allowance for Doubtful Accounts/Loan Loss Reserves</u>

- Are additions to such allowances lower or higher than in the past? For example, if the allowance for doubtful accounts must be 3% based on historical transactions, a company can report 2% instead in order to boost earnings. Analysts must verify if the allowances are justified.
- Does the collection experience justify any difference from historical provisioning?

- Is there a possibility that any lowering of the allowance may be the result of industry difficulties along with the difficulty of meeting earnings expectations?

<u>Related-Party Transactions</u>

- Is the company engaged in transactions that disproportionately benefit members of management? Does one company have control over another's destiny through supply contracts or other dealings?
- Do extensive dealings take place with non-public companies that are under management control? If so, non-public companies could absorb losses (through supply arrangements that are unfavorable to the private company) and make the public company's performance look good. This scenario may provide opportunities for an owner to cash out.

<u>Tax Asset Valuation Accounts</u>

- Tax assets, if present, must be stated at the value at which management expects to realize them, and an allowance must be set up to restate tax assets to the level expected to eventually be converted into cash. Determining the allowance involves an estimate of future operations and tax payments. Does the amount of the valuation allowance seem reasonable, overly optimistic, or overly pessimistic? For example, if a company records a DTA which expires in three years, analysts must analyze if a company can become profitable within this period.
- Are there contradictions between the management commentary and the allowance level, or the tax note and the allowance level? There cannot be an optimistic management commentary and a fully reserved tax asset, or vice versa. One of them has to be wrong.
- Look for changes in the tax asset valuation account. It may be 100% reserved at first, and then "optimism" increases whenever an earnings boost is needed. Lowering the reserve decreases tax expense and increases net income. If the valuation allowance is lower, then DTA and net income increases.

13. Warning Signs

Warning signs of information manipulation in financial reports can be seen as manipulation in:

- Biased revenue recognition.
- Biased expense recognition.

The bias may be with respect to:

- Timing of recognition: Deferring expenses by capitalizing.
- Location of recognition: Showing a loss in other comprehensive income instead of the income statement.

Analysts must look at the following for warning signs:

- Pay attention to revenue. Examine the accounting policies note for a company's revenue recognition policies. Studies have shown that most manipulations relate to revenue recognition; the largest number on the income statement.
  - Does the company recognize revenue prematurely; revenue recognition upon shipment of goods or bill-and-hold sales?
  - Does the company engage in barter transactions?
  - Are rebates offered? If yes, by how much, as these can affect revenue recognition?
  - How will revenue be recognized for multiple-deliverable arrangements of goods and services?
- Look at revenue relationships.
  - Compare a company's revenue growth with that of its competitors, industry, or the economy. If the company outperforms, then there must be a justifiable performance like superior management or product differentiation. If not, it should be a cause of concern.
  - Compare accounts receivable with revenue to see if it is increasing as a percentage of revenue over the years. It may indicate relaxed credit terms or channel stuffing.
  - Analyze asset turnover to see if the assets are efficiently used. If an asset turnover is declining, then it indicates assets may be written down in the future.
- Pay attention to signals from inventories.
  - Look at the growth in inventories relative to competitors and industry.
  - Look at the inventory turnover ratio. If the ratio is declining, it may mean obsolescent inventory.
  - US GAAP allows use of LIFO for inventory accounting. If prices increase, analysts must check to see that old inventory has not been passed through earnings (LIFO liquidation) to boost net profits.
- Pay attention to capitalization policies and deferred costs.
  - Compare a company's accounting policy for capitalization of long-term assets, interest costs, and handling of deferred costs with that of its competitors and the industry. If only this company is capitalizing costs while others are expensing, then it is a warning sign.
- Pay attention to the relationship of cash flow and net income.
  - Construct a time series of cash flow from operations divided by net income. If the ratio is consistently less than 1.0, then it indicates a problem with accrual accounts, i.e., net income is shown higher than it should be.

Other Potential Warning Signs

Other areas that require further analysis include:

- Depreciation methods and useful lives.

- Fourth-quarter surprises: For non-seasonal businesses, over- or under-performance in the fourth quarter of a year routinely is a red flag.
- Presence of related-party transactions: What is the intent behind related-party transactions, often by founding members of a company?
- Non-operating income or one-time sales included in revenue: To mask declining revenues, companies may include one-time gain as part of revenue. Ex: In 1997, Trump Hotels included a one-time gain from a lease termination as part of revenue.
- Classification of expenses as non-recurring.
- Gross/operating margins out of line with competitors or industry.
- Younger companies with an unblemished record of meeting growth projections.
- Management has adopted a minimalist approach to disclosure: Is the management withholding information from competitors?
- Management fixation on earnings reports.

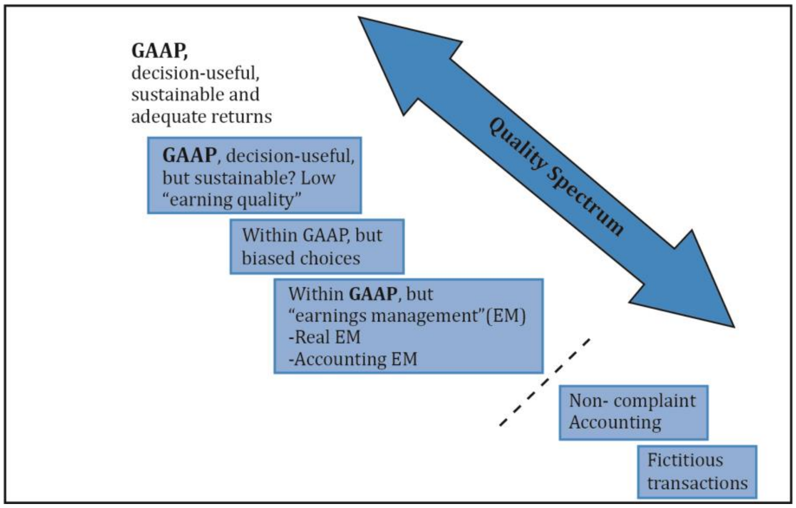
Summary

LO: Compare financial reporting quality with the quality of reported results (including quality of earnings, cash flow, and balance sheet items).

Reporting quality: It refers to the information disclosed in the firm's financial statements. High-quality reporting means that the financial statements are decision useful and represent the economic reality of the company.

Results quality (earnings quality): It refers to the earnings and cash generated by the company's actual economic activities. High-quality earnings mean that the earnings are sustainable and are expected to continue in the future.

LO: Describe a spectrum for assessing financial reporting quality explain the difference between conservative and aggressive accounting.



Aggressive accounting: It refers to biased accounting choices that aim to improve the reported earnings or financial position in the reporting period.

Conservative accounting: It refers to biased accounting choices that aim to decrease the reported earnings or financial position in the reporting period.

LO: Describe motivations that might cause management to issue financial reports that are not high quality and conditions that are conducive to issuing low-quality, or even fraudulent, financial reports.

Managers may be motivated to issue financial reports that are not high quality in order to:

- mask poor performance.
- boost the stock price.
- increase personal compensation.
- avoid violation of debt covenants.

Conditions that are conducive to issuing low-quality financial reports are:

Motivation: Covered above

Opportunity:

- Weak internal controls.
- Ineffective board of directors.
- Accounting standards that allow a range of choices.

Rationalization

- Ability to justify wrong choices to him/herself.

LO: Describe mechanisms that discipline financial reporting quality and the potential limitations of those mechanisms.

Mechanisms that discipline financial reporting quality include:

- the free market and incentives for companies to minimize cost of capital.
- auditors.
- contract provisions specifically tailored to penalize misreporting.
- enforcement by regulatory entities.

LO: Describe presentation choices, including non-GAAP measures, that could be used to influence an analyst's opinion.

- Companies may use "strange new metrics".
- Companies may present "pro forma earnings".
- Companies may construct their own version of EBITDA by excluding:
  - Rental payments for operating leases;
  - Equity-based compensation, usually justified on the grounds that it is a non-cash expense;
  - Acquisition-related charges;
  - Impairment charges for goodwill or other intangible assets;
  - Impairment charges for long-lived assets;
  - Litigation costs;
  - Loss/gain on debt extinguishments.
- IFRS requires a definition and explanation of any non-IFRS measures included in financial reports.
- If a company uses a non-GAAP financial measure, then it must also include the closest GAAP measure with prominence. It must also explain why the non-GAAP measure is a better choice to represent the company's financial condition than the GAAP measure.

LO: Describe accounting methods (choices and estimates) that could be used to manage earnings, cash flow, and balance sheet items.

Accounting methods (choices and estimates) that can be used to manage earnings and balance sheet items are shipping terms, FIFO versus weighted cost, deferred tax assets, depreciation methods, capitalization, and goodwill.

Cash flow statements can be manipulated by misclassification of cash flows (CFO, CFI, and CFF), payables management, interest capitalization, and flexibility in the classification of interest/dividends paid and received.

Choices that affect financial reporting are revenue recognition, long-lived assets (depreciation policies), intangibles (capitalization policies), allowance for doubtful accounts, inventory cost method, tax valuation accounts, goodwill, warranty reserves, and related party transactions.

LO: Describe accounting warning signs and methods for detecting manipulation of information in financial reports.

Warning signs of information manipulation in financial reports can be seen as manipulation in:

- Biased revenue recognition.
- Biased expense recognition.

The bias may be with respect to:

- Timing of recognition: Deferring expenses by capitalizing.
- Location of recognition: Showing a loss in other comprehensive income instead of the income statement.

Analysts must look at the following for warning signs:

- Pay attention to revenue.
- Look at revenue relationships.
- Pay attention to signals from inventories.
- Pay attention to capitalization policies and deferred costs.
- Pay attention to the relationship of cash flow and net income.

Other potential warning signs are:

- Depreciation methods and useful lives.
- Fourth-quarter surprises.
- Presence of related-party transactions.
- Non-operating income or one-time sales included in revenue. To mask declining revenues, companies may include one-time gain as part of revenue.
- Classification of expenses as non-recurring.
- Gross/operating margins out of line with competitors or industry.
- Younger companies with an unblemished record of meeting growth projections.
- Management has adopted a minimalist approach to disclosure. Is the management withholding information from competitors?
- Management fixation on earnings reports.

LM11 Financial Analysis Techniques

Version 1.0

1. Introduction

Financial analysis is a useful tool in evaluating a company's performance and trends. The primary source of data is the company's annual reports, financial statements, and MD&A. An analyst must be capable of using a company's financial statements along with other information such as economy/industry trends to make projections and reach valid conclusions.

2. The Financial Analysis Process

Before beginning any financial analysis, an analyst must clarify the purpose and context of why it is needed. Once the purpose is defined, an analyst can choose the right techniques for the analysis. For example, the level of detail required for a substantial long-term investment in equities will be higher than one needed for a short-term investment in fixed income securities.

The Objectives of the Financial Analysis Process

This learning module focuses on steps 3 and 4 of the financial analysis framework in detail: how to adjust financial statements, compute ratios, and produce graphs and forecasts. The processed data is then analyzed to arrive at a conclusion.

Financial Analysis Framework	
Phase	*Output of the phase*
1. *Define purpose and context* based on the analyst's function, client input, and organizational guidelines.	Objective Questions to be answered Nature and content of report to be provided Timetable and budget
2. *Collect data:* financial statements, other financial data, industry/economic data, discussions with management, suppliers, customers, and competitors.	Organized financial statements Financial tables Completed questionnaires
3. *Process data*	Adjusted financial statements Common-size statements Ratios and graphs Forecasts
4. *Analyze and interpret* processed data	Analytical results
5. *Develop and communicate* conclusions and recommendations	Report answering questions from phase 1 Recommendation regarding the purpose of the analysis
6. *Follow-up*	Updated recommendations

Distinguishing between Computations and Analysis

An effective analysis is not just a compilation of various pieces of information, tables, and graphs. It includes both calculations and interpretations. For analyzing past performance, an analyst computes several ratios, compares them against benchmarks, evaluates how the company performed, and determines the reasons behind its good/bad performance. Similarly, for a forward-looking analysis, an analyst must forecast and make recommendations after analyzing trends, management quality, etc.

Analysts often present the findings of their analysis through a written research report. For example, a report might present the following:

- The purpose of the report
- Relevant aspects of the business context, including economic environment, financial and other infrastructure, legal and regulatory environment
- Evaluation of corporate governance and assessment of management strategy
- Assessment of financial and operational data, including key assumptions in the analysis
- Conclusions and recommendations, including limitations of the analysis and risks.

3. Analytical Tools and Techniques

Various tools and techniques such as ratios, common size analysis, graphs, and regression analysis help in evaluating a company's performance. Evaluations require comparisons, but to make a meaningful comparison of a company's performance, the data needs to be adjusted first. An analyst can then compare a company's performance to other companies at any point in time (cross-section analysis) or its own performance over time (time-series analysis).

4. Financial Ratio Analysis

A ratio is an indicator of some aspect of a company's performance like profitability or inventory management that tells us what happened, but not why it happened. Ratios help in analyzing the current financial health of a company, evaluate its past performance, and provide insights for future projections. Calculating ratios is straightforward, but interpreting them is subjective.

Uses of ratio analysis

Ratios allow us to evaluate:

- operational efficiency.
- financial flexibility.
- Management's ability.
- changes in company/industry over time.
- company performance relative to industry.

Limitations of ratio analysis

Ratio analysis also has certain limitations. Some of the factors to consider include:

- The heterogeneity or homogeneity of a company's operating activities: Companies may have divisions that operate in a variety of industries. This can make it difficult to find comparable industry ratios.
- Need to use judgment: An analyst must exercise judgment when interpreting ratios. All ratios must be viewed relative to one another. For example, a current ratio of 1.1 may not necessarily be good/bad unless viewed in perspective of other companies/industry.
- Use of alternate accounting methods: Using alternate methods may require adjustments before the ratios are comparable. For example, Company A might use the LIFO method to measure inventory, while a comparable company might use the FIFO method. Similarly, one company may use the straight-line method of depreciation, while another may use an accelerated method.
- Nature of a company's business: Companies may have divisions operating in many different industries. This can make it difficult to find comparable ratios.
- Consistency of results of ratio analysis: One set of ratios may indicate a problem, while the other may indicate the problem is short-term making the results inconsistent.

Sources of Ratios

Ratios can be computed using data obtained directly from companies' financial statements or from a database such as Bloomberg, Thomson Reuters etc.

Analysts should be aware that the underlying formulas used to calculate ratios may differ by vendor. Analysts should determine whether any adjustments are necessary.

Similarly, data base providers may use judgement to classify items. For example, judgment may be used to classify income statement items as 'operating' or 'non-operating'. Variation in such judgements would affect any computations involving operating income.

A good practice is the use the same source of data when comparing different companies.

5. Common Size Balance Sheets and Income Statements

Common-size financial statements are used to compare the performance of different companies within an industry or a company's performance over time. Common size statements are prepared by expressing every item in a financial statement as a percentage of a base item.

Common-Size Analysis of the Balance Sheet

There are two types of common-size balance sheets: *vertical* and *horizontal*.

Vertical common-size balance sheet

A vertical common-size balance sheet is prepared by dividing each item on the balance sheet by the total assets for a period and expressed as a percentage. This highlights the composition of the balance sheet.

$$\text{Vertical common-size balance sheet account (in \%)} = \frac{\text{Balance sheet account}}{\text{Total assets}} * 100$$

A simple common-size vertical balance sheet for Everest Inc. is shown below:

Vertical common-size (partial) balance sheet for Everest Inc.		
ASSETS	*2015*	*2014*
Cash and cash equivalents	10.81%	13.12%
Short-term marketable securities	1.24%	0.62%
Accounts receivable	7.50%	4.80%
Inventory	25.32%	25.97%
Other current assets	3.37%	2.14%
Property, plant, and equipment (PPE)	38.76%	40.06%
…………	……	…..
Other non- current assets	0.03%	0.02%
Total	100.00%	100.00%
EQUITY and LIABILITIES		
Short-term borrowing	0.46%	0%
Deferred tax liabilities	4.01%	4.20%
…..	…	…
Stockholder's equity	58.88%	60.13%
Equity and Liabilities	100.00%	100.00%

Horizontal Common-Size Balance Sheet

In a horizontal common-size balance sheet, each balance sheet item is shown in relation to the same item in a base year. Consider the following balance sheet excerpt for Everest Inc.:

	2014 (base year)	*2015*
Cash and cash equivalents	$3,800	$3,500
Short-term marketable securities	$180	$400
Inventory	$7,520	$8,200

The corresponding horizontal common size balance sheet will look like this:

	2014 (base year)	*2015*
Cash and cash equivalents	1.0	0.9
Short-term marketable securities	1.0	2.2
Inventory	1.0	1.1

Notice that the base-year value for all balance sheet items is set to 1. This makes it easy to see the percentage change in each item relative to the base year. For the data given above, cash decreased by 10% and inventory increased by 10%. An analysis of horizontal common-size balance sheets highlights structural changes that have occurred in a business.

Common-size Analysis of the Income Statement

A vertical common-size income statement divides each income statement element by revenue.

$$\text{Vertical common-size income statement account (in \%)} = \frac{\text{Income statement account}}{\text{Revenue}} * 100$$

Exhibit 7 from the curriculum presents a hypothetical company's vertical common-size income statement for two time periods.

	Period 1 (% of total revenue)	Period 2 (% of total revenue)
Total Revenue	*100*	*100*
Operating expenses (excluding depreciation)		
Salaries and employee benefits	15	25
Administrative expenses	22	20
Rent expense	10	10
EBITDA	*53*	*45*
Depreciation and amortization	4	4
EBIT	*49*	*41*
Interest paid	7	7
EBT	*42*	*34*
Income tax provision	15	8
Net income	*27*	*26*

6. Cross-Sectional, Trend Analysis, and Relationships in Financial Statements

Time-Series Analysis (Trend Analysis)

Trend analysis or time-series analysis provides information on historical performance and growth. It indicates how a particular item is changing – whether it is improving or deteriorating – relative to total assets over multiple periods. For the data given above, we can observe that inventory decreased as a percentage of total assets in 2015, while accounts receivable increased as a percentage of total assets.

Cross-Sectional Analysis

The vertical common-size balance sheet can be used in cross-sectional analysis (also called relative analysis) to compare a specific metric of one company with the same metric for another company or companies for a single time period. As illustrated in the table below, this method allows comparison across companies which might be of significantly different sizes and/or operate in different currencies.

	Everest Inc.		*Alps Corp.*	
		2015		*2015*
Cash and cash equivalents	10.81%	$3,500	9.00%	€1,755.00
Short-term marketable securities	1.24%	$400	4.00%	€780.00
Accounts receivable	7.50%	$2,430	5.20%	€1,014.00
Inventory	25.32%	$8,200	20.10%	€3,919.50
Other non-current assets	0.03%	$10	0.50%	€97.50
Total Assets	100.00%	$32,382	100.00%	€19,500.00

This presentation makes it easy to see that Alps Corp. has lower receivables as a percentage of total assets relative to Everest Inc. Alps Corp. also has lower inventory as a percentage of total assets relative to Everest Inc.

Relationships in Financial Statements

Comparing the trend data of a horizontal common-size analysis across financial statements will give some insight into a company's financial standing. Consider the following percentage changes for a company to identify some potential issues:

Revenue: +15%, Operating income: +15%, Operating cash flow: -10%, Inventory: +60%, Receivables: +40%, Total assets: +30%

Some of the potential issues based on these numbers are:

- The assets are growing at a faster rate than revenue, which implies the company is spending more than the sales it is able to generate.
- Operating cash flow is negative whereas operating income is +15%, indicating a problem that the company is booking sales (accrual accounting) but has not realized the cash yet.
- Similarly, when inventory and receivables grow at a much faster pace than sales, it shows signs of poor inventory and receivables management.

7. The Use of Graphs and Regression Analysis

Graphs can be considered an extension of the financial analysis. It is a pictorial representation of the analysis done, be it ratio analysis or trend analysis. Analysts use appropriate graphs such as line charts and bar graphs based on the type of data to be shown. This helps in quick comparison of financial performance and structure over time.

Regression Analysis

Regression analysis, described in detail in Level II, is a statistical method of analyzing relationships (correlations) between variables.

8. Common Ratio Categories, Interpretation, and Context

A large number of ratios are used to measure various aspects of performance. Commonly used financial ratios can be categorized as follows:

Category	What they measure	Example
Activity ratios	Efficiency of a company in performing its day-to-day operations.	$\frac{\text{Revenue}}{\text{Assets}}$
Liquidity ratios	A company's ability to meet its short-term obligations.	$\frac{\text{Current assets}}{\text{Current liabilities}}$
Solvency ratios	A company's ability to meet its long-term obligations.	$\frac{\text{Assets}}{\text{Equity}}$
Profitability ratios	A company's ability to generate profit from its resources.	$\frac{\text{Net Income}}{\text{Assets}}$

Single statement ratios: Note that for some ratios, the numerator and denominator are from the same statement (Income statement, balance sheet, or cash flow statement). For example, net profit margin (net income/sales) where both items are from the income statement.

Mixed ratios: For other ratios, the numerator is from one statement and the denominator is from another statement. An example is the asset turnover ratio (sales/assets) where the numerator is from the income statement and the denominator is from the balance sheet.

Interpretation and Context

As standalone numbers, the financial ratios of a company are not meaningful. The ratios are usually industry specific. For instance, one cannot compare the ratios of Schlumberger with that of Facebook. The financial ratios should be used to periodically evaluate a company's past performance (trend analysis) and its goals and strategy; how it fares against its peers in the industry (cross-sectional analysis); and the effect of economic conditions on its business.

9. Activity Ratios

Activity ratios measure how efficiently a company manages its assets. They are also known as **asset utilization ratios** or **operating efficiency** ratios. The activity ratios usually have an element from the income statement in the numerator and one from the balance sheet in the denominator. The average of the balance sheet element is *generally* taken because the balance sheet only shows the value at the end of the period, whereas the income statement measures what happened during the period.

Activity Ratios	Formula	Interpretation
Inventory turnover	$\frac{\text{Cost of goods sold}}{\text{Average inventory}}$	• Indicates how many times per period the entire inventory was sold. • Measures the ability of a company to sell its inventory. • Higher number means greater efficiency because inventory is kept for a shorter period. It could also

		mean insufficient inventory, which in turn, might affect growth/ revenue.
Days of inventory on hand	$\frac{\text{Number of days in period}}{\text{Inventory turnover}}$	• On an average, how many days of inventory is kept on hand.
Receivables turnover	$\frac{\text{Revenue}}{\text{Average receivables}}$	• Indicates how quickly a company collects cash. • More appropriate to use credit sales instead of revenue but it is not readily available. • A higher number means greater efficiency in credit and collection. It could also mean stringent cash collection policies are hurting potential sales.
Days of sales outstanding	$\frac{\text{Number of days in period}}{\text{Receivables turnover}}$	• Elapsed time between credit sale and cash collection. • Higher number means it takes a long time to collect receivables.
Payables turnover	$\frac{\text{Purchases}}{\text{Average trade payables}}$	• Indicates how quickly a company pays suppliers. • A high number means the company is paying suppliers quickly and is possibly not making use of credit facilities. • Low number may mean the company is facing trouble making payments on time and signal liquidity issues.
Number of days of payables	$\frac{\text{Number of days in period}}{\text{Payables turnover}}$	• On an average, how many days it takes to pay suppliers.
Working capital turnover	$\frac{\text{Revenue}}{\text{Average working capital}}$	• Indicates how efficiently a company generates revenue from working capital. • Working capital = current assets (CA) – current liabilities (CL) • Higher number means greater efficiency. If CA = CL, then working capital would be zero making the ratio meaningless.
Fixed asset turnover	$\frac{\text{Revenue}}{\text{Average net fixed assets}}$	• Indicates how efficiently a company generates revenue from fixed assets.

		• A higher number means efficient use of fixed assets. • A lower number may mean inefficiency, or newer business (higher carrying value on B/S), or a capital-intensive business.
Total asset turnover	$\frac{\text{Revenue}}{\text{Average total assets}}$	• Indicates how efficiently a company generates revenue from total assets (fixed + current assets). • As with other turnover ratios, higher number means efficiency.

Purchases = Cost of goods sold + Ending inventory – Beginning inventory

Instructor's Note: How to remember the activity ratios

1. Name of the ratio indicates the balance sheet item. For example, in the receivables turnover ratio, average receivables is the balance sheet item.

2. The income statement item is in the numerator.

3. Average value of the balance sheet item is in the denominator. An income statement measures an item over a period but a balance sheet indicates values of items only at the end of a period. So, analysts typically use the average value for balance sheet items.

4. Turnover ratios except inventory turnover and payables turnover use revenue in the numerator. Inventory turnover uses cost of goods sold, while payables turnover uses purchases.

Higher number for turnover ratios = greater efficiency

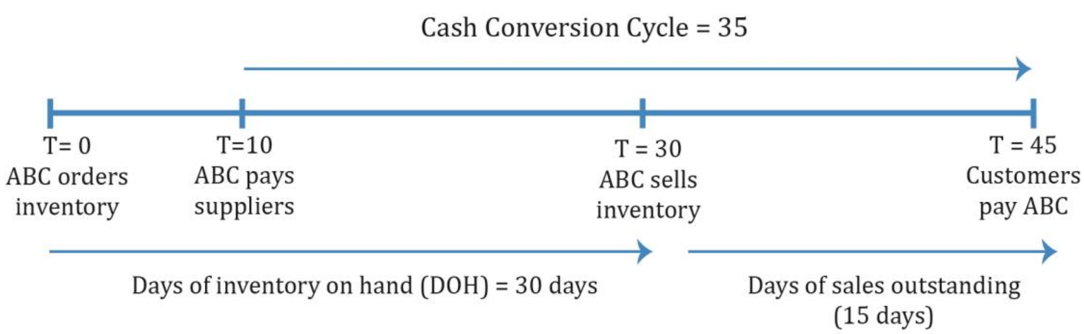
10. Liquidity Ratios

Liquidity ratios measure a company's ability to meet short-term obligations. It also indicates how quickly it turns assets into cash.

Liquidity Ratios		
Current ratio	$\frac{\text{Current assets}}{\text{Current liabilities}}$	A higher number implies greater liquidity.
Quick ratio	$\frac{\text{Cash + marketable securities + receivables}}{\text{Current liabilities}}$	A higher number implies greater liquidity. More conservative than current ratio as only more liquid current assets are included.

Cash ratio	$\frac{\text{Cash} + \text{marketable securities}}{\text{Current liabilities}}$	This is the most conservative liquidity ratio and a good measure of a company's ability to handle a crisis situation.
Defensive interval ratio	$\frac{\text{Cash} + \text{marketable securities} + \text{receivables}}{\text{Daily cash expenditures}}$	Measures the number of days a company can operate before it runs out of cash. A higher number implies greater liquidity.
Cash conversion cycle (net operating cycle)	Days of inventory on hand (DOH) + days of sales outstanding (DSO) – number of days of payables	The time between cash paid (to suppliers) and cash collected (from customers). A low number is better for the company as it means high liquidity. A long cash conversion cycle implies low liquidity

The example below for ABC Corp. illustrates the cash conversion cycle. The timeline for various events is illustrated below:



11. Solvency Ratios

Solvency ratios measure a company's ability to meet long-term obligations. In simple terms, it provides information on how much debt the company has taken and if it is profitable enough to pay the interest on debt in the long term. It has to be analyzed within an industry's perspective. Certain industries such as real estate use a higher level of leverage.

Solvency Ratios	Formula	Interpretation
Debt ratios		
Debt-to-assets ratio	$\frac{\text{Total debt}}{\text{Total assets}}$	Measures the amount of debt in total assets. Higher debt means low solvency and higher risk. A ratio of 0.5 implies 50% of assets are financed with debt.
Debt-to-capital ratio	$\frac{\text{Total debt}}{\text{Total debt} + \text{Total shareholder's equity}}$	Measures the amount of debt as a percentage of capital (debt + shareholder's equity).
Debt-to-equity ratio	$\frac{\text{Total debt}}{\text{Total shareholder's equity}}$	Measures the amount of debt as a percentage of equity.
Financial leverage ratio	$\frac{\text{Average total assets}}{\text{Average total equity}}$	Measures the amount of assets per unit of equity. A higher value means a company is more leveraged.
Debt-to-EBITDA Ratio	$\frac{\text{Total debt}}{\text{EBITDA}}$	Estimates how many years it would take to repay total debt based on earnings before income taxes, depreciation and amortization (an approximation of operating cash flow).
Coverage ratios		
Interest coverage ratio (also called 'times interest earned'	$\frac{\text{EBIT}}{\text{Interest payments}}$	Measures the company's ability to make interest payments (how many times the company can make interest payments with its EBIT). Unlike the other solvency ratios, **a higher value for this ratio is better** as it means stronger solvency.
Fixed charge coverage ratio	$\frac{\text{EBIT} + \text{lease payments}}{\text{Interest payments} + \text{lease payments}}$	Measures the ability of a company to pay interest on debt.

		Here, lease payments are added to EBIT as they are an obligation like interest payments. Like the interest coverage ratio, a higher value for this ratio implies stronger solvency. This ratio is a more meaningful measure for companies that lease a large portion of their assets. For example, airline companies.

Note that there are two categories of solvency ratios: debt (or leverage) ratios and coverage ratios.

In general, a high debt (or leverage) ratio implies a high level of debt, high risk, and low solvency. With coverage ratios, a high number is good because this indicates high income relative to interest payments.

12. Profitability Ratios

Profitability ratio	**Formula**	**Interpretation**
Return on Sales		
Gross profit margin	$\frac{\text{Gross profit}}{\text{Revenue}}$	A higher value means higher pricing and lower costs.
Operating profit margin	$\frac{\text{Operating income}}{\text{Revenue}}$	Operating profit = gross profit - operating costs. A good sign if operating profit margin grows at a faster rate than gross profit margin.
Pretax margin	$\frac{\text{EBT}}{\text{Revenue}}$	EBT = operating profit - interest related expenses. Needs further analysis if pretax income increases only because of non-operating income.
Net profit margin	$\frac{\text{Net profit}}{\text{Revenue}}$	Net profit = revenue – all expenses.

Return on Investment		
Operating ROA	$\frac{\text{Operating income}}{\text{Average total assets}}$	For return, either net income or operating income (EBIT) can be used.
Return on assets (ROA)	$\frac{\text{Net income}}{\text{Average total assets}}$	For return, either net income or operating income (EBIT) can be used.
Return on total capital	$\frac{\text{EBIT}}{\text{Average short term and long term debt } + \text{equit}}$	Like operating ROA, EBIT is used. Measures return on capital before deducting interest.
Return on equity (ROE)	$\frac{\text{Net income}}{\text{Average total equity}}$	A very important measure of return earned on equity capital. Unlike return on common equity, it includes minority and preferred equity.
Return on common equity	$\frac{\text{Net income} - \text{preferred dividend}}{\text{Average common equity}}$	Money available to common shareholders.

13. Integrated Financial Ratio Analysis

No single ratio or category of ratios in isolation can provide a good estimate of the overall position and performance of a company. Experience shows that information from one ratio category can be helpful in answering questions raised by another category. The most accurate overall picture comes from integrating information from all sources.

14. Dupont Analysis—The Decomposition of ROE

DuPont Analysis: The Decomposition of ROE

DuPont analysis decomposes a firm's ROE to better analyze a firm's performance.

Start with ROE

$$\text{ROE} = \left(\frac{\text{net income}}{\text{equity}}\right)$$

The traditional DuPont equation is:

$$\text{ROE} = \left(\frac{\text{net income}}{\text{sales}}\right)\left(\frac{\text{sales}}{\text{assets}}\right)\left(\frac{\text{assets}}{\text{equity}}\right)$$

$$\text{ROE} = (\text{net profit margin})(\text{asset turnover})(\text{leverage ratio})$$

The extended DuPont equation is:

$$\text{ROE} = \left(\frac{\text{net income}}{\text{EBT}}\right)\left(\frac{\text{EBT}}{\text{EBIT}}\right)\left(\frac{\text{EBIT}}{\text{revenue}}\right)\left(\frac{\text{revenue}}{\text{total assets}}\right)\left(\frac{\text{total assets}}{\text{total equity}}\right)$$

$$\text{ROE} = (\text{tax burden})(\text{interest burden})(\text{EBIT margin})(\text{asset turnover})(\text{financial leverage})$$

Example

The following data is available for a company:

	2010	**2011**	**2012**
ROE	19%	20%	22%
ROA	8.1%	8%	7.9%
Total asset turnover	2	2	2.1

Based only on the information above, the most appropriate conclusion is that over the period 2010 to 2012, the company's:

A. Net profit margin and financial leverage have decreased.

B. Net profit margin and financial leverage have increased.

C. Net profit margin has decreased but its financial leverage has increased.

Solution:

A quick glance at the data says profitability is going up and asset turnover has slightly increased from 2010 to 2012. ROA is going down from the second year.

First, break down ROE into: $\frac{\text{Net income}}{\text{Assets}}$ x $\frac{\text{Assets}}{\text{Equity}}$ = ROA x Leverage.

ROE is going up (first row). Since ROA is going down, leverage must increase for ROE to increase. So A is incorrect.

Next, to determine if net profit margin increased or decreased, break down ROA into $\frac{\text{Net income}}{\text{Sales}}$ x $\left(\frac{\text{Sales}}{\text{Assets}}\right)$. Since $\left(\frac{\text{Sales}}{\text{Assets}}\right)$ or asset turnover is increasing, net profit margin has to decrease for return on assets to decrease. So, the correct answer is C.

15. Industry-Specific Financial Ratios

Ratios serve as indicators of some aspect of a company's performance and value. Aspects of performance that are important in one industry may be irrelevant in another. These differences are reflected through industry-specific ratios. For example, companies in the retail industry may report same-store sales changes because in the retail industry it is important to distinguish between growth that results from opening new stores and growth that results from generating more sales at existing stores.

Other examples of industry specific ratios include:

- Service companies: revenue per employee, net income per employee
- Hotels: Average daily rate, occupancy rate

16. Model Building and Forecasting

Analysts use several methods to forecast future performance. One commonly used method is to project sales and to combine the forecasted sales numbers with expected values for key ratios. For example, by using sales numbers and gross profit margin, one can determine cost of goods sold and gross profit. This method is particularly useful for mature companies with stable margins.

Besides ratio analysis, techniques such as sensitivity analysis, scenario analysis, and simulations are often used as part of the forecasting process.

- Sensitivity analysis shows a range of possible outcomes as specific assumptions or input variables are changed.
- With scenario analysis, a number of different scenarios are defined and outcomes are estimated for each outcome.
- Simulations involve the use of computer models and input variables which are based on a pre-defined probability distribution.

Summary

LO: Describe tools and techniques used in financial analysis, including their uses and limitations.

A ratio is an indicator of some aspect of a company's performance like profitability or inventory management.

Uses of ratio analysis	Limitations of ratio analysis
• Evaluate operational efficiency and financial flexibility. • Compare company performance relative to industry and peer companies. • Compare across companies irrespective of size and currency.	• An analyst must exercise judgment when interpreting ratios. • Use of alternate accounting methods may require adjustments before the ratios are comparable. • Companies may have divisions operating in many different industries. This can make it difficult to find comparable ratios.

Common-size financial statements are used to compare the performance of different companies within an industry or a company's performance over time. The vertical common-size balance sheet helps in cross-sectional and time-series analysis. An analysis of horizontal common-size balance sheets highlights structural changes that have occurred in a business.

A graph is a pictorial representation of the analysis done, be it ratio analysis or trend analysis. It helps in quick comparison of financial performance and structure over time.

Regression analysis is a statistical method of analyzing relationships (correlations) between variables.

LO: Calculate and interpret activity, liquidity, solvency, and profitability ratios.

Activity ratios measure the efficiency of a company's operations, such as collection of receivables or management of inventory. They include inventory turnover, days of inventory on hand, receivables turnover, days of sales outstanding, payables turnover, number of days of payables, working capital turnover, fixed asset turnover, and total asset turnover.

Liquidity ratios measure the ability of a company to meet short-term obligations. They include the current ratio, quick ratio, cash ratio, and defensive interval ratio.

Solvency ratios measure the ability of a company to meet long-term obligations. They include debt-to-assets ratio, debt-to-capital ratio, debt-to-equity ratio, financial leverage ratio, debt-to-EBITDA ratio, interest coverage ratio and fixed charge coverage ratio.

Profitability ratios measure the ability of a company to generate profits from revenue and assets. They include gross profit margin, operating profit margin, pretax margin, net profit margin, ROA, return on total capital, ROE, and return on common equity.

LO: Describe relationships among ratios and evaluate a company using ratio analysis.

To evaluate the overall position and performance of a company, a single ratio or a single category of ratios is not examined in isolation. The information from one ratio category can be helpful in answering questions raised by another category and the most accurate overall picture comes from integrating information from all sources.

LO: Demonstrate the application of DuPont analysis of return on equity and calculate and interpret effects of changes in its components.

The traditional DuPont equation is:

$$\text{ROE} = \left(\frac{\text{net income}}{\text{sales}}\right)\left(\frac{\text{sales}}{\text{assets}}\right)\left(\frac{\text{assets}}{\text{equity}}\right)$$

The extended DuPont equation is:

$$\text{ROE} = \left(\frac{\text{net income}}{\text{EBT}}\right)\left(\frac{\text{EBT}}{\text{EBIT}}\right)\left(\frac{\text{EBIT}}{\text{revenue}}\right)\left(\frac{\text{revenue}}{\text{total assets}}\right)\left(\frac{\text{total assets}}{\text{total equity}}\right)$$

LO: Describe the uses of industry-specific ratios used in financial analysis.

Aspects of performance that are important in one industry may be irrelevant in another. These differences are reflected through industry-specific ratios. For example, companies in the retail industry may report same-store sales changes because in the retail industry it is important to distinguish between growth that results from opening new stores and growth that results from generating more sales at existing stores.

LO: Describe how ratio analysis and other techniques can be used to model and forecast earnings.

Analysts often use common-size analysis and ratio analysis to prepare pro forma financial statements. They forecast future sales and combine the forecasted sales numbers with expected value for key ratios.

LM12 Introduction to Financial Statement Modeling

Version 1.0

1. Introduction

The learning module introduces us to financial modeling; we will learn how to forecast each item of the income statement, balance sheet, and cash flow statement, and subsequently construct a pro forma income statement, balance sheet and cash flow statement.

Then, we will discuss five key behavioral biases that influence the modeling process and strategies to mitigate them.

We will also look at the nuances required to build forecasting models such as: the impact of competitive forces on prices and costs, the effects of inflation and deflation, and long-term forecasting considerations.

2. Building a Financial Statement Model

Financial modeling typically begins with forecasting items on the income statement.

Income Statement Modeling: Revenue

Most companies have multiple sources of revenue. A company's revenue sources can be broken down by:

- Geographical regions
- Business segments
- Product lines

Once the sources of revenue are known, an analyst can use the following approaches to forecast revenue:

- The 'top-down' approach begins at the level of the overall economy. We then move down to sector, industry, and market for a specific product to forecast the revenue for an individual company.
- The 'bottom-up' approach begins at the level of the individual company or a unit within the company (e.g., a product line). We then sum up the projections for the individual products to forecast the total revenue for the company.
- The 'hybrid' approach combines top-down and bottom-up approaches. By using elements from both approaches, a hybrid approach can reveal implicit assumptions or errors that may arise from using a single approach.

<u>Top-down Approaches to Modeling Revenue</u>

Two common top-down approaches to modeling revenue are:

Growth relative to GDP growth approach: In this approach, we first forecast the growth rate of nominal GDP. We then forecast the company's revenue growth relative to GDP growth. For example, we can assume that the company's revenue will grow at a rate of 150 bps above the nominal GDP growth rate.

The forecast may also be in relative terms. For example, if we forecast that the GDP will grow at 4%, and we believe that the company's revenue will grow at a 20% faster rate, then the forecasted increase in the company's revenue is 4% x (1 + 0.20) = 4.8%.

Market growth and market share: In this approach, we combine forecasts of growth in particular markets with forecasts of a company's market share. For example, assume Tesla is expected to maintain a market share of 1% in the automobile market. If the automobile market is expected to grow to $30 billion in annual revenue, then Tesla's annual revenue is forecasted to grow to 1% * $30 billion = $300 million.

<u>Bottom-Up Approaches to Modeling Revenue</u>

Examples of bottom-up approaches include:

- *Time-series:* Forecasts based on historical growth rates or time-series analysis. For example, we may assume that the historical growth rate will continue, or that the growth rate will decline linearly from current rates to some long-run rate.
- *Return on capital:* Forecasts based on balance sheet accounts. For example, a bank's interest revenue can be calculated as loans multiplied by the average interest rate.
- *Capacity-based measure:* For example, a retailer's revenue may be forecasted based on same-store sales growth and sales related to new stores.

<u>Hybrid Approaches to Modeling Revenue</u>

Hybrid approaches are the most commonly used approaches in practice. They combine elements of both top-down and bottom-up approaches. For example, we may use a market growth and market share approach to model individual product lines or business segments (top-down), and then combine the individual projections to arrive at a forecast for the overall company (bottom-up).

Income Statement Modeling: Operating Costs

Operating costs include cost of goods sold (COGS) and selling, general, and administrative expenses (SG&A). Disclosures about operating costs are often less detailed than revenue. If information is available, then we can match cost analysis to revenue analysis. For example, costs may be modeled separately for different geographic regions, business segments or product lines.

Similar to revenue forecasting, costs can also be forecasted using a top-down, bottom-up, or hybrid approach:

- *Top-down approach*: Consider factors such as overall level of inflation, or industry-specific costs.
- *Bottom-up approach:* Consider factors such as segment-level margins, historical cost-growth rates, historical margin levels, etc.

- *Hybrid view:* Incorporate elements from both top-down and bottom-up approaches.

Some points that analysts must consider when projecting operating costs are:

- Since variable costs are linked to revenue growth, they can be forecasted as a percentage of revenue.
- Since fixed costs are not directly related to revenue, they are assumed to grow at their own rate or at the rate of future PP&E growth.
- Determine whether a company has economies of scale at the current level of output. Economies of scale means the average costs per unit of a good produced falls as volume increases. Gross and operating margins tend to be positively related to sales, if there are economies of scale.
- Be aware of the uncertainty related to cost estimates such as reserve accounts, competitive factors and technological developments. For example, banks and insurance companies create reserves against estimated future losses. However, the actual losses may not be known for many years.

Cost of Goods Sold

Since COGS are directly related to sales, we can forecast future COGS as a percentage of future sales. Analysts should understand the historical relationship between COGS and sales, and determine if this relationship is likely to decrease, increase, or remain unchanged.

Sales – COGS = gross margin. Therefore, COGS and gross margin are inversely related.

Some factors that analysts should consider while forecasting COGS are:

- Forecasting accuracy can be improved by forecasting COGS for the company's various segments or product categories separately.
- Determine if a company has employed hedging strategies to protect its gross margins. When input prices increase, COGS increase and gross margin decreases. However, hedging strategies can help mitigate this impact.
- Examine gross profit margins of competitors. This can be a useful cross check for estimating a realistic gross margin. Any difference in gross margins for companies in the same segment must relate to differences in their business operations.

SG&A Expenses

As compared to COGS, SG&A expenses are less directly related to revenue.

SG&A can be broken down into two components – fixed and variable. The fixed components such as R&D expenses, management salaries, head office expenses, supporting IT and administrative operations tend to increase and decrease gradually over time. They do not fluctuate with sales.

On the other hand, the variable components such as selling and distribution costs are more directly related to sales. For example, when sales increase, the company may pay out higher bonuses to its salesforce, it may also hire additional salespersons.

Income Statement Modelling: Non-Operating Costs and Other Items

Non-operating costs appear below operating profit on the income statement. The two significant non-operating costs that need to be forecasted are financing expenses (i.e. interest) and taxes.

<u>Financing Expenses</u>

Interest expense depends on the amount of debt on the balance sheet, as well as the interest rate associated with the debt. To forecast financing expenses, we first determine the capital structure of the company. The projected level of debt and the corresponding interest rates are then used to forecast the interest expense. To improve the quality of forecasts, we can use information provided in the notes to the financial statements about the maturity structure of the company's debt.

Some companies may also have interest income from investments. Interest income depends on the amount of cash and investments on the balance sheet, and the rates of return earned on investments. It is a key component of revenue for banks and insurance companies, but less important for non-financial companies.

<u>Corporate Income Tax</u>

There are three types of tax rates:

- *Statutory tax rate*: The tax rate established by law.
- *Effective tax rate:* Tax amount reported on the income-tax statement divided by the pre-tax income.
- *Cash tax rate:* Tax actually paid divided by pre-tax income.

Differences between cash taxes and reported taxes occur due to timing differences between accounting and tax calculations. These differences are reflected as a deferred tax asset or a deferred tax liability on the balance sheet.

Statutory tax rate and effective tax rates can be different for a number of reasons, such as tax credits, adjustments to previous years, withholding tax on dividends, etc.

The effective tax rate of a company that operates in multiple countries is the weighted average of the effective tax rates in each country. If the company is expected to report higher profits in a country with a high tax rate and lower profits in a country with a low tax rate, then its effective tax rate will increase. The effective tax rate is used to forecast earnings on the income statement and the cash tax rate is used to forecast cash flows.

To improve the quality of forecasts, the analyst should adjust for one-time events. Information provided in the notes to the financial statements can be used to identify such events.

If a company's effective tax rates are consistently lower than statutory rates or the effective tax rates reported by competitors, then analysts must scrutinize the reason when forecasting tax expense.

Income Statement Modeling: Other Items

A few other items on the income statement that can be forecasted are:

Dividends: Dividends are often forecasted to grow each year by a certain dollar amount or as a proportion of net income.

Income from affiliates: There are two possibilities here, depending on a company's ownership in an affiliate:

- More than 50%: consolidate the affiliate's results with its own. Report the portion of income that does not belong to the parent as minority interest.
- Less than 50%: Report its share of income from the affiliate under the equity method.

Share count: Share count can change under three circumstances: dilution related to stock options and convertible bonds, issuance of new shares, and share repurchases. Analysts must consider the market price of a stock and the capital structure of the company when projecting share count.

Unusual charges: These are difficult to predict and are usually not forecasted.

Balance Sheet and Cash Flow Statement Modeling

Income statement modeling is the foundation for modeling balance sheet and cash flow statements. Analysts can choose whether to focus on the balance sheet or the cash flow statement. Here we focus on the balance sheet. Cash flow statement can be derived based on the income statement and balance sheet.

Balance sheet modeling

Some balance sheet items flow directly from the forecasted income statement. For example, net income minus dividends will flow through to retained earnings.

Other balance sheet items are closely linked to the forecasted income statement and can be projected based on their historical relationship with income statement items. For example, working capital accounts are forecasted through efficiency ratios.

- Projected accounts receivable = forecasted sales x days sales outstanding/365
- Projected inventory = forecasted COGS / inventory turnover

If efficiency ratios are held constant, then working capital accounts will grow at the same rate as income statement accounts.

Long-term assets such as net PP&E are less directly related to the income statement. PP&E depends on capital expenditures and depreciation. Depreciation forecasts are based on historical depreciation, whereas capital expenditure forecasts are based on the analyst's estimate of the future need for new PP&E. To improve the quality of forecasts, capital expenditure can be broken down into maintenance capital expenditure (needed to sustain current business) and growth capital expenditure (needed to expand the business). Each component can be forecasted separately. Due to inflation, maintenance capital expenditure forecasts should normally be higher than historical deprecation.

Finally, to forecast the capital structure, analysts use leverage ratios (such as debt to equity, debt to capital).

Summary

The general steps while building a financial statement model are summarized below.

1. Perform an industry overview using the Porter's five forces model.
2. Perform a company overview to account for company specific factors. If applicable, divide the company's revenue sources into different segments.
3. Construct a pro-forma income statement.
    a. Revenue can be forecasted using a top-down, bottom-up or hybrid approach.
    b. COGS can be forecasted as a percentage of sales using historical relationships.
    c. In SG&A, selling and distribution costs such as wages are variable and can be estimated as a percentage of sales. General and administrative expenses are more or less fixed or increase/decrease gradually.
    d. Financing costs can be forecasted using the expected level of debt and the corresponding interest rates.
    e. Income tax expense can be forecasted based on the trends in the historical effective tax rate.
4. Construct a pro-forma balance sheet.
    a. Retained earnings will flow from the income statement.
    b. Working capital accounts such as inventory, accounts receivables and accounts payable can be forecasted through efficiency ratios and corresponding income statement accounts.
    c. PP&E can be forecasted based on projected capital expenditures (both maintenance and growth) and depreciation.
5. Construct a pro-forma cash flow statement using the income statement and the balance sheet.
6. Determine the intrinsic value of the company using an appropriate valuation model.
    a. Forecast earnings or some other cashflow measure over the forecast horizon.
    b. Determine a terminal value using a multiples-based approach or DFC approach.

c. Determine an appropriate discount rate for the company.

3. Behavioral Finance and Analyst Forecasts

Five key biases that influence analyst forecasts are:

Bias	Description	Effect	Remedial action
Overconfidence Bias	A bias in which people demonstrate unwarranted faith in their own abilities.	Overconfidence arises more frequently when making contrarian predictions that others do not expect.	Analysts should record and share their forecasts and review them regularly, identifying both the correct and incorrect forecasts they have made.
Illusion of Control Bias	Tendency to overestimate the ability to control what cannot be controlled and to take ultimately fruitless actions in pursuit of control.	Analysts believe that forecasts can be made more accurate in two ways: by acquiring more information and opinions from experts and by creating more granular and complex models.	Restrict modeling variables to those that are regularly disclosed by the company, focus on the most important or impactful variables, and speak only with those who are likely to have unique or significant perspectives.
Conservatism Bias	People maintain their prior views or forecasts by inadequately incorporating new information.	Analysts do not update their forecasts after receiving conflicting information, such as disappointing earnings results or a competitor action.	Periodic reviews of forecasts and models by an investment team at a regular interval, such as each quarter. Create flexible models with fewer variables, to make changing assumptions easier
Representativeness Bias	Tendency to classify information based on past experiences and known classification.	Leads to base-rate neglect in which a phenomenon's rate of incidence in a larger population, or characteristics of a larger class to which a specific member belongs—its base rate—is neglected in favor of	Start with the base rate but determine which factors make the target company different from the base rate or class average and what the implications of those differences are, if any.

		situation- or member-specific information.	
Confirmation Bias	Tendency to look for and notice what confirms prior beliefs and to ignore or undervalue whatever contradicts them.	Example: An analyst who has a positive view on a company might speak only to other analysts who share that view and the company's management, all of whom will likely tell the analyst what they want to hear and already know.	Speak to or read research from analysts with a negative opinion on the security under scrutiny. Seek perspectives from colleagues who are not economically or psychologically invested in the subject security.

4. The Impact of Competitive Factors in Prices and Costs

Competitive factors affect a company's ability to negotiate lower input prices with suppliers and to raise prices for products and services. Porter's five forces framework can be used as a basis for identifying such factors. The table below summarizes this framework.

Force	**Comment**
Threat of substitutes	Fewer substitutes or higher switching costs increase industry profitability.
Internal rivalry	Lower rivalry increases industry profitability.
Supplier power	Presence of many suppliers limits their pricing power, which in turn does not put a downward pressure on the industry's profitability.
Customer power	Presence of many buyers limits their negotiating power, which in turn does not put a downward pressure on the industry's profitability.
Threat of new entrants	High barriers to entry increase industry profitability.

5. Modeling Inflation and Deflation

Inflation is an increase in the price of goods and services, whereas deflation is a decrease in the prices of goods and services. They can significantly affect the accuracy of forecasts of revenue, profit and cash flow. However, the degree of impact varies from company to company. Some companies are better able to pass on higher input costs to their customers by selling their output at higher prices. Such companies are more likely to have higher and more stable profits and cash flow, relative to competitors.

Sales Projections with Inflation and Deflation

We will now analyze how inflation and deflation affect industry and company sales forecasts.

Industry sales and inflation/deflation

- Higher input costs usually result in higher prices for end products. The industry structure determines how fast the increase in input prices is transmitted to output prices.
- Suppose input costs go up by 5%. To offset this, the selling price is increased by 5% and the same sales volume is maintained. Under this scenario, the gross profit margin percentage would be the same but the absolute amount of gross profit would increase.
- However, most products have an elastic demand. Therefore, when prices are increased, sales volume decreases.
- The decline in demand will not only depend on the elasticity of demand but also on the availability of substitutes, competitor behavior, etc. In an inflationary environment, companies that raise prices too soon will experience volume losses. Whereas companies that raise prices too late will experience declining profit margins.

Company sales and inflation/deflation

When forecasting the revenue of a company, identify a company's main input costs and the company's strategy for how it responds to rising input prices. This will help in determining the pricing. Will it pass on the increase in costs to customers, or accept margin reduction to retain market share? Revenue projections are based on expected price and volume development.

We consider the following factors to analyze the effect of inflation:

- *Price elasticity of the products:* If demand is relatively price inelastic, revenue will benefit from inflation. If demand is price elastic, revenues will decrease even if unit prices are increased because volumes decline.
- *Different rates of cost inflation:* Consider the geographic mix of a company's operations to reflect different rates of the cost of inflation among countries. High inflation in export market might increase sales in foreign currency terms, but that gain might be lost if the foreign currency depreciates.
- *Inflation in costs of a company's individual product category:* For example, an increase in grain prices is likely to impact a bakery more than a supermarket.

Cost Projections with Inflation and Deflation

We will now analyze how inflation and deflation affect industry and company cost forecasts.

Industry costs and inflation/deflation

- Understanding the purchasing characteristics of an industry can be useful in forecasting costs. For example, if the industry uses hedging instruments such as long-term price-fixed forward contracts, then input price increases will be gradual instead of a sudden hike.
- Analysts should monitor the underlying drivers of input prices. For example, weather conditions can have a significant impact on the price of agricultural products and consequently on the input costs of industries that rely on agricultural products.
- The impact of inflation or deflation on an industry's cost structure depends on the competitive structure. For example, if companies have access to alternative inputs or are vertically integrated, the impact of volatility of input costs will be low.
- An analyst may also examine if a company can cut costs in other areas such as marketing and advertising to maintain margins in the short term.

<u>Company costs and inflation/deflation</u>

- While forecasting a company's costs, break down the cost structure by category and geography. For each cost element, assess the impact of inflation and deflation on input prices.
- Consider the company's ability to use cheaper alternatives for expensive inputs when input prices rise.

6. The Forecast Horizon and Long-Term Forecasting

Forecast time horizon is influenced by factors such as:

- <u>Investment strategy for which the stock is being considered</u>: For most investors, the forecast horizon is usually the expected holding period of the stock. Holding period corresponds to the portfolio turnover: average holding period = 1/portfolio turnover. For example, an annual portfolio turnover of 20% indicates an average holding period of 5 years.
- <u>Cyclicality of the industry</u>: The forecast period should be long enough for the business to reach the mid-cycle level of sales and profitability.
- <u>Company specific factors</u>: Factors such as recent acquisition or restructuring can also influence the forecast horizon. The forecast period should be long enough to allow the expected benefits from such an activity to be realized.
- <u>Analyst's employer preferences</u>: An analyst's employer may specify what forecast horizon to use.

Longer-term projections often provide a better representation of the normalized earnings potential of a company than a short-term forecast, especially when certain temporary factors are present. **Normalized earnings** are the expected level of mid-cycle earnings for a company in the absence of any unusual or temporary factors that impact profitability (either positively or negatively).

"Growth relative to GDP growth" and "market growth and market share" methods can be applied to develop longer term revenue projections. Once revenue is projected, previously described methods are used to forecast costs and to complete the income statement, balance sheet and cash flow statement.

Along with earnings projections over the forecast period, analysts also determine the terminal value of the stock at the end of the forecast period. Terminal values are usually estimated using a multiples-based approach or a DCF approach.

When using a multiples-based approach, the choice of the multiple should be consistent with the long-run expectations for growth and required return. Generally, a historical multiple is used as a basis for the target multiple. If the future growth is likely to differ from the historical average, then the target multiple should include a premium or discount to account for this difference. For example, say a stock's median P/E over the last decade was 12. If the company is expected to perform very well in the future, then the P/E multiple should be adjusted upward while calculating the terminal value (and vice versa).

When using a DCF approach, two important parameters are the terminal year cash flow projections and the long-term growth rate. The terminal year cash flow should be normalized to a mid-cycle value to remove the impact of short-term events. An appropriate long-term perpetual growth rate should be determined that matches the future outlook of the company. Using an unrealistic long-term growth rate will make the terminal value projections incorrect. The greatest challenge that analysts face while using a DCF approach is anticipating inflection points, i.e., when the future will look significantly different from the past. Inflection points can occur due to: business cycles, economic disruptions, regulatory and technological changes, etc.

Summary

LO: Demonstrate the development of a sales-based pro forma company model

The general steps while building a financial statement model are summarized below.

1. Perform an industry overview using the Porter's five forces model.
2. Perform a company overview to account for company specific factors. If applicable, divide the company's revenue sources into different segments.
3. Construct a pro-forma income statement.
   a. Revenue can be forecasted using a top-down, bottom-up or hybrid approach.
   b. COGS can be forecasted as a percentage of sales using historical relationships.
   c. In SG&A, selling and distribution costs such as wages are variable and can be estimated as a percentage of sales. General and administrative expenses are more or less fixed or increase/decrease gradually.
   d. Financing costs can be forecasted using the expected level of debt and the corresponding interest rates.
   e. Income tax expense can be forecasted based on the trends in the historical effective tax rate.
4. Construct a pro-forma balance sheet.
   a. Retained earnings will flow from the income statement.
   b. Working capital accounts such as inventory, accounts receivables and accounts payable can be forecasted through efficiency ratios and corresponding income statement accounts.
   c. PP&E can be forecasted based on projected capital expenditures (both maintenance and growth) and depreciation.
5. Construct a pro-forma cash flow statement using the income statement and the balance sheet.
6. Determine the intrinsic value of the company using an appropriate valuation model.
   a. Forecast earnings or some other cashflow measure over the forecast horizon.
   b. Determine a terminal value using a multiples-based approach or DFC approach.
   c. Determine an appropriate discount rate for the company.

LO: Explain how behavioral factors affect analyst forecasts and recommend remedial actions for analyst biases.

Five key behavioral biases that influence analyst forecasts are overconfidence, illusion of control, conservatism, representativeness, and confirmation bias.

LO: Explain how the competitive position of a company based on a Porter's five forces analysis affects prices and costs.

Competitive factors affect a company's ability to negotiate lower input prices with suppliers and to raise prices for products and services. Porter's five forces framework can be used as a basis for identifying such factors.

Force	Comment
Threat of substitutes	Fewer substitutes or higher switching costs increase industry profitability.
Internal rivalry	Lower rivalry increases industry profitability.
Supplier power	Presence of many suppliers limits their pricing power, which in turn does not put a downward pressure on the industry's profitability.
Customer power	Presence of many buyers limits their negotiating power, which in turn does not put a downward pressure on the industry's profitability.
Threat of new entrants	High barriers to entry increase industry profitability.

LO: Explain how to forecast industry and company sales and costs when they are subject to price inflation or deflation.

<u>Effect on sales</u>:

- Higher input costs usually result in higher prices for end products. Since most products have an elastic demand, when prices are increased sales volume decrease.
- In an inflationary environment, companies that raise prices too soon will experience volume losses. Whereas, companies that raise prices too late will experience declining profit margins.

<u>Effect on costs</u>:

- If the company uses hedging instruments such as long-term forward contracts, then input price increases will be gradual instead of a sudden hike.
- If companies have access to alternative inputs or are vertically integrated, the impact of volatility of input costs will be low.

LO: Explain considerations in the choice of an explicit forecast horizon and an analyst's choices in developing projections beyond the short-term forecast horizon.

The factors to consider when choosing an explicit time horizon include the projected holding period, an investor's average portfolio turnover, cyclicality of an industry, company specific factors, and employer preferences.

Along with earnings projections over the forecast period, analysts also determine the terminal value of the stock at the end of the forecast period. Terminal values are usually estimated using a multiples-based approach or a DCF approach.

- While using the multiples-based approach, if the future growth is likely to differ from the historical average, then the target multiple should include a premium or discount to account for this difference.

- While using the DCF approach, the terminal year cash flow should be normalized to a mid-cycle value, to remove the impact of short-term events. An appropriate long-term perpetual growth rate should be determined that matches the future outlook of the company.

